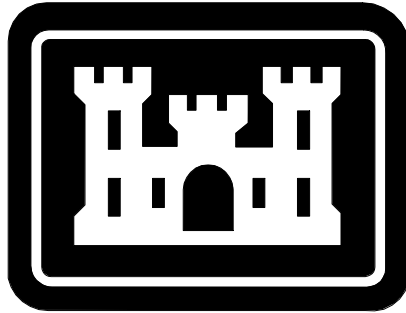




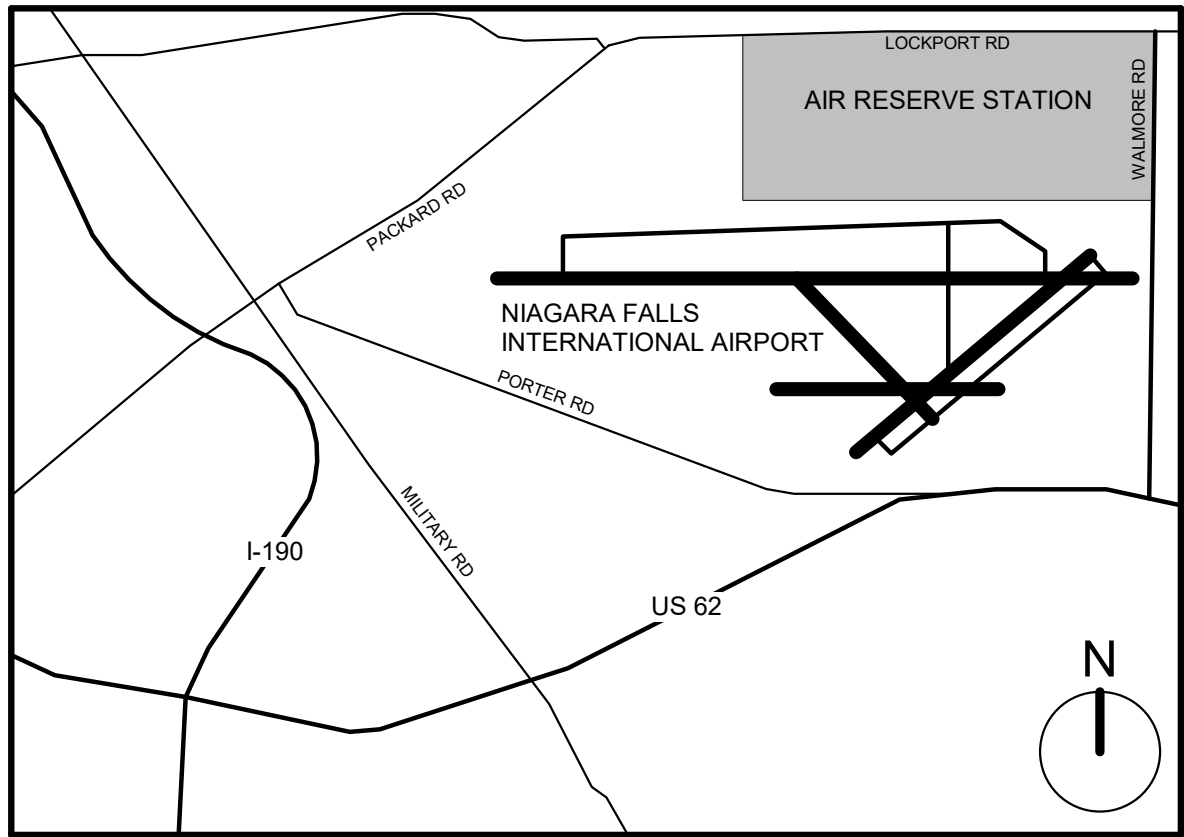
US Army Corps
of Engineers ®
LOUISVILLE DISTRICT

15 March 2025
W912QR25BA008

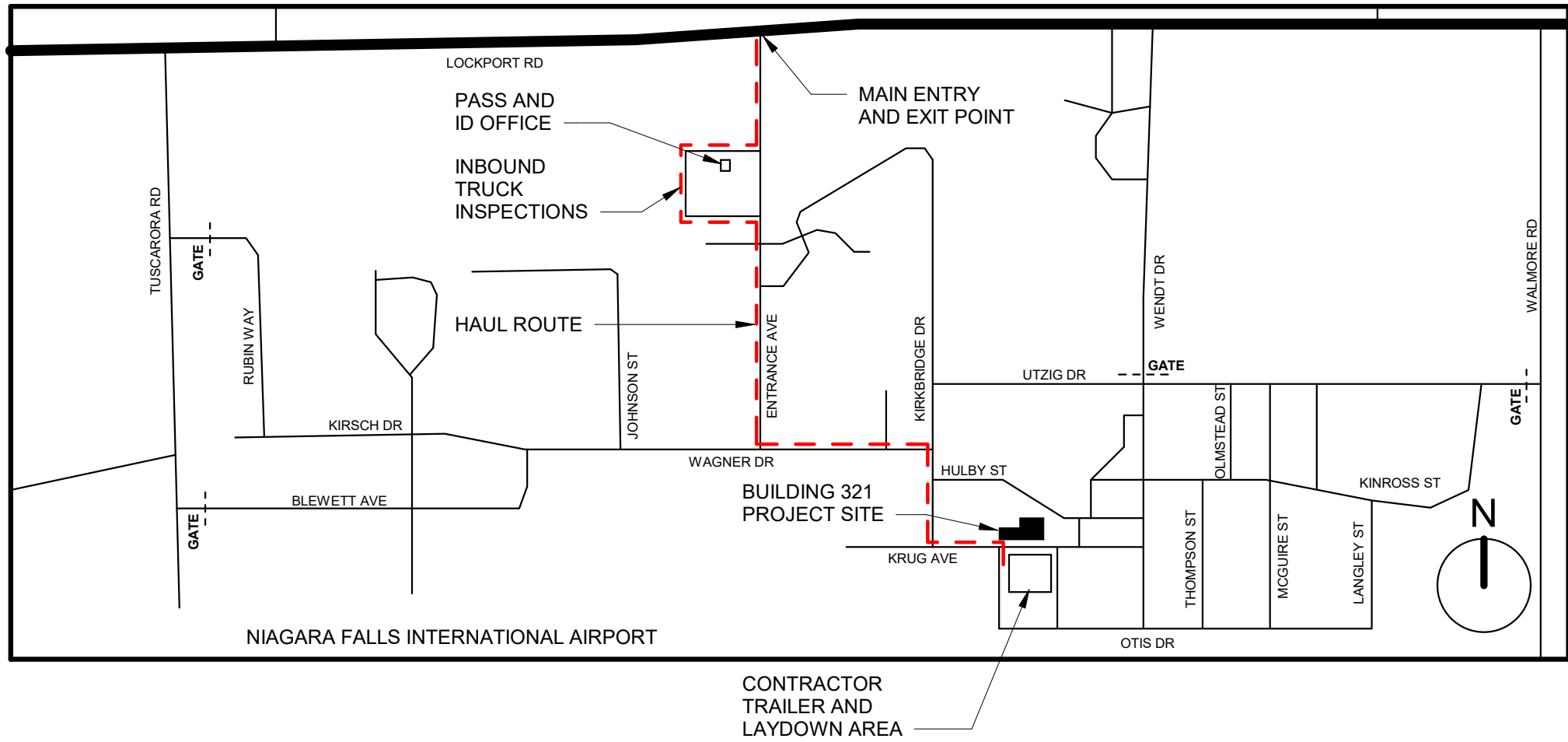
BOWST ADDITION AND RENOVATIONS TO B-321
NIAGARA FALLS AIR RESERVE STATION, NY

P2#: 508014

AREA MAP



NFARS MAP



SOLICITATION NO.:
CONTRACT NO.:
ISSUE DATE: JUNE 26, 2024
VOLUME:



CERTIFIED FINAL



MARK	DESCRIPTION	DATE

DESIGNED BY: A. LUGNBILL	ISSUE DATE: JUNE 26, 2024
DRAWN BY: A. MORRISON	SOLICITATION NO.:
CHECKED BY: S. DUFFY	CONTRACT NO.:
SUBMITTED BY:	
SIZE: ANSI D	

US ARMY CORPS OF ENGINEERS
LOUISVILLE DISTRICT
600 DR. MARTIN LUTHER KING, JR. PLACE
LOUISVILLE KY 40202-2239

AECemersion
joint venture

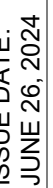
BOWST ADDITION AND RENOVATIONS TO B-321 NIAGARA FALLS AIR RESERVE STATION, NY P2#: 508014 COVER SHEET
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SHEET ID G-001

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US Army Corps
of Engineers®

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	US ARMY CORPS OF ENGINEERS LOUISVILLE DISTRICT 600 DR. MARTIN LUTHER KING, JR. PLACE LOUISVILLE KY 40202-2239		DESIGNED BY: P. BOWMAN	ISSUED DATE: JUNE 26, 2024
			DRAWN BY: P. BOWMAN	SOLICITATION NO.:
			CHECKED BY: B. JOHNSON	CONTRACT NO.:
			SUBMITTED BY:	
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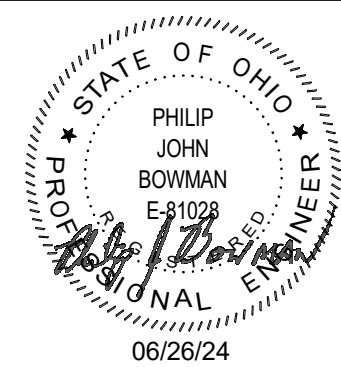
**BOWST ADDITION AND RENOVATIONS TO B-321
NIAGARA FALLS AIR RESERVE STATION, NY**

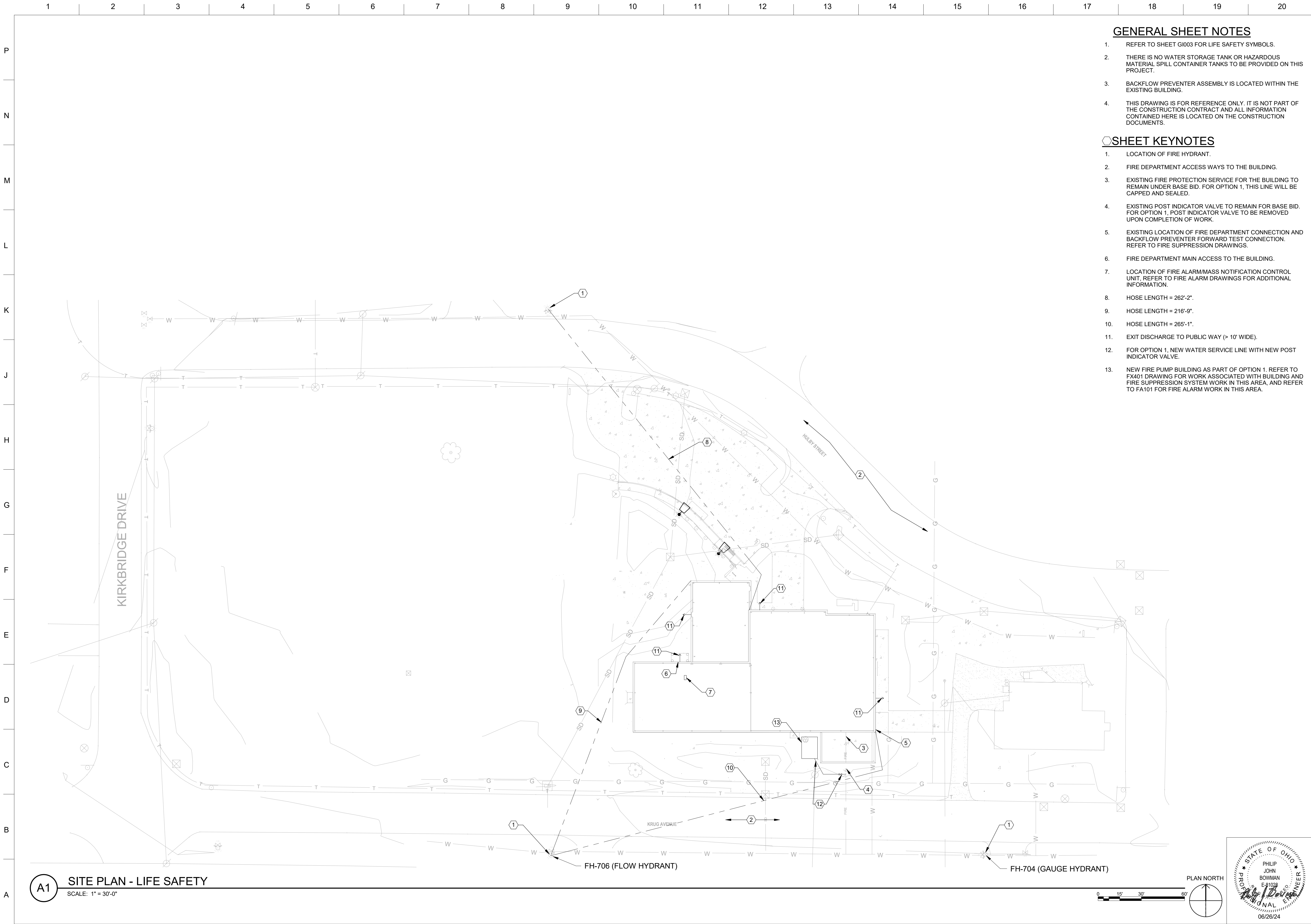
P2#: 508014

GENERAL INFORMATION - LIFE SAFETY

SHEET ID

GI002



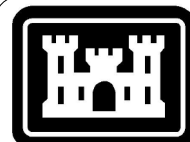


GENERAL SHEET NOTES

- 1. REFER TO SHEET G1003 FOR LIFE SAFETY SYMBOLS.
- 2. THERE IS NO WATER STORAGE TANK OR HAZARDOUS MATERIAL SPILL CONTAINER TANKS TO BE PROVIDED ON THIS PROJECT.
- 3. BACKFLOW PREVENTER ASSEMBLY IS LOCATED WITHIN THE EXISTING BUILDING.
- 4. THIS DRAWING IS FOR REFERENCE ONLY. IT IS NOT PART OF THE CONSTRUCTION CONTRACT AND ALL INFORMATION CONTAINED HERE IS LOCATED ON THE CONSTRUCTION DOCUMENTS.

SHEET KEYNOTES

- 1. LOCATION OF FIRE HYDRANT.
- 2. FIRE DEPARTMENT ACCESS WAYS TO THE BUILDING.
- 3. EXISTING FIRE PROTECTION SERVICE FOR THE BUILDING TO REMAIN UNDER BASE BID. FOR OPTION 1, THIS LINE WILL BE CAPPED AND SEALED.
- 4. EXISTING POST INDICATOR VALVE TO REMAIN FOR BASE BID. FOR OPTION 1, POST INDICATOR VALVE TO BE REMOVED UPON COMPLETION OF WORK.
- 5. EXISTING LOCATION OF FIRE DEPARTMENT CONNECTION AND BACKFLOW PREVENTER FORWARD TEST CONNECTION. REFER TO FIRE SUPPRESSION DRAWINGS.
- 6. FIRE DEPARTMENT MAIN ACCESS TO THE BUILDING.
- 7. LOCATION OF FIRE ALARM/MASS NOTIFICATION CONTROL UNIT. REFER TO FIRE ALARM DRAWINGS FOR ADDITIONAL INFORMATION.
- 8. HOSE LENGTH = 262'-2".
- 9. HOSE LENGTH = 216'-9".
- 10. HOSE LENGTH = 265'-1".
- 11. EXIT DISCHARGE TO PUBLIC WAY (> 10' WIDE).
- 12. FOR OPTION 1, NEW WATER SERVICE LINE WITH NEW POST INDICATOR VALVE.
- 13. NEW FIRE PUMP BUILDING AS PART OF OPTION 1. REFER TO FX401 DRAWING FOR WORK ASSOCIATED WITH BUILDING AND FIRE SUPPRESSION SYSTEM WORK IN THIS AREA, AND REFER TO FA101 FOR FIRE ALARM WORK IN THIS AREA.



US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

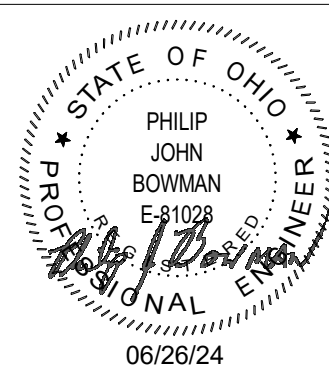
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CHECKED BY: B. JOHNSON	CONTRACT NO.:
SUBMITTED BY:	
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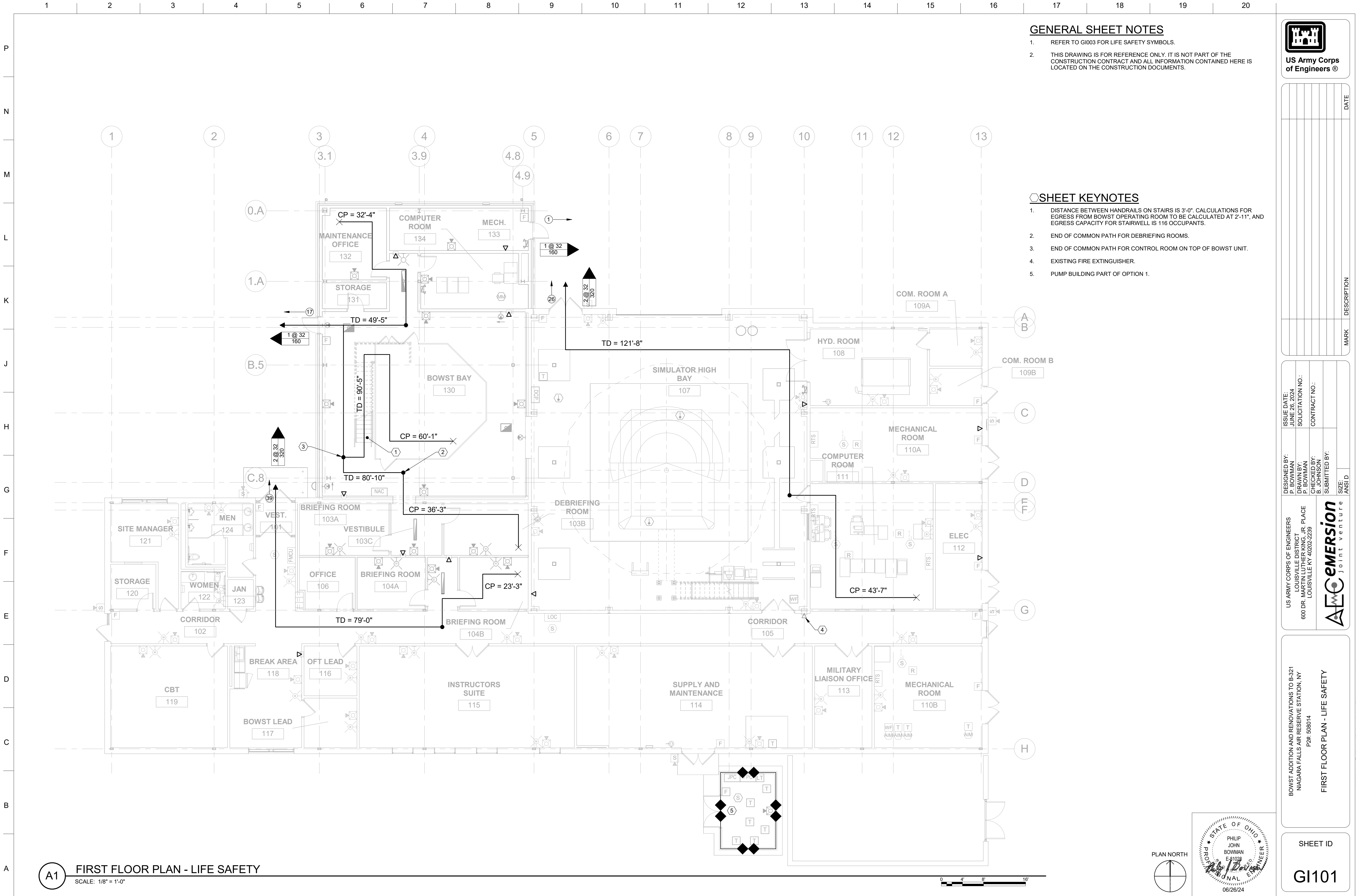
US ARMY CORPS OF ENGINEERS
LOUISVILLE DISTRICT
600 DR. MARTIN LUTHER KING, JR. PLACE
LOUISVILLE KY 40202-2239

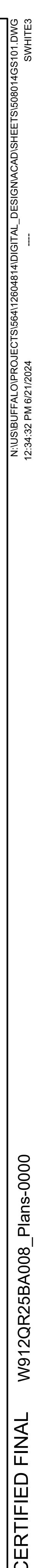
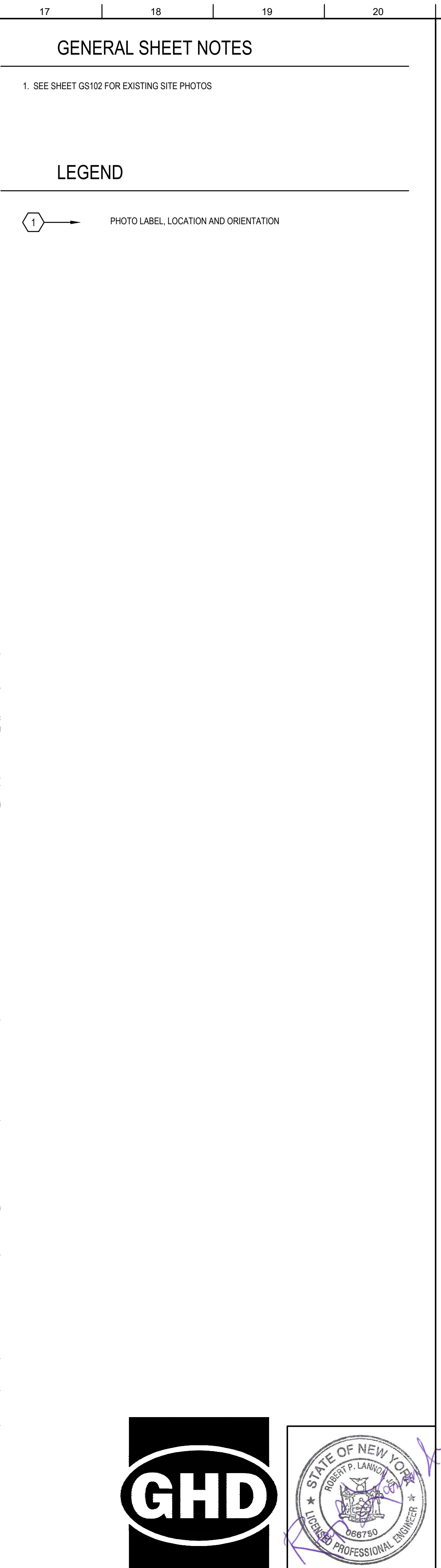
AECemersion
joint venture

BOWST ADDITION AND RENOVATIONS TO B-321 NIAGARA FALLS AIR RESERVE STATION, NY P2# 508014	SITE PLAN - LIFE SAFETY
--	-------------------------

SHEET ID
G100







1. SEE SHEET GS101 FOR EXISTING SITE PHOTO LOCATIONS.



SITE PHOTOGRAPHS PLAN

SCALE: 1"=20'

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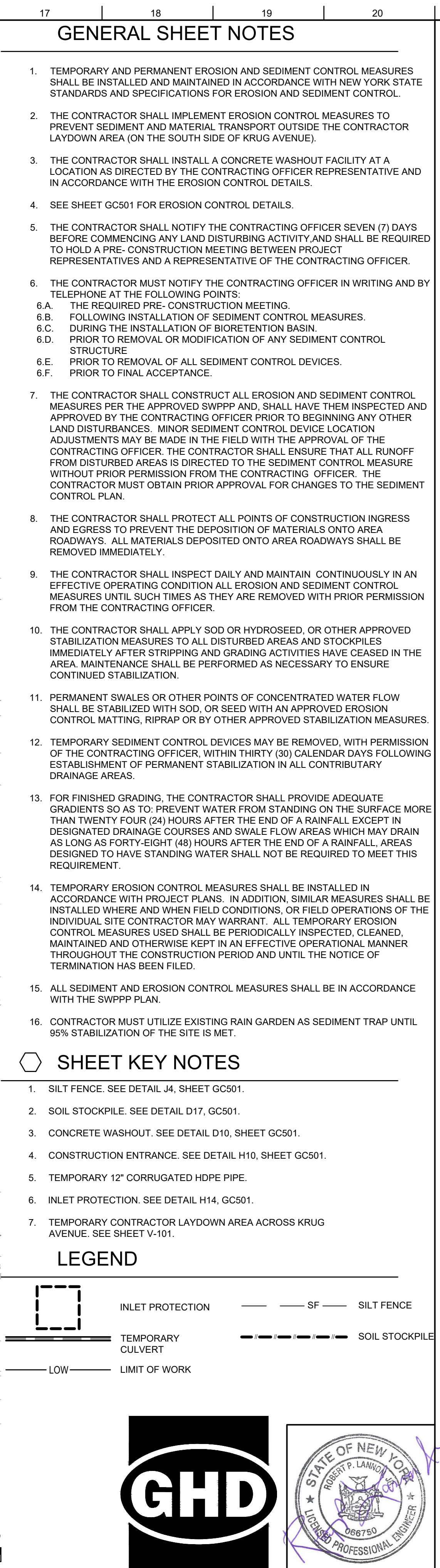
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DRAWN BY:	S.WHITE	SOLICITATION NO.:	
CHECKED BY:	M.BISIGNANI	CONTRACT NO.:	
SUBMITTED BY:			
SIZE:	ANSI D		

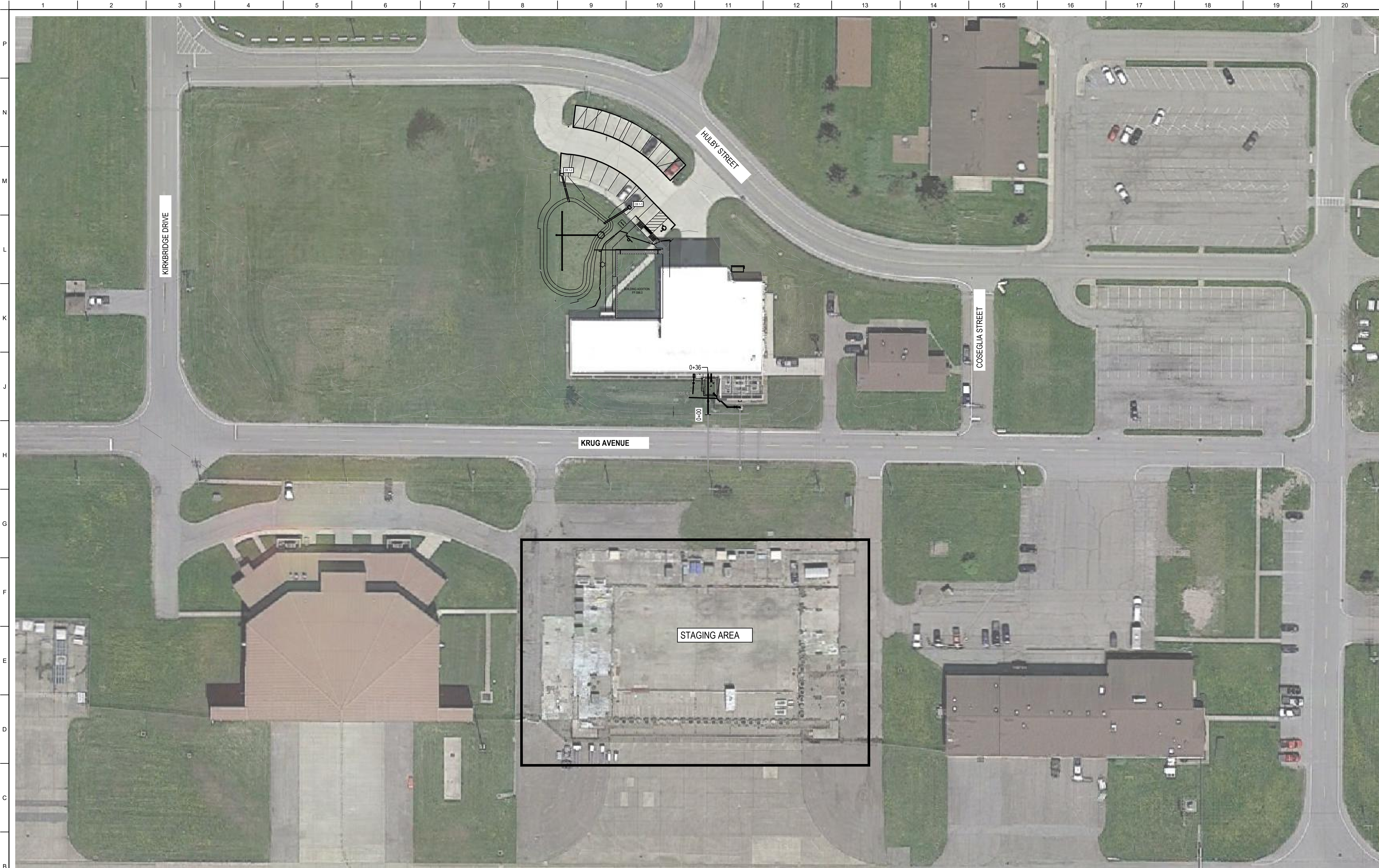


SITE PHOTOGRAPHS

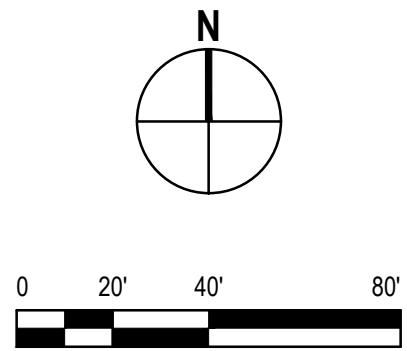
GS102



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A1 AERIAL PHOTOGRAPH
SCALE: 1"=20'



NIAGARA FALLS ARS, NIAGARA FALLS NY
BOWST ADDITION TO B321
PS# 508014

AERIAL PHOTOGRAPH

U.S. ARMY CORPS OF ENGINEERS
LOUISVILLE DISTRICT
600 MARTIN LUTHER KING, JR. PLACE
LOUISVILLE KY 40202-2239



DESIGNED BY:
M. GATES
DRAWN BY:
M. BISHOP
CHECKED BY:
M. BISHOP
SUBMITTED BY:

SIZE:
ANSI D

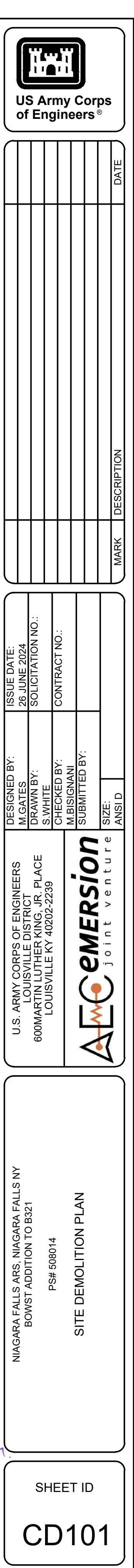
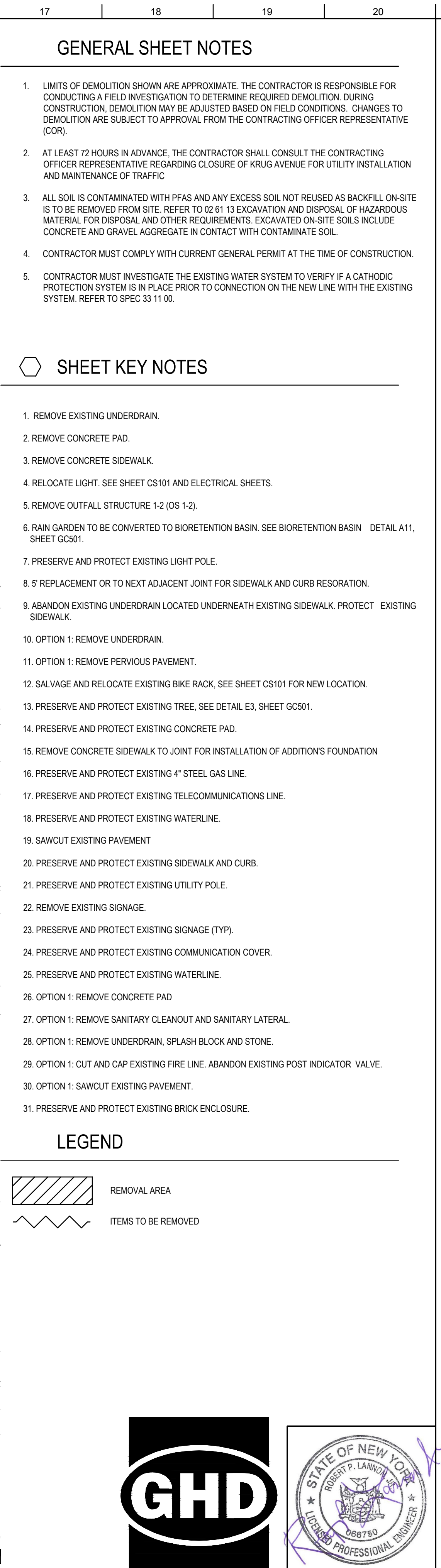
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SOLICITATION NO.:

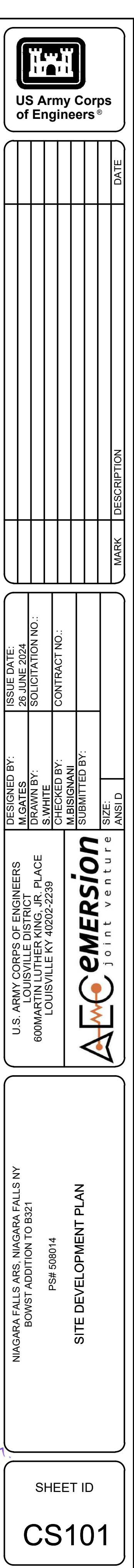
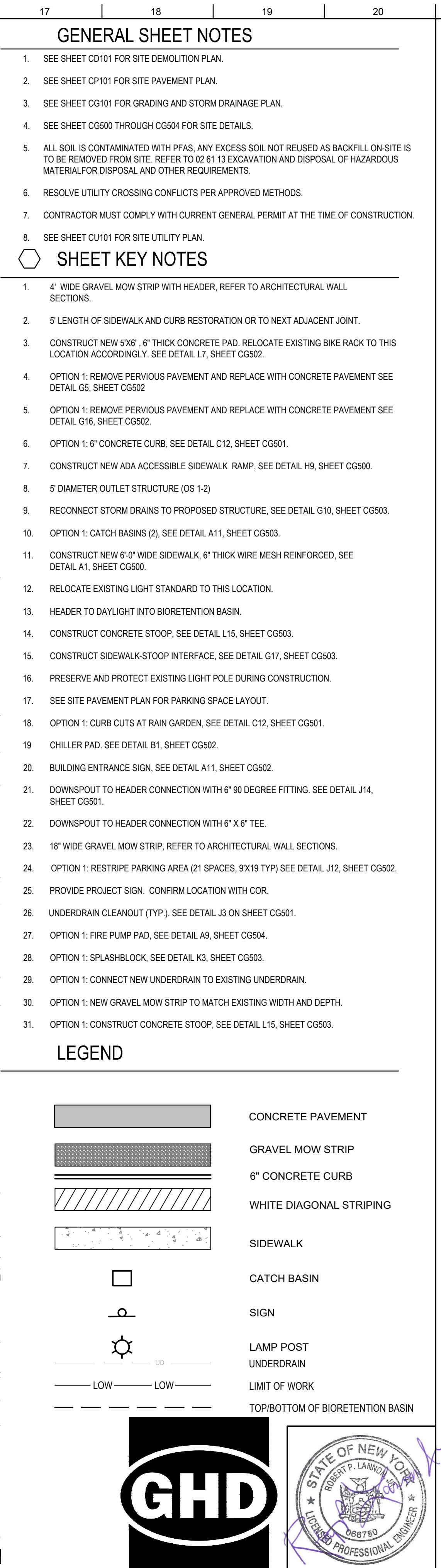
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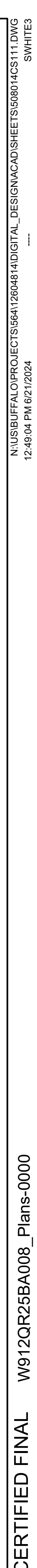
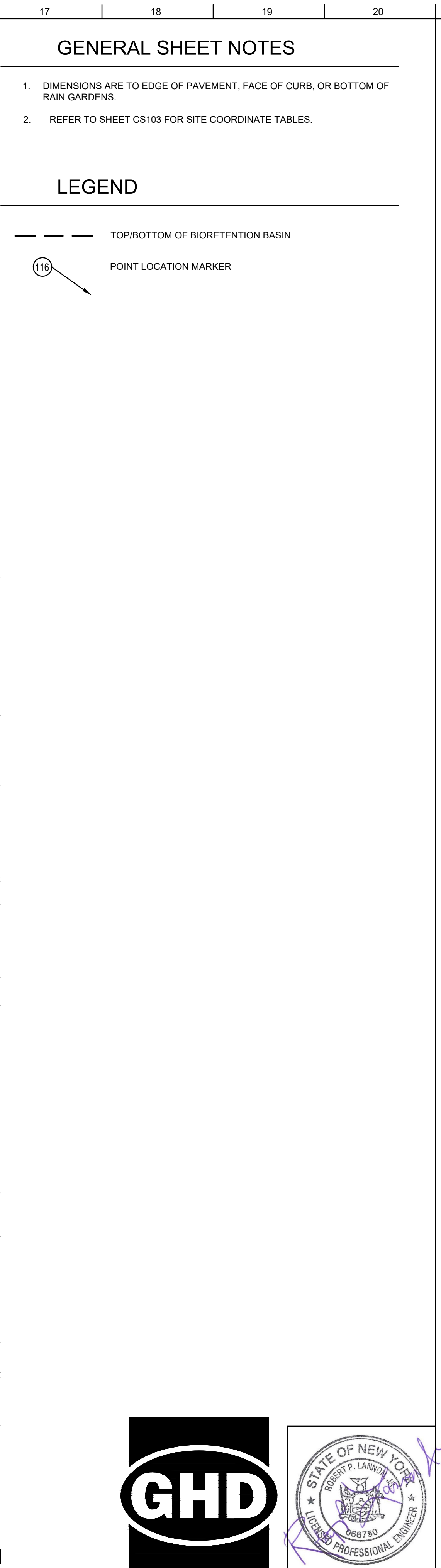
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DESCRIPTION
DATE

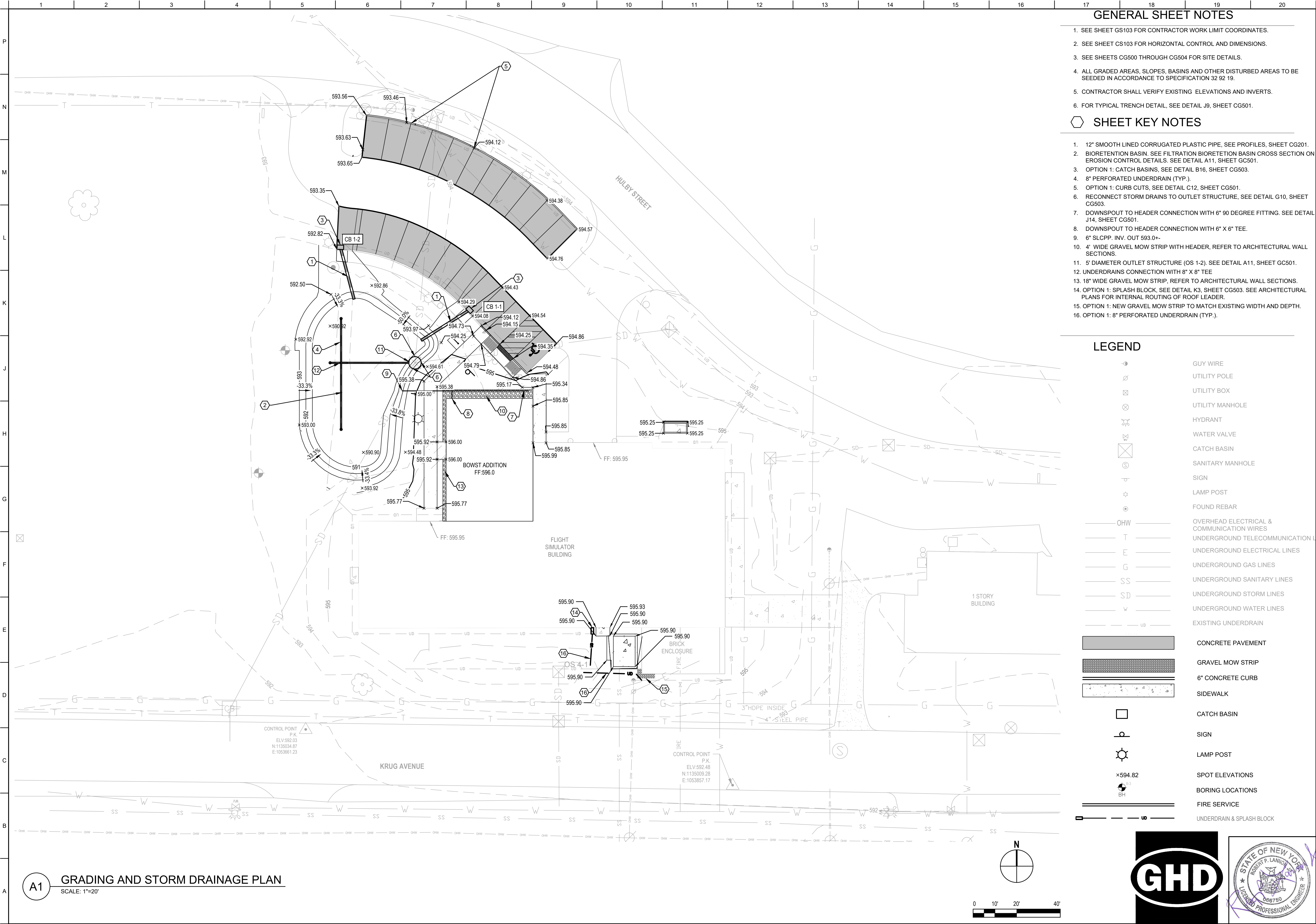


US Army Corps of Engineers®









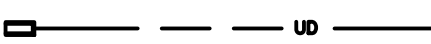
GENERAL SHEET NOTES

1. SEE SHEET GS103 FOR CONTRACTOR WORK LIMIT COORDINATES.
2. SEE SHEET CS103 FOR HORIZONTAL CONTROL AND DIMENSIONS.
3. SEE SHEETS CG500 THROUGH CG504 FOR SITE DETAILS.
4. ALL GRADED AREAS, SLOPES, BASINS AND OTHER DISTURBED AREAS TO BE SEEDED IN ACCORDANCE TO SPECIFICATION 32 92 19.
5. CONTRACTOR SHALL VERIFY EXISTING ELEVATIONS AND INVERTS.
6. FOR TYPICAL TRENCH DETAIL, SEE DETAIL J9, SHEET CG501.

SHEET KEY NOTES

1. 12" SMOOTH LINED CORRUGATED PLASTIC PIPE, SEE PROFILES, SHEET CG201.
2. BIORETENTION BASIN. SEE FILTRATION BIORETENTION BASIN CROSS SECTION ON EROSION CONTROL DETAILS. SEE DETAIL A11, SHEET GC501.
3. OPTION 1: CATCH BASINS, SEE DETAIL B16, SHEET CG503.
4. 8" PERFORATED UNDERDRAIN (TYP.).
5. OPTION 1: CURB CUTS, SEE DETAIL C12, SHEET CG501.
6. RECONNECT STORM DRAINS TO OUTLET STRUCTURE, SEE DETAIL G10, SHEET CG503.
7. DOWNSPOUT TO HEADER CONNECTION WITH 6" 90 DEGREE FITTING. SEE DETAIL J14, SHEET CG501.
8. DOWNSPOUT TO HEADER CONNECTION WITH 6" X 6" TEE.
9. 6" SLCPP. INV. OUT 593.0+.
10. 4' WIDE GRAVEL MOW STRIP WITH HEADER, REFER TO ARCHITECTURAL WALL SECTIONS.
11. 5' DIAMETER OUTLET STRUCTURE (OS 1-2). SEE DETAIL A11, SHEET GC501.
12. UNDERDRAINS CONNECTION WITH 8" X 8" TEE
13. 18" WIDE GRAVEL MOW STRIP, REFER TO ARCHITECTURAL WALL SECTIONS.
14. OPTION 1: SPLASH BLOCK, SEE DETAIL K3, SHEET CG503. SEE ARCHITECTURAL PLANS FOR INTERNAL ROUTING OF ROOF LEADER.
15. OPTION 1: NEW GRAVEL MOW STRIP TO MATCH EXISTING WIDTH AND DEPTH.
16. OPTION 1: 8" PERFORATED UNDERDRAIN (TYP.).

LEGEND

	GUY WIRE
	UTILITY POLE
	UTILITY BOX
	UTILITY MANHOLE
	HYDRANT
	WATER VALVE
	CATCH BASIN
	SANITARY MANHOLE
	SIGN
	LAMP POST
	FOUND REBAR
	OVERHEAD ELECTRICAL & COMMUNICATION WIRES
	UNDERGROUND TELECOMMUNICATION LINES
	UNDERGROUND ELECTRICAL LINES
	UNDERGROUND GAS LINES
	UNDERGROUND SANITARY LINES
	UNDERGROUND STORM LINES
	UNDERGROUND WATER LINES
	EXISTING UNDERDRAIN
	CONCRETE PAVEMENT
	GRAVEL MOW STRIP
	6" CONCRETE CURB
	SIDEWALK
	CATCH BASIN
	SIGN
	LAMP POST
	SPOT ELEVATIONS
	BORING LOCATIONS
	FIRE SERVICE
	UNDERDRAIN & SPLASH BLOCK

ISSUE DATE: 26 JUNE 2024	SOLICITATION NO.:	CONTRACT NO.:
DESIGNED BY: MGATES	DRAWN BY: M.BISIGNANI	SUBMITTED BY: M.BISIGNANI

U.S. ARMY CORPS OF ENGINEERS LOUISVILLE DISTRICT 600 MARTIN LUTHER KING, JR. PLACE LOUISVILLE, KY 40202-2239	
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NIAGARA FALLS ARS, NIAGARA FALLS NY BOWST ADDITION TO B321 PS# 508014	GRADING AND STORM DRAINAGE PLAN
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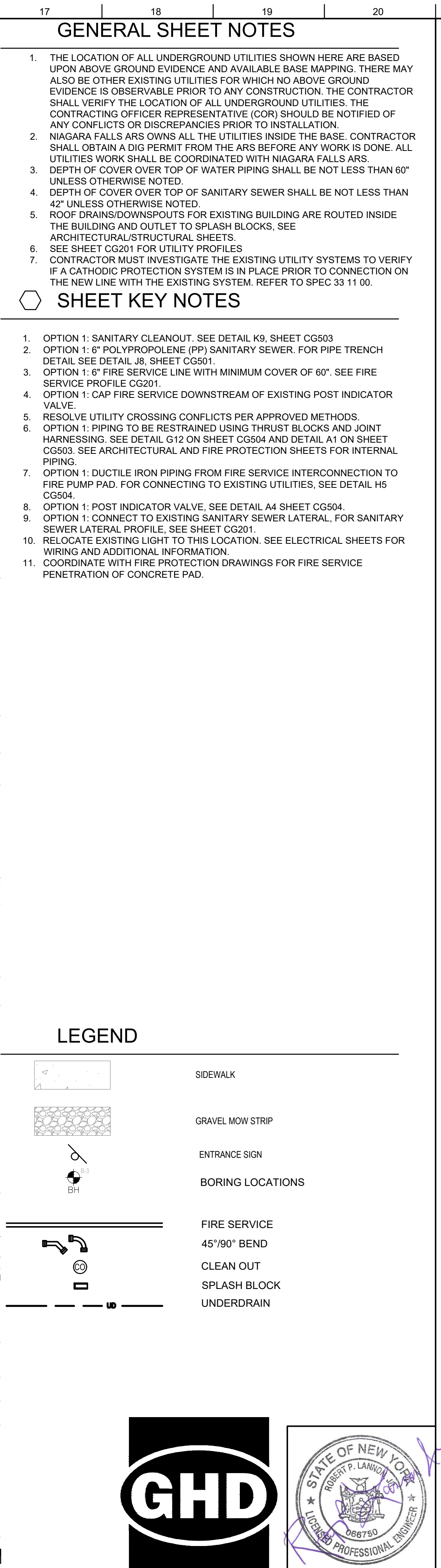
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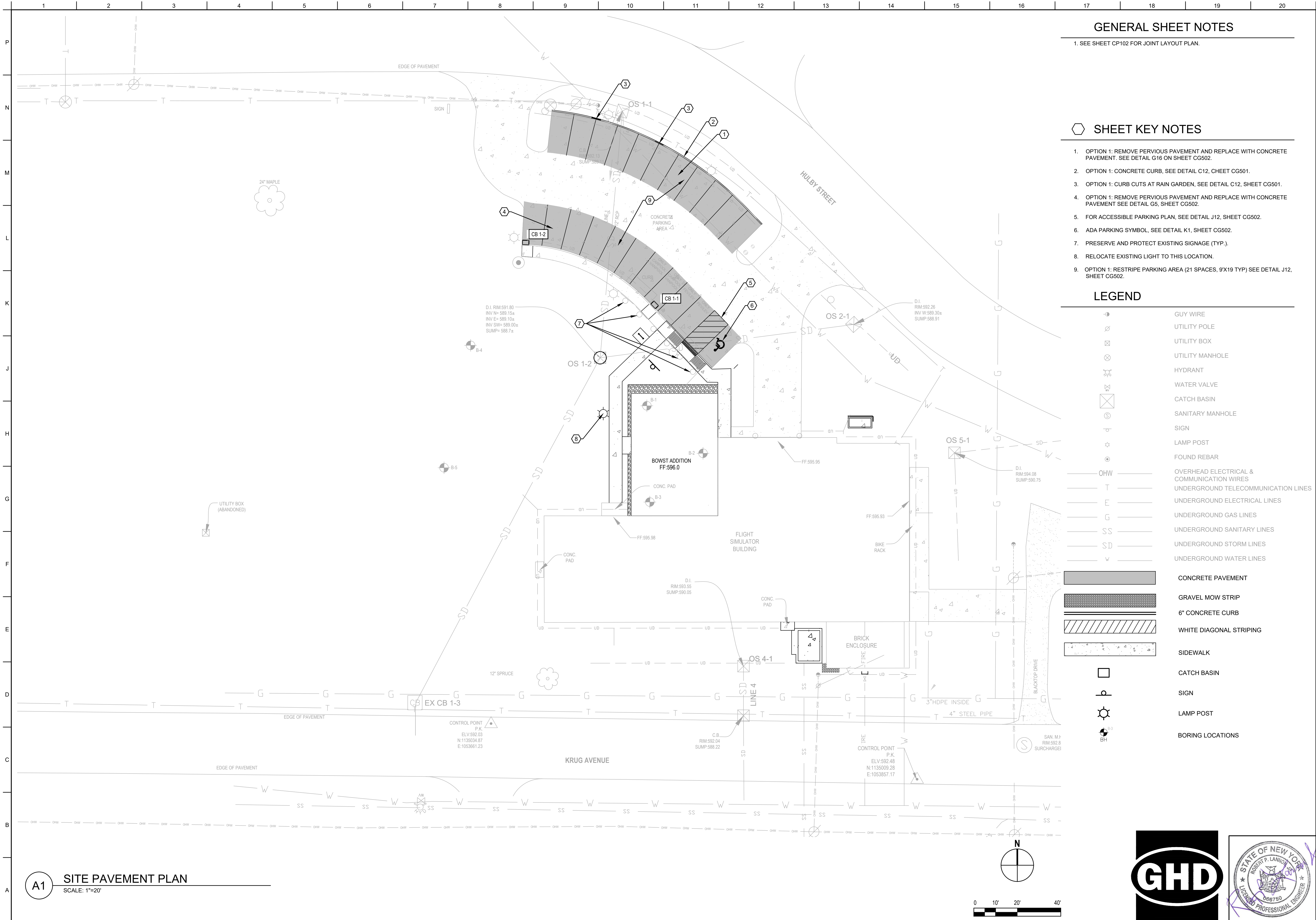
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GENERAL SHEET NOTES

1. SEE SHEET CP102 FOR JOINT LAYOUT PLAN.

SHEET KEY NOTES

- OPTION 1: REMOVE PERVIOUS PAVEMENT AND REPLACE WITH CONCRETE PAVEMENT. SEE DETAIL G16 ON SHEET CG502.
- OPTION 1: CONCRETE CURB, SEE DETAIL C12, CHEET CG501.
- OPTION 1: CURB CUTS AT RAIN GARDEN, SEE DETAIL C12, SHEET CG501.
- OPTION 1: REMOVE PERVIOUS PAVEMENT AND REPLACE WITH CONCRETE PAVEMENT SEE DETAIL G5, SHEET CG502.
- FOR ACCESSIBLE PARKING PLAN, SEE DETAIL J12, SHEET CG502.
- ADA PARKING SYMBOL, SEE DETAIL K1, SHEET CG502.
- PRESERVE AND PROTECT EXISTING SIGNAGE (TYP.).
- RELOCATE EXISTING LIGHT TO THIS LOCATION.
- OPTION 1: RESTRIPE PARKING AREA (21 SPACES, 9'X19 TYP) SEE DETAIL J12, SHEET CG502.

LEGEND

	GUY WIRE
	UTILITY POLE
	UTILITY BOX
	UTILITY MANHOLE
	HYDRANT
	WATER VALVE
	CATCH BASIN
	SANITARY MANHOLE
	SIGN
	LAMP POST
	FOUND REBAR
	OVERHEAD ELECTRICAL & COMMUNICATION WIRES
	UNDERGROUND TELECOMMUNICATION LINES
	UNDERGROUND ELECTRICAL LINES
	UNDERGROUND GAS LINES
	UNDERGROUND SANITARY LINES
	UNDERGROUND STORM LINES
	UNDERGROUND WATER LINES
	CONCRETE PAVEMENT
	GRAVEL MOW STRIP
	6" CONCRETE CURB
	WHITE DIAGONAL STRIPING
	SIDEWALK
	CATCH BASIN
	SIGN
	LAMP POST
	BORING LOCATIONS

ISSUE DATE: 26 JUNE 2024	SOLICITATION NO.:	CONTRACT NO.:
DESIGNED BY: MGATES	DRAWN BY: M.BISIGNANI	SUBMITTED BY: M.BISIGNANI

U.S. ARMY CORPS OF ENGINEERS LOUISVILLE DISTRICT 600 MARTIN LUTHER KING, JR. PLACE LOUISVILLE, KY 40202-2239	 joint venture
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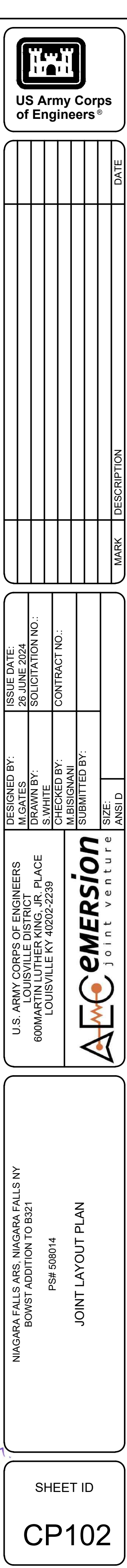
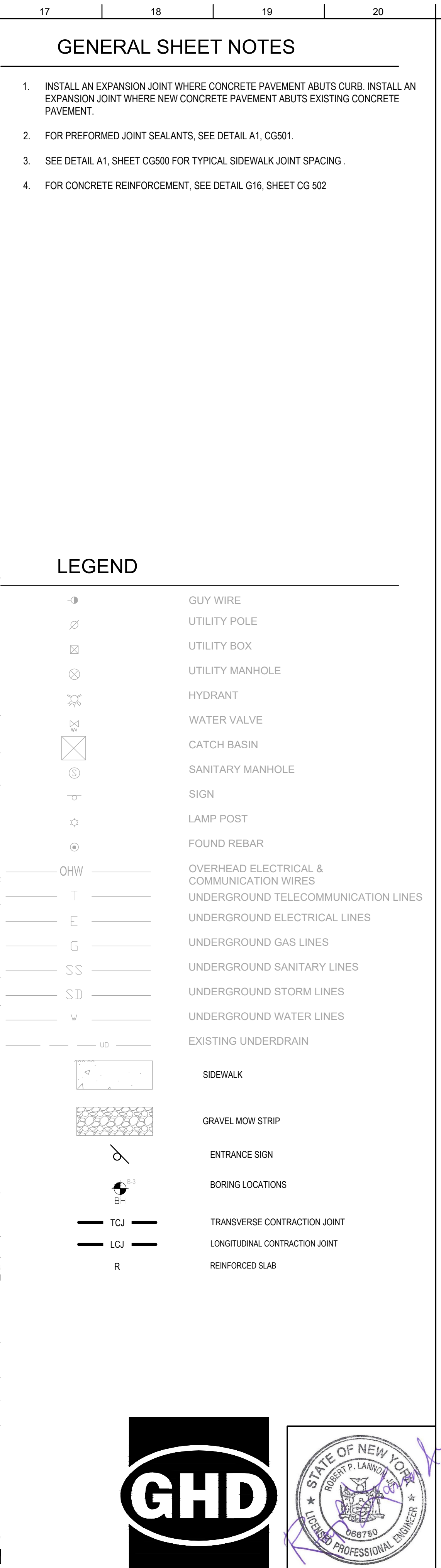
NIAGARA FALLS ARS, NIAGARA FALLS NY BOWST ADDITION TO B321	PS# 508014
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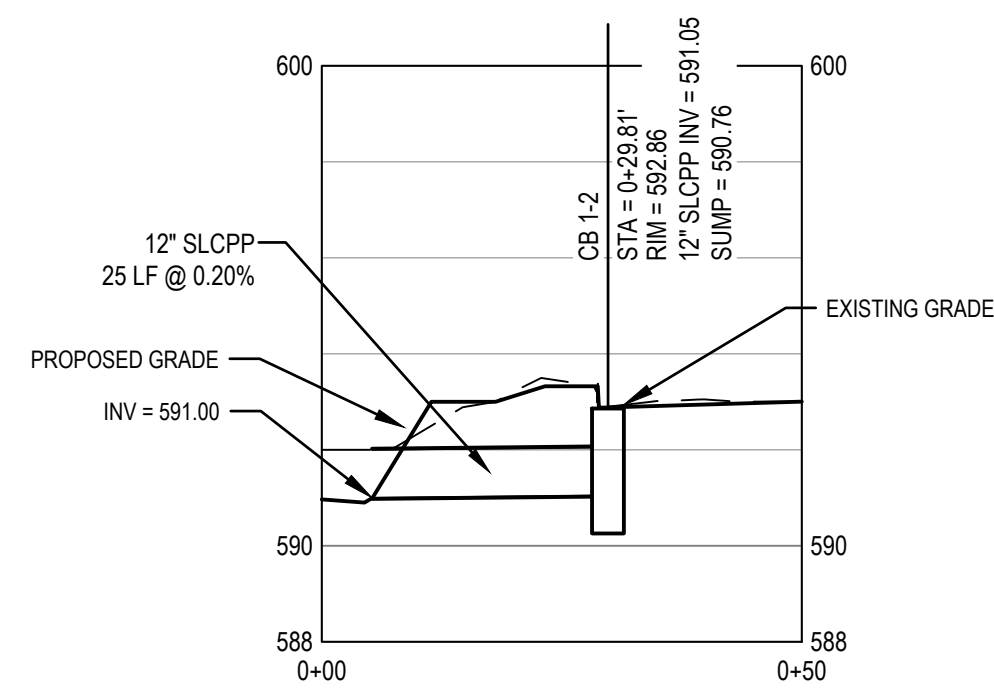
SITE PAVEMENT PLAN

SHEET ID CP101

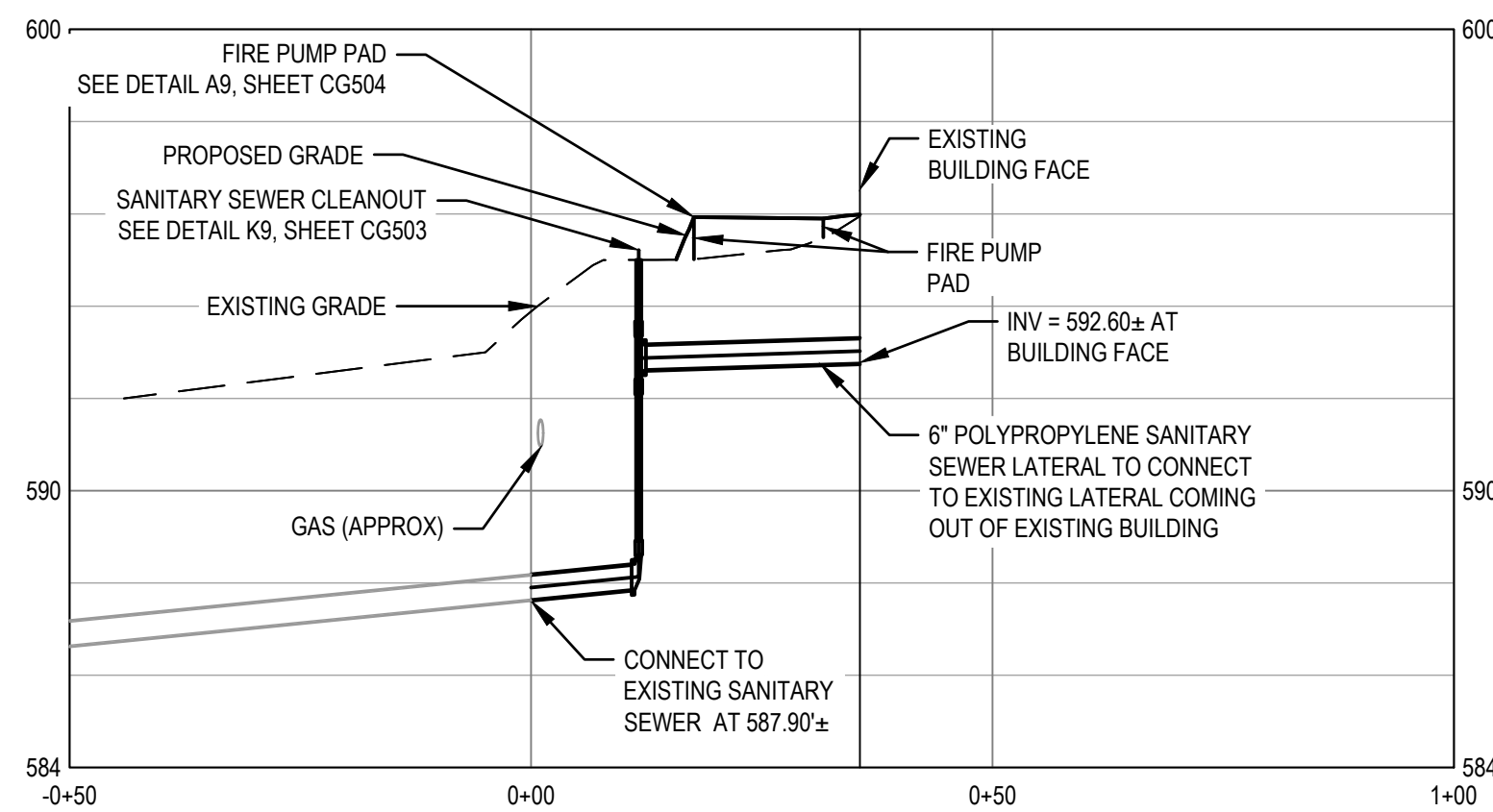
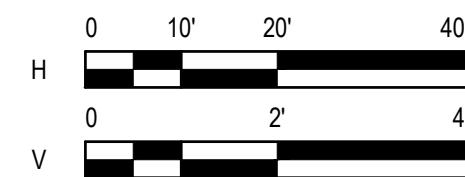
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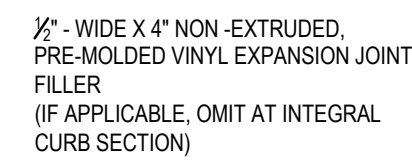


PROFILE VIEW OF CB1-2 TO BIORETENTION BASIN

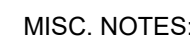


PROFILE VIEW OF SANITARY LATERAL CONNECTION



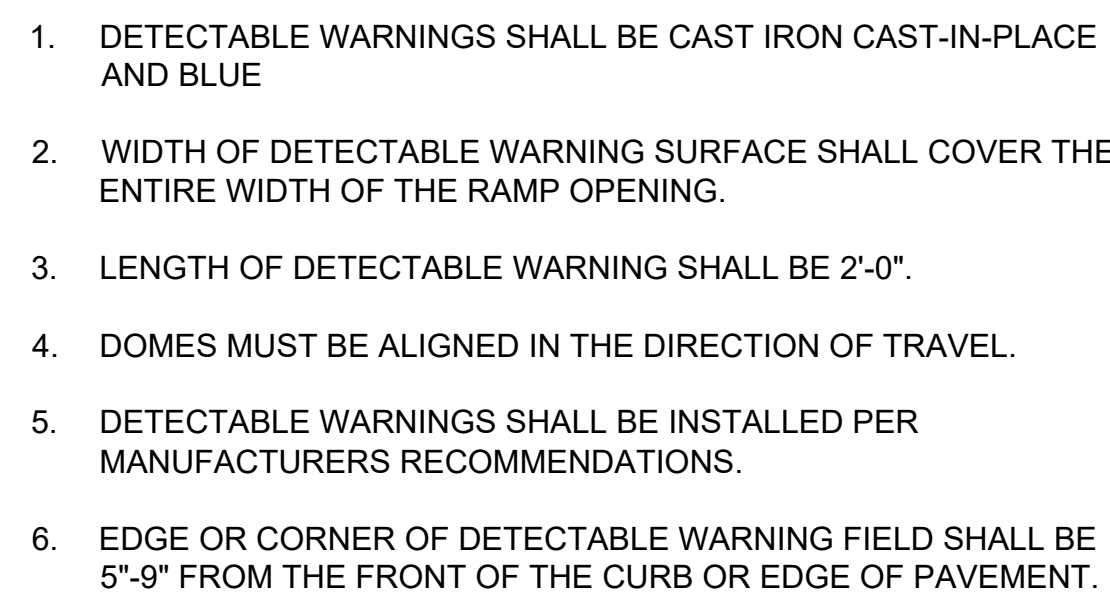


SECTION A-A



-

CONTRACTION JOINT



PLAN

EDGE OF WARNING FIELD



DOME SPACING



DOVE PROFILE



NIAGARA FALLS ARS, NIAGARA FALLS NY
BOWST ADDITION TO B321

PS# 508014

TYPICAL DETAILS-CIVIL

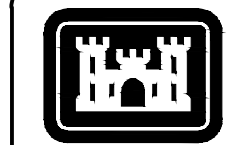
U.S. ARMY CORPS OF ENGINEERS
LOUISVILLE DISTRICT
6000 MARTIN LUTHER KING, JR. PLACE
LOUISVILLE KY 40202-2239



DESIGNED BY:	ISSUE DATE:
M.GATES	26 JUNE 2024
DRAWN BY:	CALCULATIONS:

DRAWN BY: S.WHITE	SOLICITATION NO.
CHECKED BY:	CONTRACT NO.:

CHILD OF: M.BISIGNANI	CONTROL NO.:
SUBMITTED BY:	

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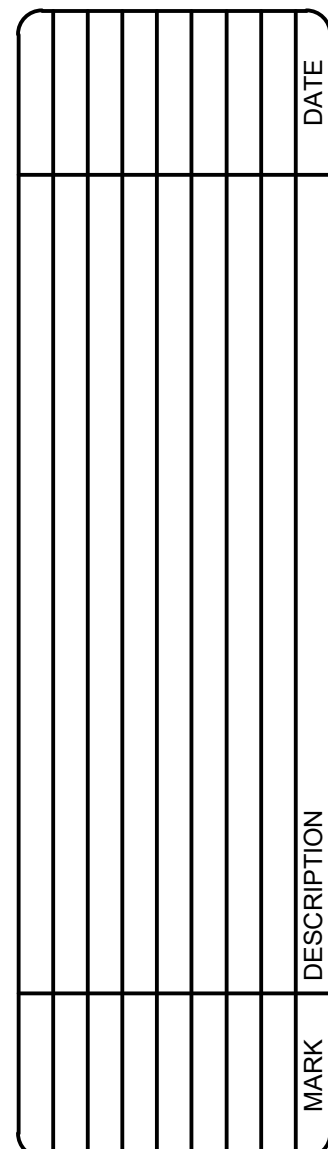
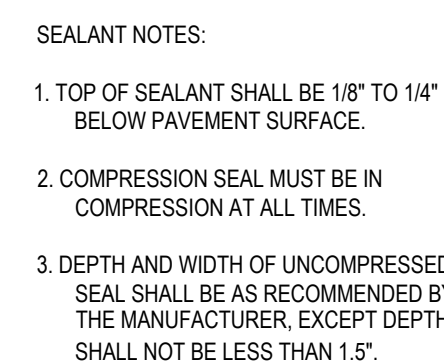
**US Army Corps
of Engineers®**

CERTIFIED FINAL

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DESIGNED BY: M.GATES	ISSUE DATE: 26 JUNE 2024
DRAWN BY: S.WHITE	SOLICITATION NO.:
CHECKED BY: M.BISIGNANI	CONTRACT NO.:
SUBMITTED BY:	
SIZE: ANSI D	

U.S. ARMY CORPS OF ENGINEERS
LOUISVILLE DISTRICT
3000 MARTIN LUTHER KING, JR. PLACE
LOUISVILLE KY 40202-2239

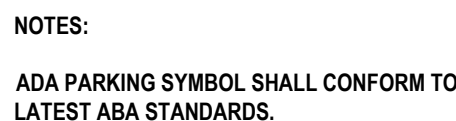


PS# 508014

TYPICAL DETAILS-CIVIL

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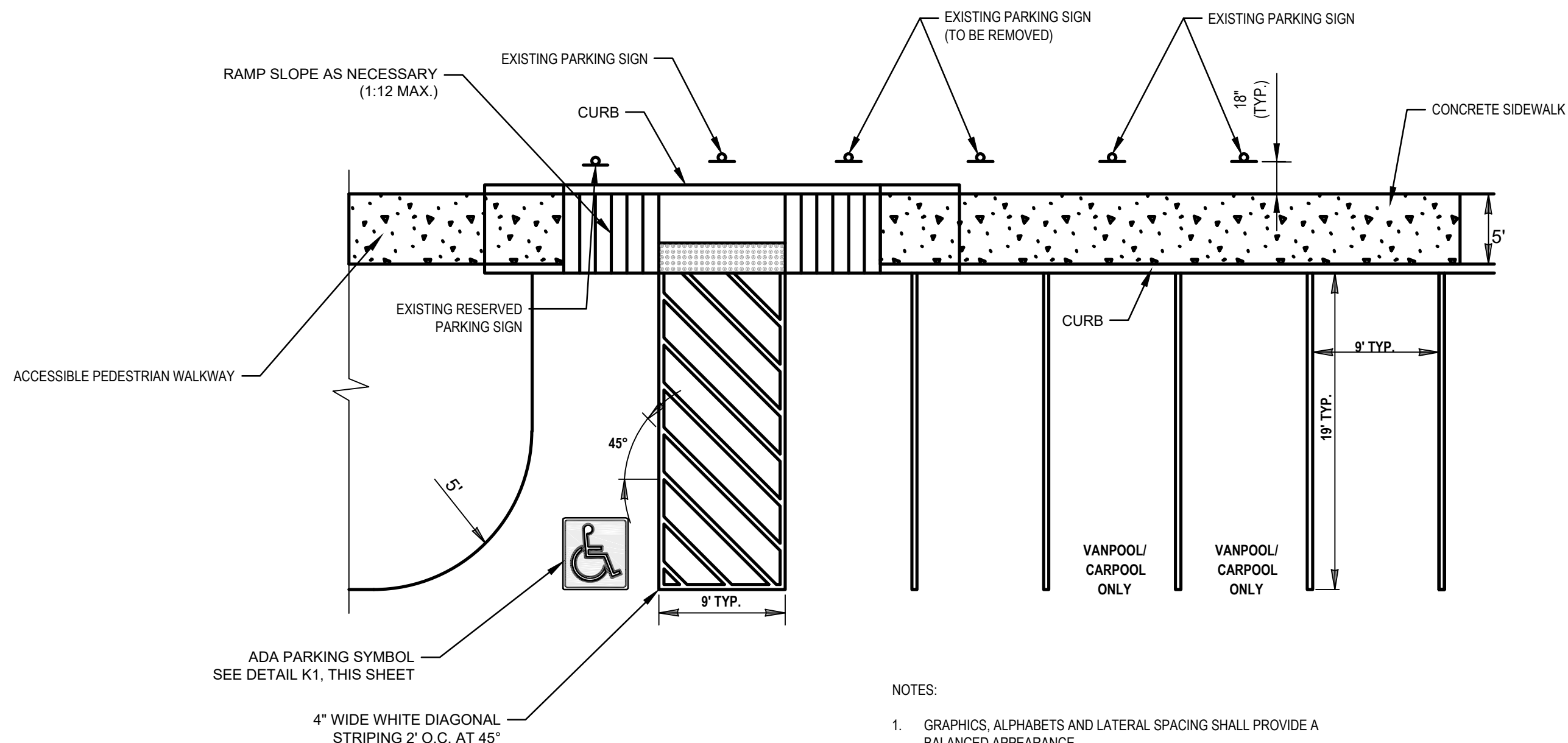
CG501



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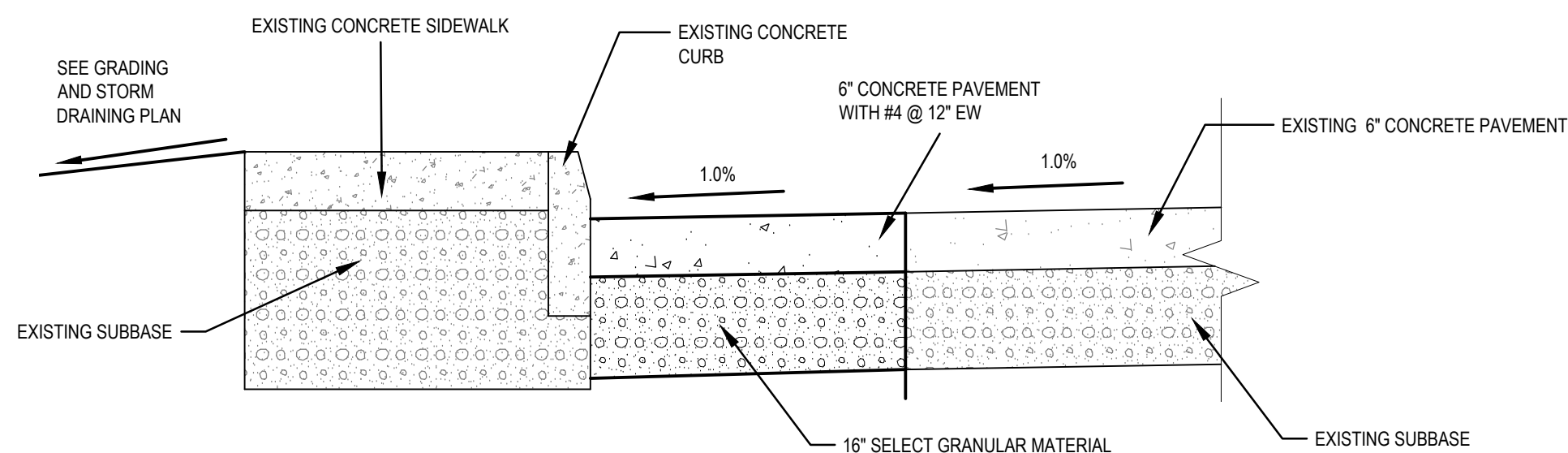


- NOTES:
1. SURFACE MTD. RACK ATOP 6" MIN. PLAIN CONCRETE SLAB VERIFY ADEQUATE PAD DIMENSIONS WITH MANUFACTURER.
 2. EXISTING BIKE RACK MUST BE SALVAGED AND RELOCATED AS SHOWN ON DRAWING CS101.
 3. CONTRACTOR MUST INSTALL APPROPRIATE NUMBER OF 3/8 INCH BY 3 INCH LONG EXPANSION ANCHORS INTO CONCRETE PAD.



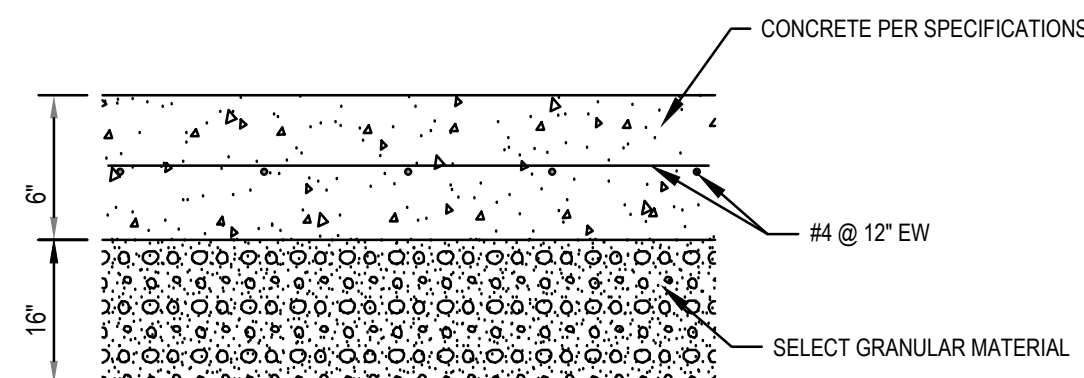
- NOTES:
1. GRAPHICS, ALPHABETS AND LATERAL SPACING SHALL PROVIDE A BALANCED APPEARANCE.
 2. LEED PARKING STALL LETTERING SHALL BE 12" SOLID AND CONTINUOUS. PAINT FEDERAL GREEN COLOR NO. FS 24190.

SCALE: NTS

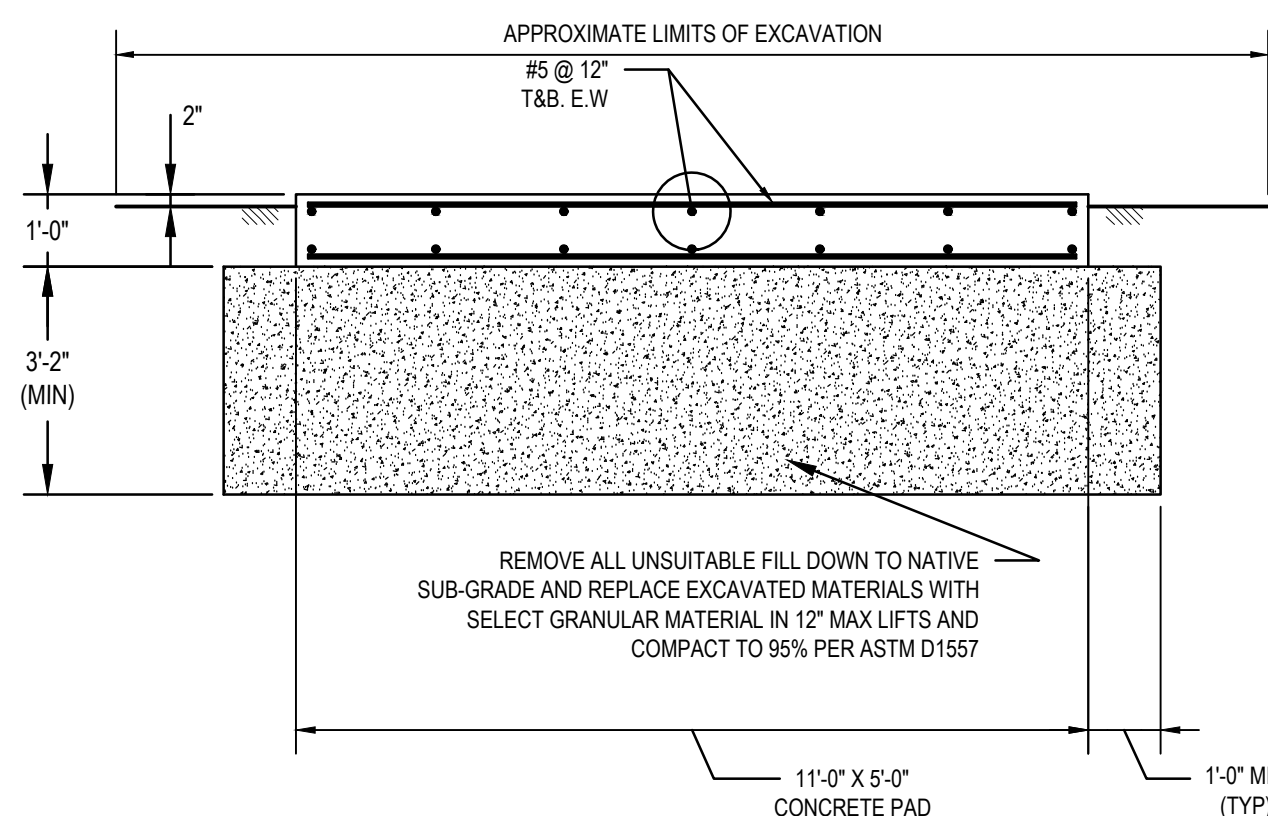


SCALE: NTS

- NOTES:
1. FOR CONCRETE PAVEMENT REINFORCEMENT, SEE DETAIL G16, SHEET CG502.
 2. FOR SLABS THAT RECIEVE REINFORCEMENT, SEE SHEET CP102

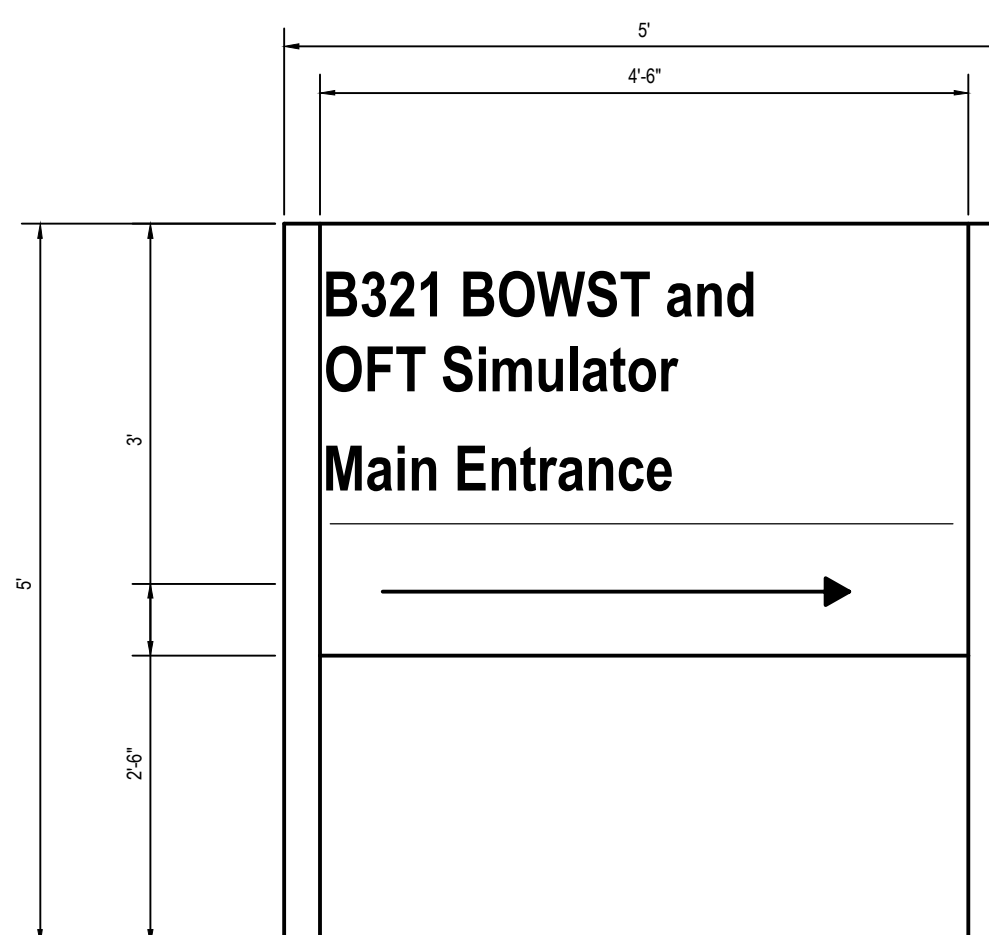


G16

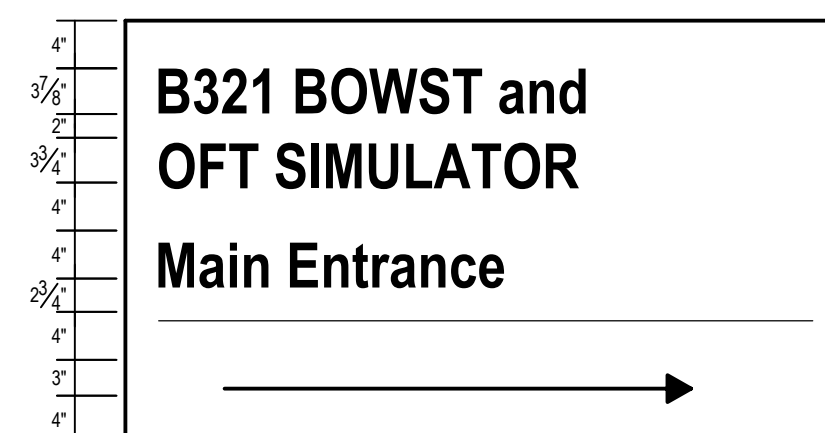


- NOTES:
1. DESIGN IS BASED ON PRELIMINARY EQUIPMENT INFORMATION. THE CONTRACTOR MUST VERIFY THE ADEQUACY AND SIZE OF THE STRUCTURAL DESIGN AT NO ADDITIONAL COST TO THE GOVERNMENT TO PROVIDE A MINIMUM DIMENSION OF 12' BEYOND THE FOOTPRINT SIZE OF THE EQUIPMENT, AND 12" IN FRONT OF THE ACCESS DOORS.
 2. CONTRACTOR TO SUBMIT SHOP DRAWINGS FOR THE EQUIPMENT SELECTED
 3. PROVIDE 2" WIDE CONCRETE PAD UNDERNEATH CHILLER SUPPLY AND RETURN PIPES FROM CHILLER CONNECTION TO BUILDING EXTERIOR WALL. COORDINATE LOCATION AND DIMENSIONS WITH DRAWING M-402 AND MECHANICAL CONTRACTOR.
 4. RESEED EXCAVATED AREA PER SPECIFICATIONS.

B1



SCALE: NTS



- NOTES:**
1. HELVETICA NEUE 75 BOLD
 2. HELVETICA NEUE 65 MEDIUM
 3. BACKGROUND OF MESSAGE PANEL AND SUPPORT POSTS TO BE DARK BROWN PATED FINISH
 4. SIGN CONTENT TO BE WHITE REFLECTIVE ENGINEERING GRADE VINYL.
 5. SIGN POST AND PANEL MUST BE ENGINEERED BY THE SIGN CONTRACTOR ACCORDING TO WIND LOAD IN SPECIFICATION 10 14 00 10.
 6. CONTRACTOR MUST VERIFY THE ADEQUACY AND SIZE OF THE STRUCTURAL DESIGN AT NO ADDITIONAL COST TO THE GOVERNMENT.
 7. CONTRACTOR TO SUBMIT SHOP DRAWINGS FOR THE SIGN SELECTED

[illegible]

	U.S. ARMY CORPS OF ENGINEERS LOUISVILLE DISTRICT 6000 MARTIN LUTHER KING, JR. PLACE LOUISVILLE KY 40202-2239		ISSUE DATE: 26 JUNE 2024	
			DESIGNED BY: M.GATES	
			DRAWN BY: S.MCCOY	
			CHECKED BY: M.BISIGNANI	
			SUBMITTED BY:	
			DATE: ANSID:	

NIAGARA FALLS ARS, NIAGARA FALLS NY
BOWST ADDITION TO B321

PS# 508014

TYPICAL DETAIL S-CIVIL

SHEET ID

CG502



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GOVERNING DESIGN SPECIFICATION

2021 INTERNATIONAL BUILDING CODE AND REFERENCED CODES, AS MODIFIED BY UFC 3-301-01 (11 APRIL 2023) AND UFC 1-200-01 (01 SEPTEMBER 2022, CHANGE 2, 12 JUNE 2023)

DEAD LOADS

ROOF:

- FRAMING MEMBERS
 - BEAMS5.0 PSF
 - METAL DECK2.0 PSF
 - INSULATION (6" RIGID)3.0 PSF
- ROOF TYPE
 - TPO w/ COVER BOARD2.5 PSF
 - SPRINKLER5.0 PSF
 - MECHANICAL/ELECTRICAL/LIGHTS/MISC.5.0 PSF

ROOF & FLOOR LIVE LOADS

ROOF LIVE LOAD:20 PSF

GROUND FLOOR LOAD (NEW ADDITION):150 PSF OR EQUIPMENT WEIGHTS (AS NOTED BELOW):

- * BOWST MOUNTING PADS - 2000 LBS (EA. PAD LOCATION)
- * COMPUTER ROOM - 1000 LBS ROLLING LOAD AND 250 LBS CONCENTRATED LOAD

ROOF RAIN LOAD

RAIN INTENSITY:2.31 in/hr

STRUCTURAL DESIGN DATA

WIND LOAD DESIGN DATA

ULTIMATE DESIGN WIND SPEED, Vult (3-SECOND GUST):109 MPH

NOMINAL DESIGN WIND SPEED, Vasd (3-SECOND GUST):85 MPH

RISK CATEGORY:II

WIND EXPOSURE CATEGORY:C

INTERNAL PRESSURE COEFFICIENT, (GCpi):+/- 0.18

SEISMIC DESIGN DATA

SITE CLASS D (PER GEOTECH REPORT)

RISK CATEGORY:II

SEISMIC IMPORTANCE FACTOR, Ie:1.0

Ss = 0.163

S1 = 0.044

SDS = 0.173

SD1 = 0.071

SEISMIC DESIGN CATEGORY: B

RESPONSE MODIFICATION FACTOR, R: 3.25 (STEEL ORDINARY CONCENTRICALLY BRACED FRAMES)

SEISMIC RESPONSE COEFFICIENT, Cs: 0.053

ANALYSIS PROCEDURE: EQUIVALENT LATERAL FORCE PROCEDURE

SEISMIC DESIGN BASE SHEAR, V:

- NORTH/SOUTH DIRECTION = 9.5k
- EAST/WEST DIRECTION = 9.5k

SNOW LOAD DESIGN DATA

GROUND SNOW LOAD, (Pg):50 PSF

FLAT-ROOF SNOW LOAD, (Pf):35 PSF

MINIMUM SNOW LOAD, (Pm): (LOW SLOPED ROOFS)20 PSF

SNOW EXPOSURE FACTOR, (Ce):1.0

SNOW LOAD IMPORTANCE FACTOR, (Is):1.0

THERMAL FACTOR, (Ct):1.0

SLOPE FACTOR, (Cs):1.0

SLOPED ROOF SNOW LOAD, (Ps):33.3 PSF

DRIFT CRITERIA (DOES NOT INCLUDE Pb):

- AGAINST EXISTING HIGH BAY BUILDING WALL (LEEWARD FROM EXISTING HIGH BAY ROOF ONTO NEW ADDITION ROOF):
Pd = 65 PSF, w = 12.5 FT
- AGAINST NEW ADDITION BUILDING WALL (LEEWARD FROM NEW ADDITION ROOF ONTO EXISTING BUILDING ROOF):
Pd = 69 PSF, w = 13.5 FT

ABBREVIATIONS

ABV
ACI
ADDL
AFF
ALT
APPROX
ASTM
BL
BLDG
BOT
BRG
BTWN
CJ
CLR
COL
CONC
CONN
CONT
CONTR
COORD
DEG
DEMO
DIA
DIMS
DIR
DR
DWGS

ABOVE
AMERICAN CONCRETE INSTITUTE
ADDITIONAL
ABOVE FINISHED FLOOR
ALTERNATE
APPROXIMATE
AMERICAN SOCIETY FOR TESTING AND MATERIALS
BUILDING LINE
BUILDING
BOTTOM
BEARING
BETWEEN
CONTROL JOINT
CLEAR
COLUMN
CONCRETE
CONNECTION
CONTINUOUS
CONTRACT(OR)
COORDINATE
DEGREE
DEMOLISH
DIMENSION
DIRECTION
DRAIN
DRAWINGS

EA
EL
ELEV
ENTR
EQ
EQUIP
EXIST
EXP
EXT
FDN
FLR
FOC
FTG
GA
GALV
GD
GEN
HORIZ
INSUL
LG
LLH
LLV
MANUF
MAX
MECH
MIN
MISC
MSL
MTS

EACH
ELEVATION
ELEVATOR
ENTRANCE
EQUAL
EQUIPMENT
EXISTING
EXPANSION
EXTERIOR
FOUNDATION
FLOOR
FACE OF CONCRETE
FOOTING
GAUGE
GALVANIZE
GRADE, GRADING
GENERAL
HORIZONTAL
INSULATION
LONG
LONG LEG HORIZONTAL
LONG LEG VERTICAL
MANUFACTURER
MAXIMUM
MECHANIC(AL)
MINIMUM
MISCELLANEOUS
MEAN SEA LEVEL
METAL

NTS
OPNG
PCF
PERIM
PROJ
RC
REINF
REQD
SOG
SPA
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W.W.F.

NOT TO SCALE
OPENING
POUNDS PER CUBIC FOOT
PERIMETER
PROJECTION
REINFORCED CONCRETE
REINFORCEMENT
REQUIRED
SLAB ON GRADE
SPACING
STANDARD
STEEL
SYSTEM
THICK(NESS)
TYPICAL
UNLESS NOTED
OTHERWISE
VERTICAL
VERIFY IN FIELD
WITH
WITHOUT
WEIGHT
WELDED WIRE FABRIC

GROSS COMPONENT & CLADDING WIND LOAD - ADDITION (PSF)

EFFECTIVE AREA (SF)		<10	20	50	100	200	>500
ROOF ZONE	1	+16.0 -32.3	+16.0 -32.3	+16.0 -32.3	+16.0 -32.3	+16.0 -32.3	+16.0 -32.3
	2	+16.0 -37.4	+16.0 -36.7	+16.0 -35.2	+16.0 -34.9	+16.0 -34.9	+16.0 -34.9
	2'	+16.0 -45.0	+16.0 -44.5	+16.0 -43.8	+16.0 -42.5	+16.0 -42.5	+16.0 -42.5
	3	+16.0 -50.1	+16.0 -45.3	+16.0 -39.5	+16.0 -34.9	+16.0 -34.9	+16.0 -34.9
	3'	+16.0 -70.3	+16.0 -62.7	+16.0 -52.6	+16.0 -45.0	+16.0 -45.0	+16.0 -45.0
OVERHANG	1, 2e	-63.3	-63.3	-48.6	-38.0	-38.0	-38.0
	2n, 2r	-88.6	-79.7	-69.1	-61.0	-50.6	-50.6
	3e	-103.7	-88.8	-69.1	-55.7	-38.0	-38.0
	3r	-118.9	-100.7	-75.9	-58.2	-58.2	-58.2
WALL ZONE	4	+29.9 -32.4	+28.1 -30.9	+26.6 -29.4	+25.3 -28.1	+24.3 -26.6	+22.3 -24.8
	5	+29.9 -40.0	+28.1 -37.2	+26.6 -33.7	+25.3 -30.9	+24.3 -28.4	+22.3 -24.8

NOTE: GROSS COMPONENT & CLADDING VALUES PER TABLE HAVE NOT BEEN FACTORED FOR ALLOWABLE STRESS DESIGN OR LOAD RESISTANCE FACTOR DESIGN. APPLICABLE FACTORS SHALL BE APPLIED TO VALUES FOR ALL CALCULATIONS

WALL WIND LOADING - BUILDING ADDITION (C&C)

ROOF WIND LOADING - BUILDING ADDITION (C&C)

US Army Corps of Engineers

DESIGNED BY:
S. RICCI

DRAWN BY:
S. RICCI

CHECKED BY:
S. DUFFY

SUBMITTED BY:
S. DUFFY

ISSUE DATE:
JUNE 26, 2024

SOLICITATION NO.:

CONTRACT NO.:

US ARMY CORPS OF ENGINEERS

AECOM

emersion joint venture

BOWST ADDITION AND RENOVATIONS TO B-321
NIAGARA FALLS AIR RESERVE STATION, NY
P#H: 508014

DESIGN LOADS AND ABBREVIATIONS -
STRUCTURAL

SHEET ID

S-002

STATE OF OHIO
PROFESSIONAL ENGINEER
STEVEN RICCI
E-76269

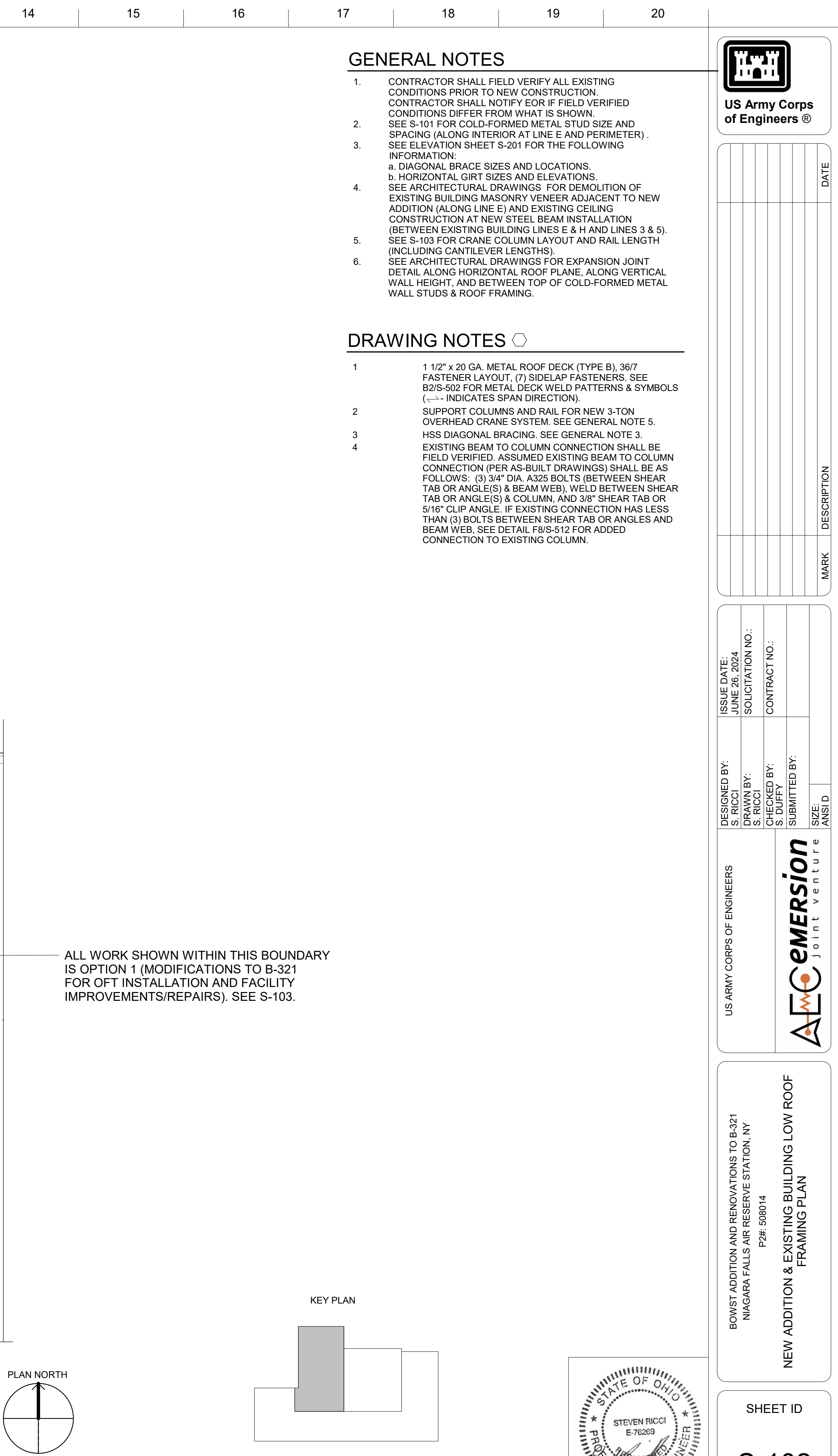
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DRAWING NOTES

- 1 1 1/2" x 20 GA. METAL ROOF DECK (TYPE B), 36/7
FASTENER LAYOUT. (7) SIDELAP FASTENERS. SEE
B2/S-502 FOR METAL DECK WELD PATTERNS & SYMBOLS.
(→ INDICATES SPAN DIRECTION).
- 2 SUPPORT COLUMNS AND RAIL FOR NEW 3-TON
OVERHEAD CRANE SYSTEM. SEE GENERAL NOTE 5.
- 3 HSS DIAGONAL BRACING. SEE GENERAL NOTE 3.
- 4 EXISTING BEAM TO COLUMN CONNECTION SHALL BE
FIELD VERIFIED. ASSUMED EXISTING BEAM TO COLUMN
CONNECTION (PER AS-BUILT DRAWINGS) SHALL BE AS
FOLLOWS: (3) 3/4" DIA. A325 BOLTS (BETWEEN SHEAR
TAB OR ANGLE(S) & COLUMN, AND 3/8" SHEAR TAB OR
5/16" CLIP ANGLE. IF EXISTING CONNECTION HAS LESS
THAN (3) BOLTS BETWEEN SHEAR TAB OR ANGLES AND
BEAM WEB, SEE DETAIL F8/S-512 FOR ADDED
CONNECTION TO EXISTING COLUMN.

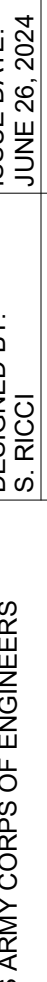
ALL WORK SHOWN WITHIN THIS BOUNDARY
IS OPTION 1 (MODIFICATIONS TO B-321
FOR OBT INSTALLATION AND FACILITY
IMPROVEMENTS/REPAIRS). SEE S-103.

KEY PLAN

KEY PLAN - ADDITION

**US Army Corps
of Engineers ®**

[illegible]

	US ARMY CORPS OF ENGINEERS		DESIGNED BY: S. RICCI	ISSUE DATE: JUNE 26, 2024
			DRAWN BY: S. RICCI	SOLICITATION NO.:
		CHECKED BY:	CONTRACT NO.:	
		S. DUFFY	SUBMITTED BY:	
			SIZE: ANSI D	

BOWST ADDITION AND RENOVATIONS TO B-321
NIAGARA FALLS AIR RESERVE STATION, NY

P2#: 508014

NEW ADDITION & EXISTING BUILDING LOW ROOF
FRAMING PLAN

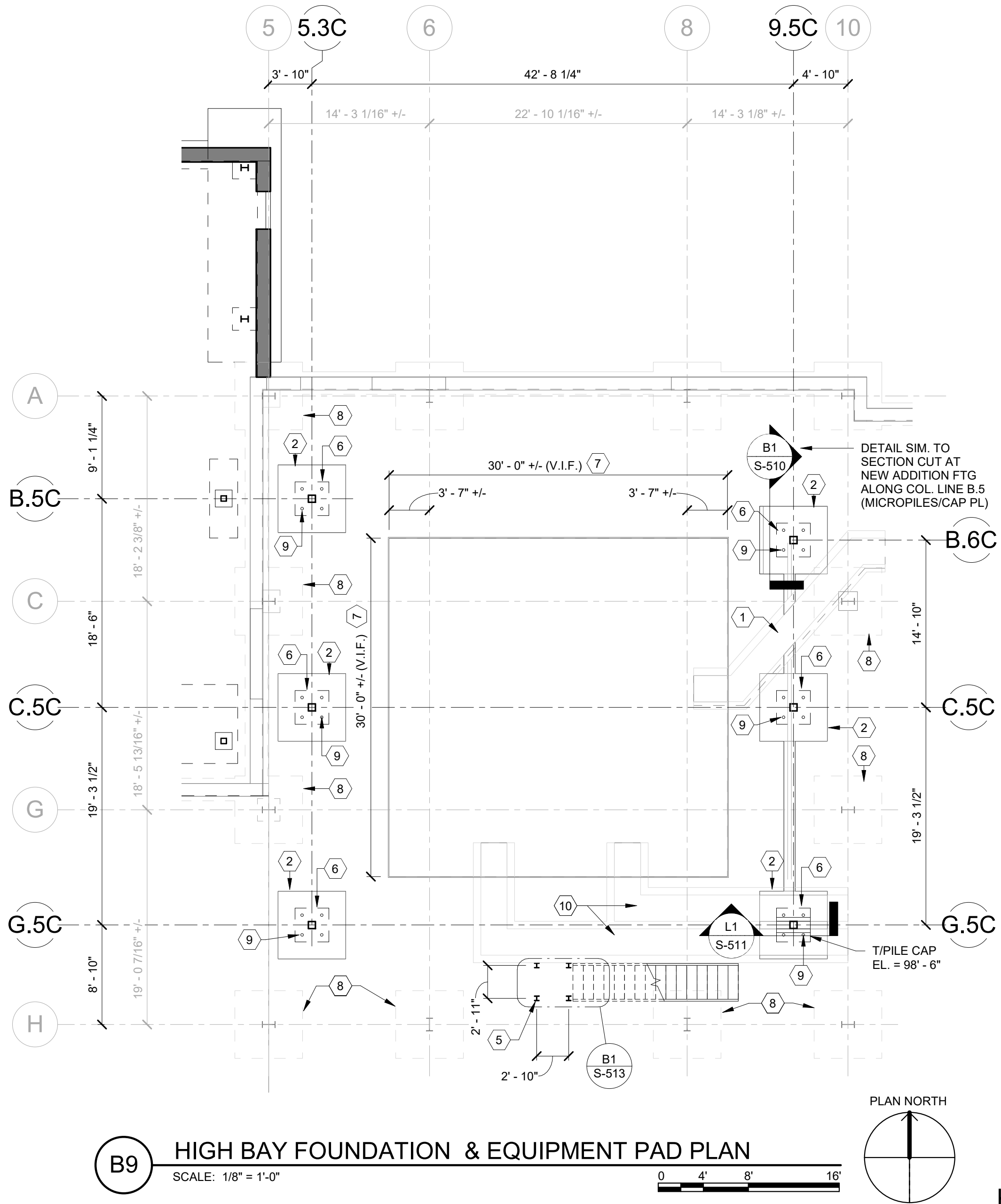
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KEY PLAN

The key plan shows a street grid with several lots. The subject property is a rectangular lot shaded in gray, located at the intersection of a north-south street and an east-west street. It is situated between two other lots on the same block.



DRAWING NOTES

- 1 EXISTING UTILITY TRENCH (LOCATION OF UTILITY TRENCH SHOWN PER EXISTING DRAWINGS AND FIELD VERIFICATION) - TO REMAIN.
- 2 EXISTING CONCRETE SLAB ON GRADE (8" THK. V.I.F.) REMOVAL EXTENTS FOR NEW CRANE FOUNDATION SYSTEM INSTALLATION. SEE GENERAL NOTE 4.
- 3 NEW SUPPORT COLUMNS (HSS8x8 ASSUMED) AND RAIL FRAMING (W16 ASSUMED) FOR OVERHEAD CRANE. CRANE SYSTEM FRAMING SIZE, TYPE, AND CONFIGURATION PER CONTRACTOR'S DELEGATED DESIGN. LOCATION OF COLUMNS SHOWN FOR CLEARANCE PURPOSES. TOP OF CRANE COLUMN EL. = 30' - 0".
- 4 ASSUMED BAY FOR LATERAL CROSS BRACING (AS REQUIRED PER CONTRACTOR'S DELEGATED DESIGN OF CRANE SYSTEM) ALONG LINES 5.3C AND 9.5C.
- 5 W5x16 COLUMNS FOR DRAWBRIDGE APPARATUS AND STAIR SUPPORT. SEE CONTRACTOR'S DRAWINGS FOR LOCATION OF DRAWBRIDGE APPARATUS AND STAIR.
- 6 CRANE COLUMN FOUNDATION DESIGN CONCEPT: NEW 3' - 0"x3' - 0"x1' - 6" (THK.) CONCRETE PILE CAP (MIN., FINAL DIMENSIONS PER MICROPILE DESIGN PARAMETERS). TOP OF CONCRETE CAP EL. = 99' - 4" (UNLESS NOTED OTHERWISE ON PLAN). SEE GENERAL NOTE 4 FOR REMOVAL OF EXISTING SLAB. FINAL DESIGN OF CRANE COLUMN FOUNDATION PER CONTRACTOR'S DELEGATED DESIGN, BASED ON THE CRANE SYSTEM LOADS AND FRAMING SYSTEM AS DETERMINED BY THE CONTRACTOR'S DELEGATED DESIGN.
- 7 EXISTING 20" (THK.) CONCRETE PAD FOR O&T EQUIPMENT - TO REMAIN. ARCHITECTURAL DRAWINGS FOR TOP OF SLAB FINISH REQUIREMENTS.
- 8 EXISTING CONCRETE FOUNDATION SYSTEM (SPREAD FOOTING/PEDESTAL AT COLUMNS & FOUNDATION WALL/STRIP FOOTING AT WALLS) - TO REMAIN. CONTRACTOR SHALL NOT INTERFERE OR MODIFY EXISTING FOUNDATIONS. NOTIFY EOR IF EXISTING FOUNDATIONS ARE DIFFERENT THAN SHOWN ON DRAWINGS.
- 9 CRANE COLUMN FOUNDATION DESIGN CONCEPT: NEW MICROPILE FOUNDATION SYSTEM - QUANTITY (4) AT EACH COLUMN LOCATION. MICROPILES SHALL BE DESIGNED BY SPECIALTY CONTRACTOR. MICROPILE DEPTH SHALL BE PER REQUIRED TORQUE AS DETERMINED BY MICROPILE DESIGNER. INDICATES MICROPILE). FINAL DESIGN OF CRANE COLUMN FOUNDATION PER CONTRACTOR'S DELEGATED DESIGN, BASED ON THE CRANE SYSTEM LOADS AND FRAMING AS DETERMINED BY THE CONTRACTOR'S DELEGATED DESIGN.
- 10 EXISTING UTILITY TRENCHES (LOCATION OF UTILITY TRENCH SHOWN PER EXISTING DRAWINGS AND FIELD VERIFICATION). PORTION OF TRENCH SLAB AND WALLS SHALL BE REMOVED FOR INSTALLATION OF NEW CONCRETE PILE CAP AND CRANE COLUMN.

ALL WORK NOTED ON THIS PAGE IS
OPTION 1 (MODIFICATIONS TO B-321
FOR OFT INSTALLATION AND FACILITY
IMPROVEMENTS/REPAIRS).

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CHECKED BY: S. DUFFY	SUBMITTED BY:	CONTRACT NO.:	
SIZE: ANSI D			

US ARMY CORPS OF ENGINEERS



BOWST ADDITION AND RENOVATIONS TO B-321
NIAGARA FALLS AIR RESERVE STATION, NY

HIGH BAY FOUNDATION & EQUIPMENT PAD PLAN
 AND OVERHEAD CRANE SUPPORT FRAMING
 DI AN

SHEET ID

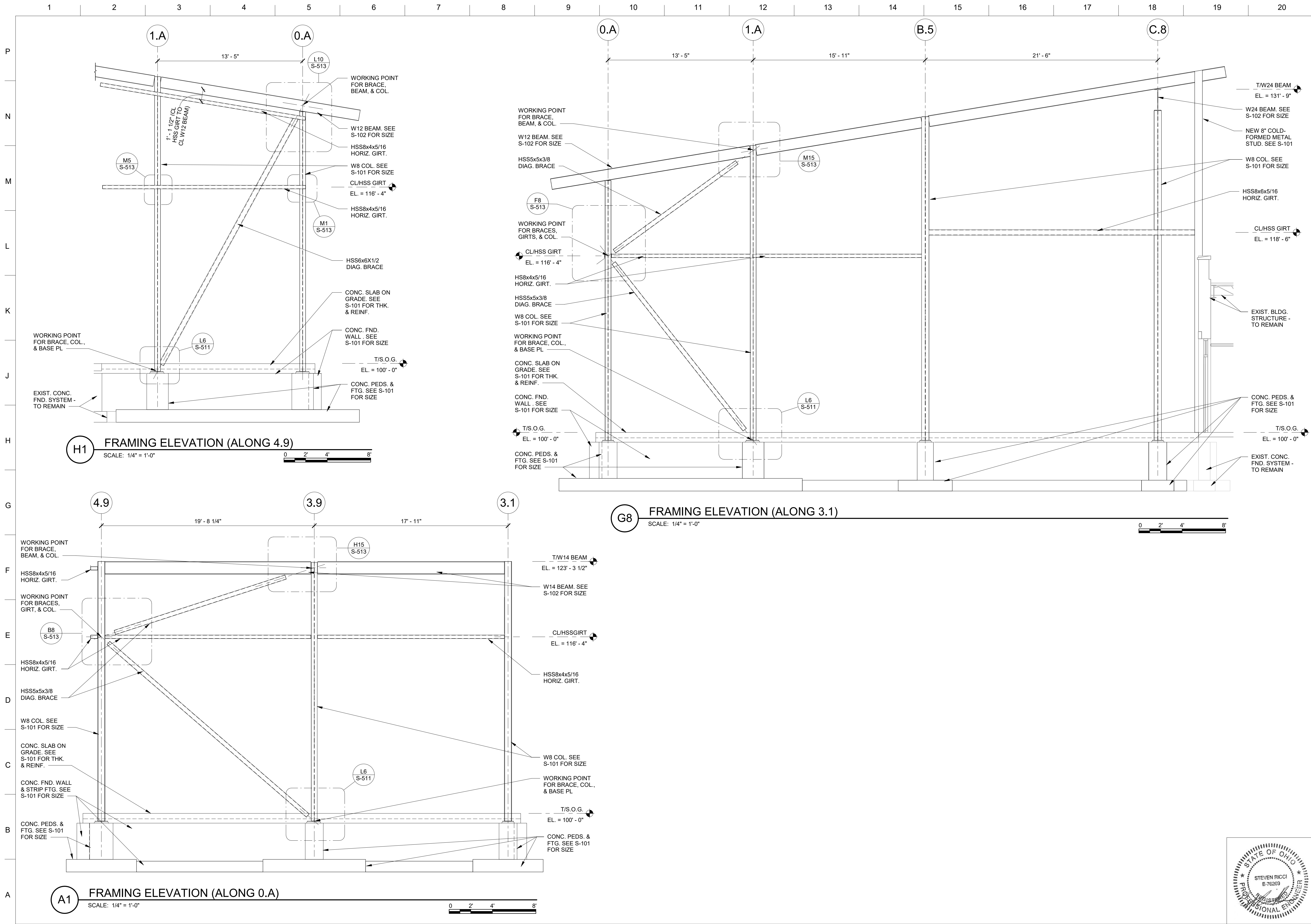
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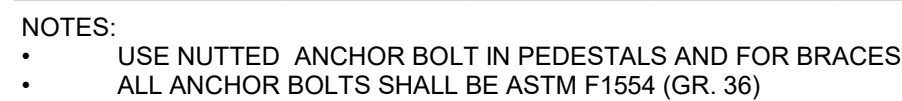
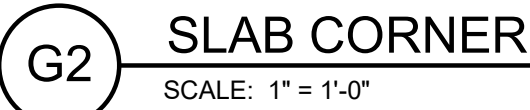
DESIGNED BY: S. RICCI	ISSUE DATE: JUNE 26, 2024
DRAWN BY: A. MORRISON	SOLICITATION NO.:
CHECKED BY: S. DUFFY	CONTRACT NO.:
SUBMITTED BY:	
SIZE: ANSI D	

US ARMY CORPS OF ENGINEERS

BOWST ADDITION AND RENOVATIONS TO B-321
NIAGARA FALLS AIR RESERVE STATION, NY
P2# 506014
FRAMING ELEVATIONS - STRUCTURAL



SHEET ID
S-201

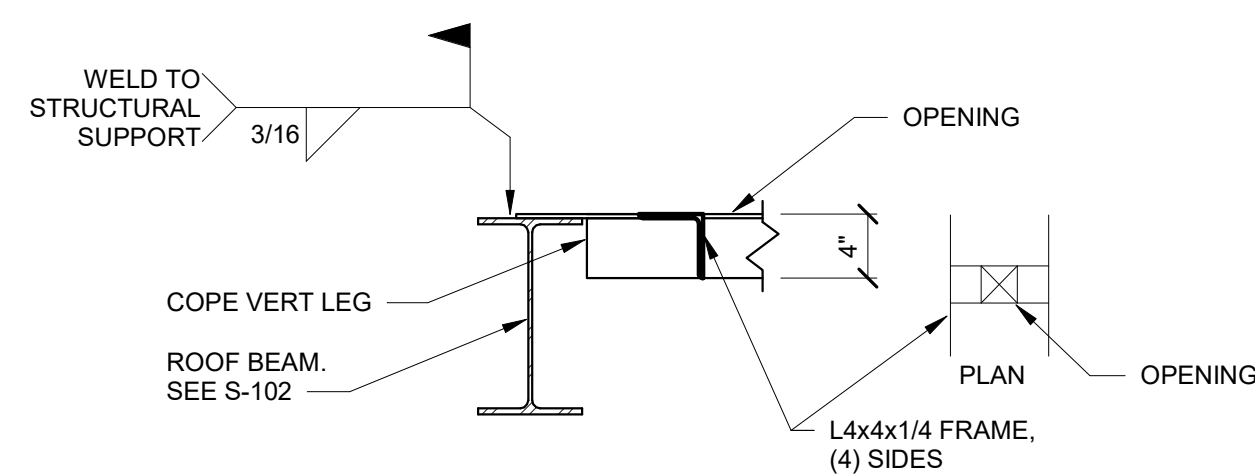


LOOSE STEEL ANGLE LINTEL SCHEDULE			
LINTEL DESIGNATION	STEEL ANGLE SIZE	MASONRY OPENING MAXIMUM SPAN (FT)	MINIMUM END BEARING (INCHES)
SL1	L3 1/2 x 3 1/2 x 1/4	5' - 0"	4
SL2	L4 x 3 1/2 X 1/4	6' - 0"	5
SL3	L6 x 3 1/2 x 3/8	8' - 0"	6
SL4	L8 x 4 x 1/2	10' - 0"	7

LINTEL NOTES:
 1. ALL LINTELS EXPOSED TO EXTERIOR SHALL BE HOT-DIPPED GALVANIZED.
 2. THE ANGLE LINTELS FIRST LEG SIZE GIVEN ABOVE SHALL BE THE VERTICAL LEG.
 3. UNLESS OTHERWISE SHOWN, SELECT LINTEL TO SUIT INDICTED MASONRY OPENING
 AT OR BELOW MAXIMUM ALLOWABLE SPAN INDICATED. LINTEL TYPE IS AT CONTRACTORS
 OPTION EXCEPT AS NOTED OR DETAILED.
 4. REFER TO ARCHITECTURAL DRAWINGS FOR OPENING SIZES.

12 LOOSE ANGLE LINTEL SCHEDULE

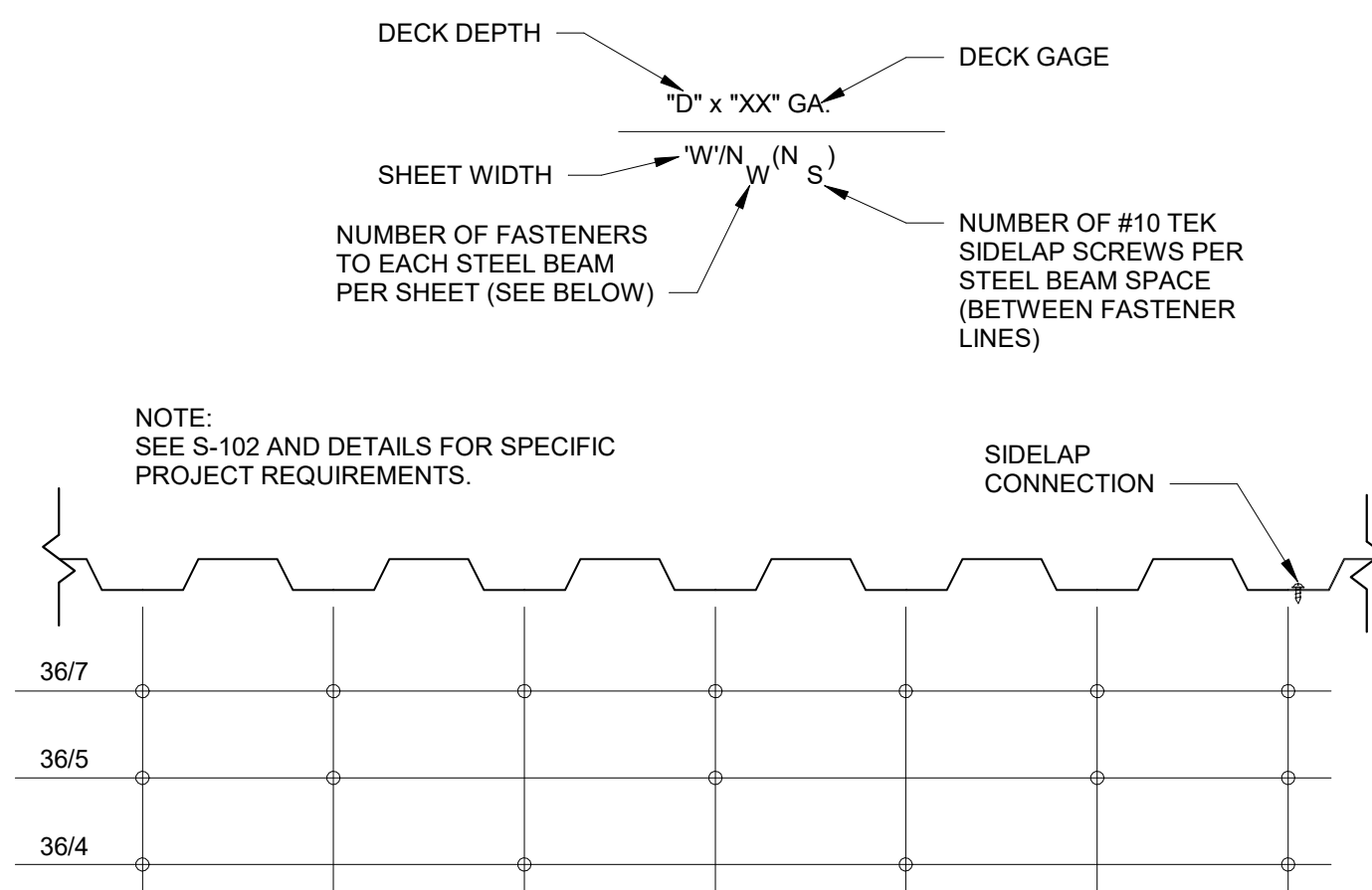
SCALE: 1" = 1'-0"



NOTE:
1. ALL CONNECTIONS SHALL BE SHOP AND OR FIELD WELDED
2. COORDINATE OPENING SIZE AND LOCATION WITH ARCHITECTURAL & MECHANICAL DRAWINGS

G2 ROOF OPENINGS

SCALE: 1" = 1'-0"

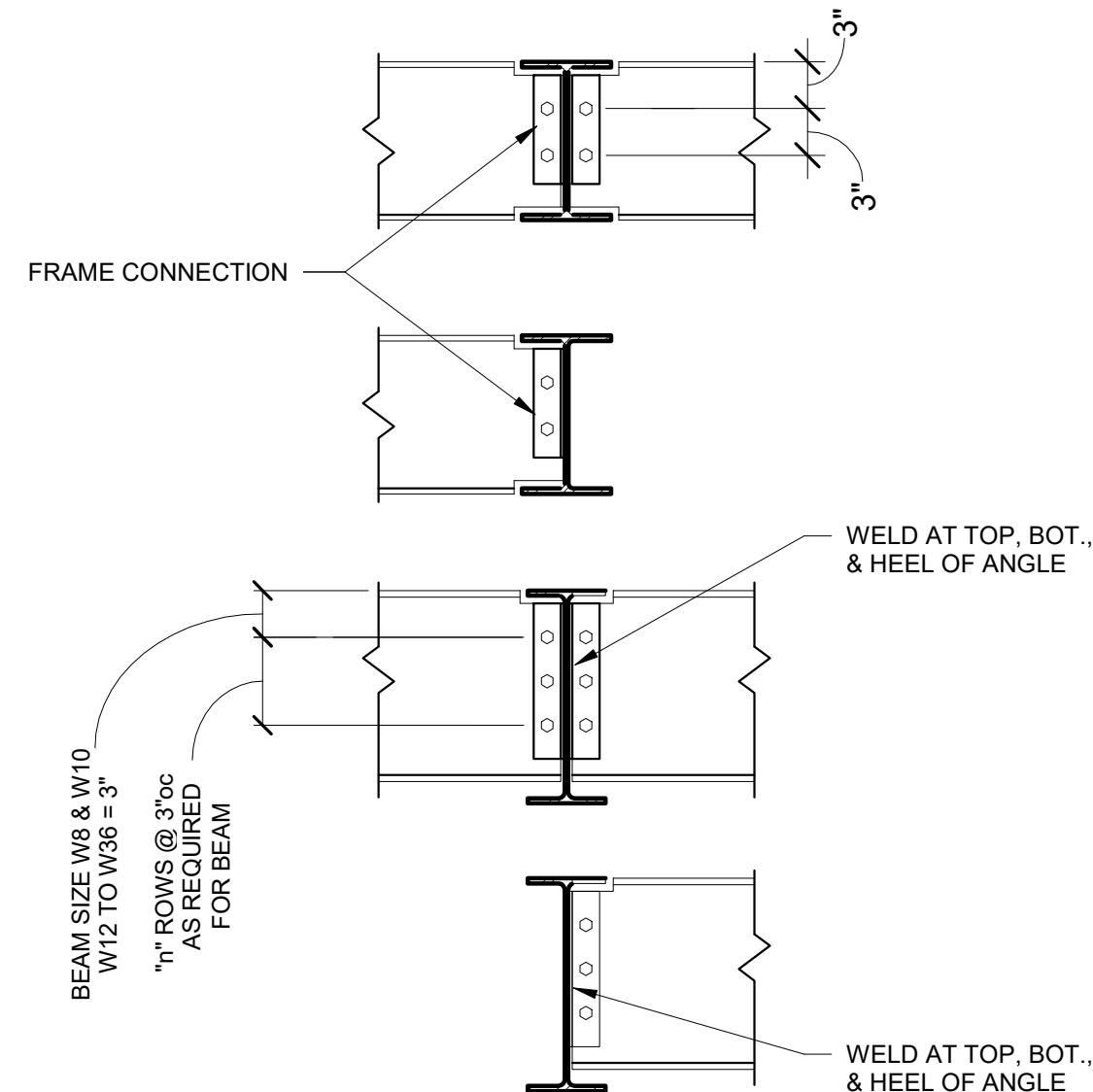


DECK FASTENERS:
1) WELDS SHALL BE 5/8" DIA PUDDLE
WELDS. TOUCH UP ALL WELDS WITH
COLD GALVANIZING COMPOUND
2) SCREWS SHALL BE #10 TEK SCREWS

NOTE:
FASTEN EVERY FLUTE ON ANY
PARTIAL WIDTH DECK SHEET AND 6" ON
AROUND ALL FRAMES AND OPENINGS

B2 DECK SYMBOLS AND WELD PATTERNS

SCALE: 1" = 1'-0"

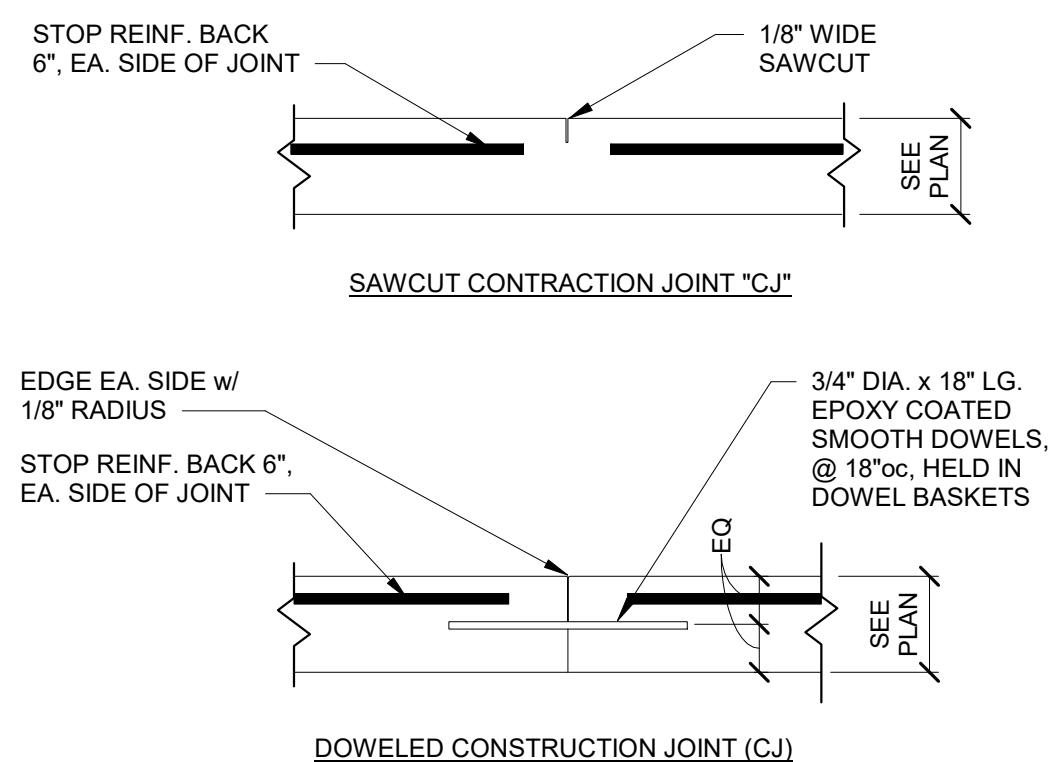


DEPTH OF BEAM	MIN. NUMBER OF BOLTS FOR EA. CONNECTION
UP TO 10"	2
12" TO 16"	3
18" TO 21"	4
24" TO 27"	5

NOTE:
THE MIN. NUMBER OF BOLTS FOR EACH BEAM CONNECTION SHALL BE AS CALLED FOR IN ADJACENT TABLE, REGARDLESS OF SIZE OF BOLTS. IF DESIGN DICTATES THAT A GREATER NUMBER OF BOLTS IS REQ'D., THEN THE GREATER SHALL BE USED.

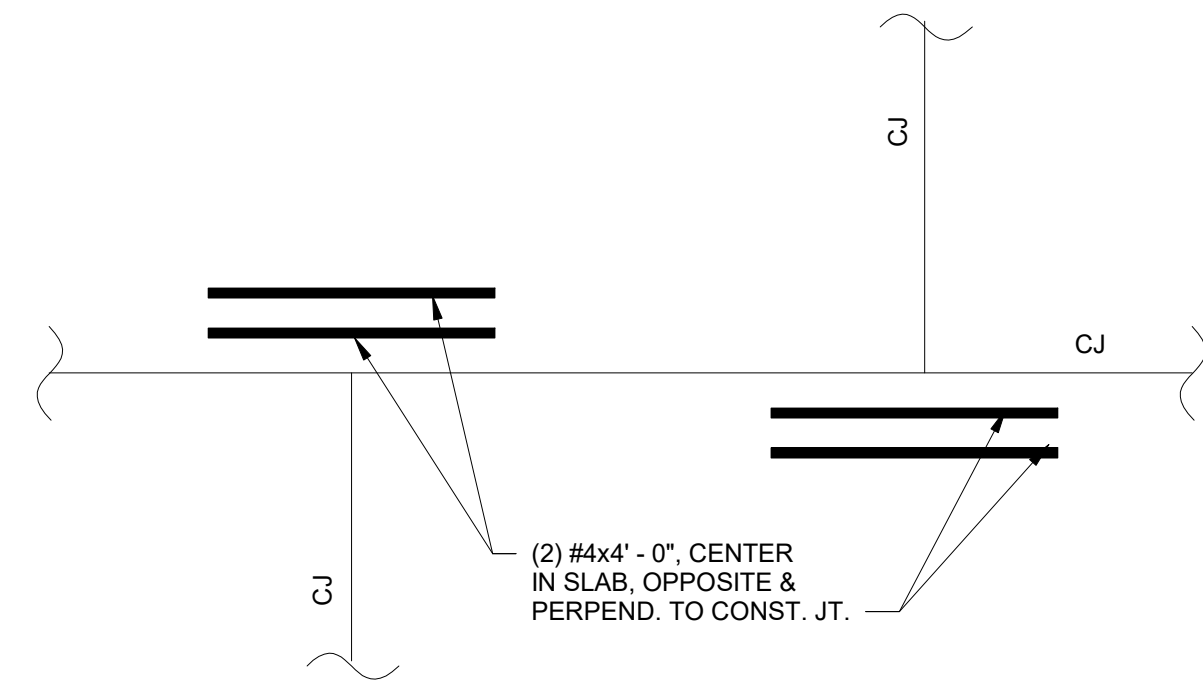
G9 BEAM CONNECTIONS

SCALE: 1" = 1'-0"



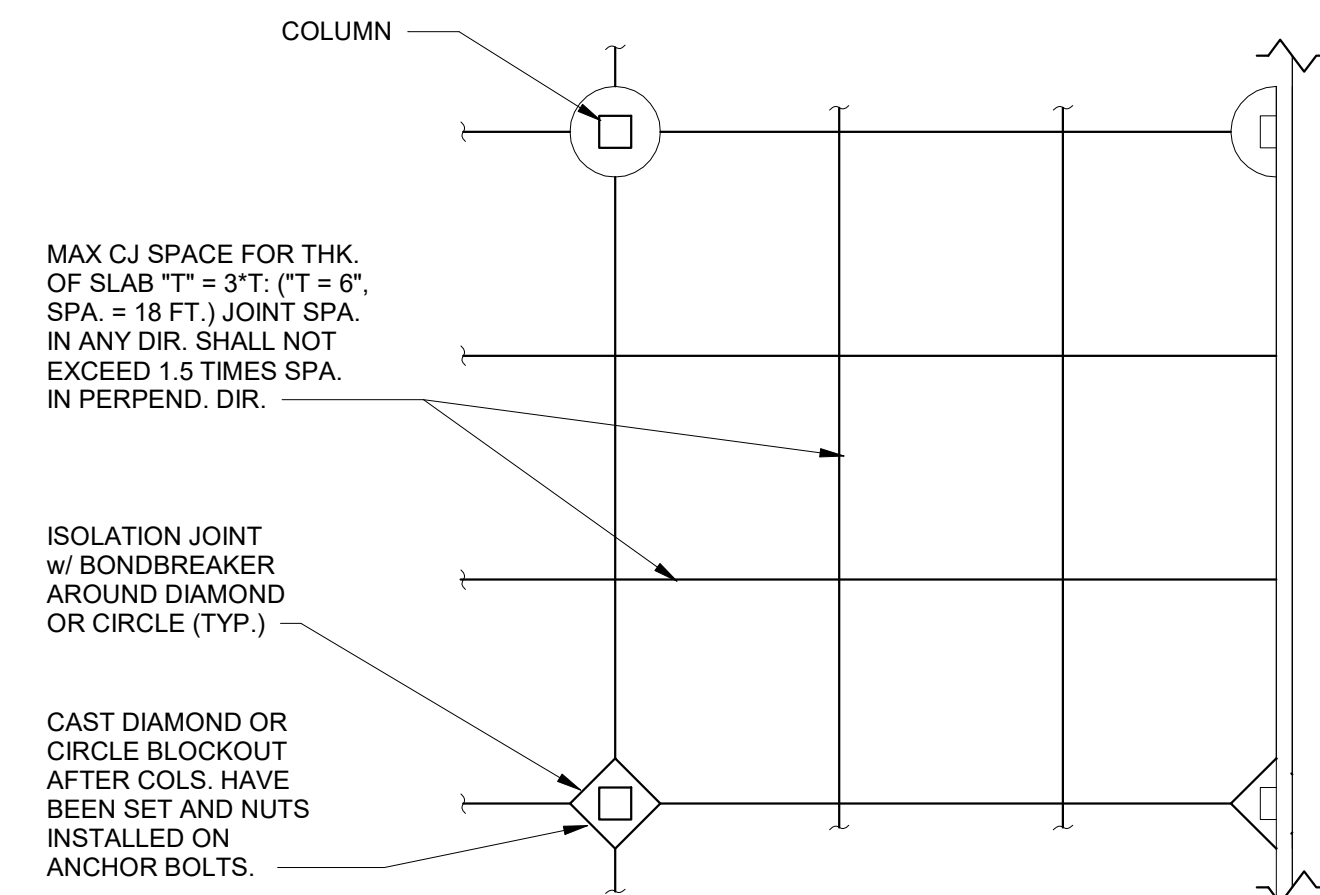
B9 INTERIOR SLAB ON GRADE JOINT DETAILS

SCALE: 1" = 1'-0"



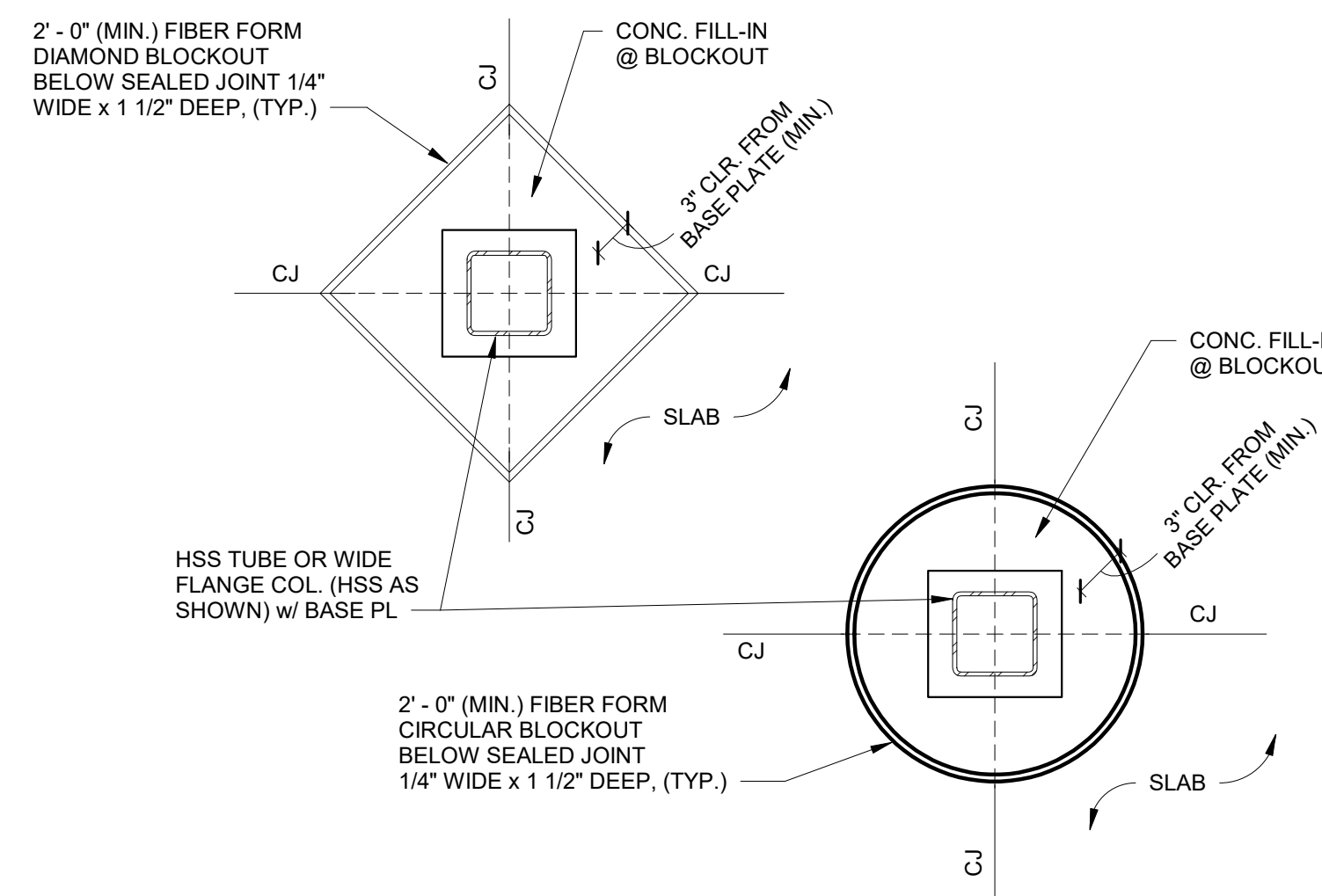
**SLAB REINFORCING
AT CONSTRUCTION JOINT ENDS**

SCALE: 1" = 1'-0"



C15 SLAB ON GRADE JOINT PATTERN

SCALE: 1" = 1'-0"



B14 SLAB BLOCKOUT AT COLUMN

SCALE: 1" = 1'-0"



**US Army Corps
of Engineers ®**

[illegible]

DESIGNED BY: S. RICCI	ISSUE DATE: JUNE 26, 2024
DRAWN BY: S. RICCI	SOLICITATION NO.:
CHECKED BY: S. DUFFY	CONTRACT NO.:
SUBMITTED BY:	
SIZE: ANSI D	

U.S. ARMY CORPS OF ENGINEERS

BOWST ADDITION AND RENOVATIONS TO B-321
NIAGARA FALLS AIR RESERVE STATION, NY

P2#: 508014

SHEET ID

S-502



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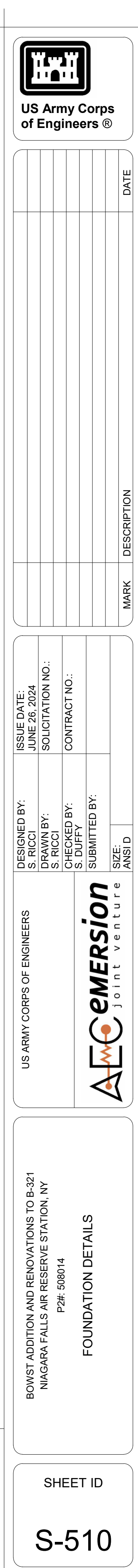
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AD101

FIRST FLOOR DEMOLITION FLOOR PLAN

Conversion

SUBMITTED BY

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==== EXISTING TO REMAIN
 - - - - - DEMOLITION

- 02-018 EXISTING BUILDING TO REMAIN
- 02-019 REMOVE CONCRETE CABLE TRENCH WITH 3/8" THICK DIAMOND PLATE STEEL COVER
- 02-032 EXISTING EYE WASH
- 02-050 REMOVE PORTION OF WALL TO ACCOMMODATE NEW CONSTRUCTION. COORDINATE OPENING WITH NEW CONSTRUCTION
- 02-051 REMOVE DOOR AND FRAME
- 02-052 REMOVE FLOOR SLAB, CUT VAPOR BARRIER AND PERIMETER INSULATION
- 02-053 CUT OPENING IN SHEATHING AND METAL FRAMING
- 02-054 REMOVE BRICK, BRICK TIES AND INSULATION - SHEATHING AND FRAMING TO REMAIN
- 02-055 REMOVE MILLWORK AND COUNTERTOP
- 02-056 REMOVE STEEL STAIR, CUT ANCHORS FLUSH WITH FLOOR SLAB
- 02-057 REMOVE CONCRETE SLAB, VAPOR BARRIER, CONCRETE TRENCHESS, STEEL PLATES AND STEEL ANGLES AS NECESSARY FOR INSTALLATION OF NEW FOUNDATIONS
- 02-060 REMOVE OPERABLE PARTITION
- 02-061 REMOVE CARPET AND ADHESIVE - REMOVE RESILIENT BASE
- 02-064 REMOVE PORTION OF CONCRETE TRENCH AS NECESSARY TO INSTALL CRANE FOUNDATION - SAWCUT CONCRETE TO HAVE A CLEAN JOINT
- 02-065 REMOVE TRENCH DRAIN
- 02-066 REMOVE GYPSUM BOARD AND METAL FRAMING PARTITION
- 02-068 REMOVE AND SALVAGE EMERGENCY EYE WASH
- 02-069 REMOVE CARPET AND ADHESIVE - REMOVE RESILIENT BASE - RESILIENT FLOORING UNDER CARPET TO REMAIN
- 02-070 REMOVE AND SALVAGE RESILIENT FLOORING AS NECESSARY TO CUT SLAB AND INSTALL SPRINKLER PIPING BELOW GRADE
- 02-071 REMOVE AND SALVAGE ROOF LEADER OUTLET THROUGH BRICK
- 02-072 REMOVE BRICK AND CUT HOLE IN INSULATION, AIR BARRIER AND SHEATHING AS NECESSARY TO INSTALL ROOF LEADER OUTLET
- 02-073 REMOVE AND SALVAGE CONCRETE SPLASH BLOCK

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AD111

FIRST FLOOR DEMOLITION CEILING PLAN



ALCemersion
joint venture

02-060	REMOVE OPERABLE PARTITION
02-062	REMOVE LAY-IN CEILING AND GRID - REMOVE LIGHT FIXTURES, DIFFUSERS, SPRINKLER HEADS AND OTHER DEVICES ON CEILING
02-074	REMOVE AND SALVAGE LAY-IN CEILING, GRID, AND DEVICES AS NECESSARY TO REMOVE ROOF LEADER AND INSTALL NEW LINE

 2' x 2' ACT CEILING

 2' x 2' ACT CEILING TO BE REMOVED

 GYP. BD. CEILING

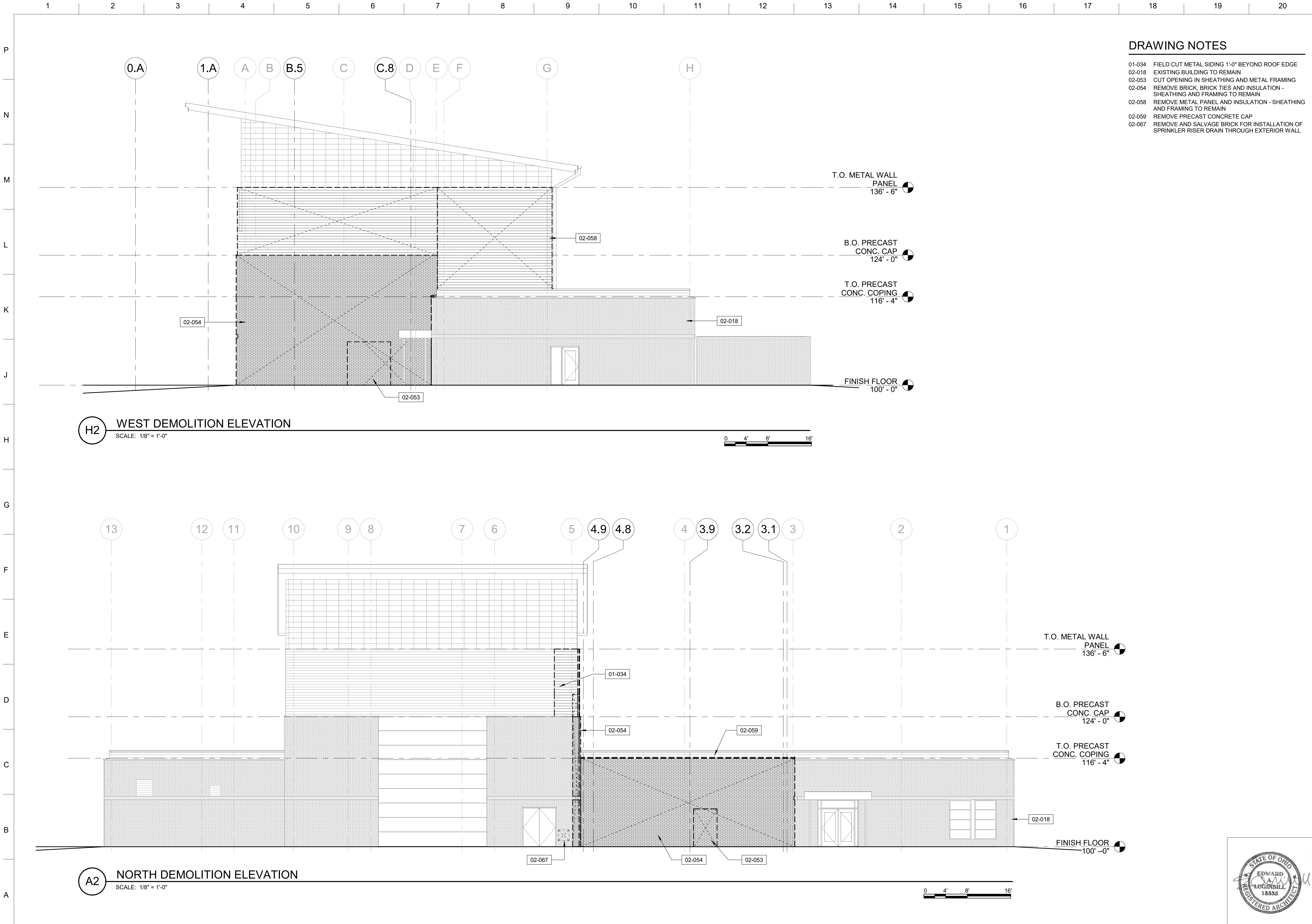
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W9120R25BA008 Plans-0000

CERTIFIED FINANCIAL



- DRAWING NOTES
- 01-034

FIELD CUT METAL SIDING 1'-0" BEYOND ROOF EDGE
- 02-018

EXISTING BUILDING TO REMAIN
- 02-053

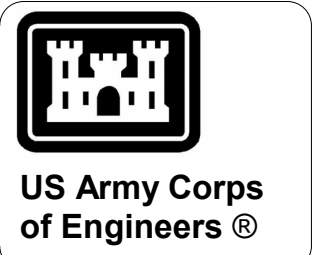
CUT OPENING IN SHEATHING AND METAL FRAMING
- 02-054

REMOVE BRICK, BRICK TIES AND INSULATION - SHEATHING AND FRAMING TO REMAIN
- 02-058

REMOVE METAL PANEL AND INSULATION - SHEATHING AND FRAMING TO REMAIN
- 02-059

REMOVE PRECAST CONCRETE CAP
- 02-067

REMOVE AND SALVAGE BRICK FOR INSTALLATION OF SPRINKLER RISER DRAIN THROUGH EXTERIOR WALL



DATE	
MARK	DESCRIPTION

ISSUE DATE: JUNE 26, 2024	SOLICITATION NO.:	CONTRACT NO.:
DESIGNED BY: A. LUGNBILL	DRAWN BY: A. MORRISON	CHECKED BY: S. DUFFY
SUBMITTED BY:		
SIZE: ANSI D		

US ARMY CORPS OF ENGINEERS

LOUISVILLE DISTRICT

600 DR. MARTIN LUTHER KING, JR. PLACE

LOUISVILLE KY 40202-2239

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BOWST ADDITION AND RENOVATIONS TO B-321

NIAGARA FALLS AIR RESERVE STATION, NY

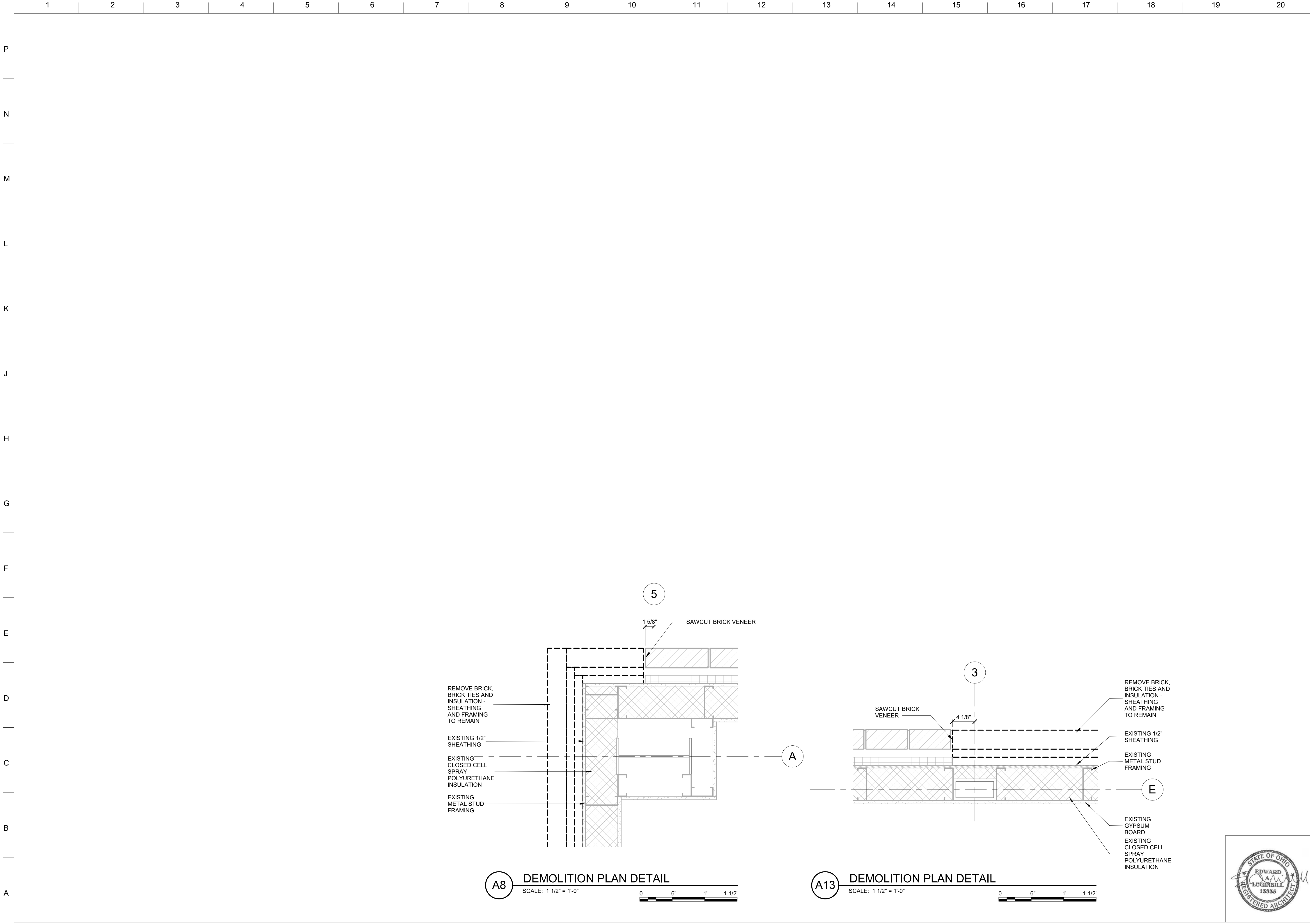
P2#: 508014

DEMOLITION ELEVATIONS



SHEET ID

AD201



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MARK	DESCRIPTION	DATE

DESIGNED BY: A. LUGNIBILL	ISSUE DATE: JUNE 26, 2024
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CHECKED BY: S. DUFFY	CONTRACT NO.:
SUBMITTED BY:	
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US ARMY CORPS OF ENGINEERS
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LOUISVILLE KY 40202-2239

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P2#-508014

DEMOLITION PLAN DETAILS

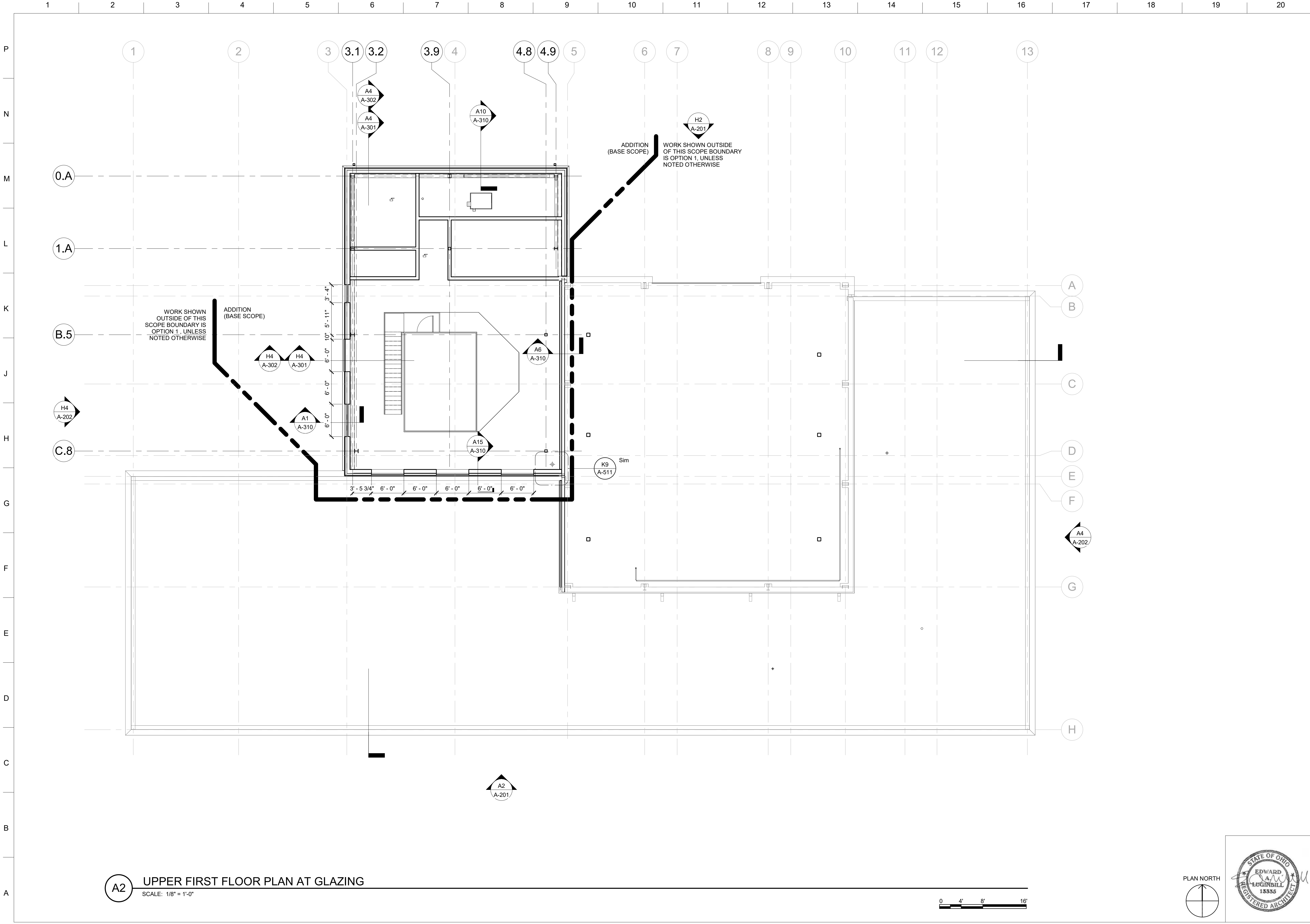


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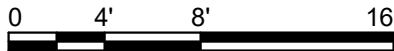


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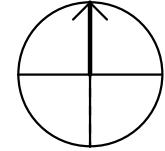
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A2 UPPER FIRST FLOOR PLAN AT GLAZING
SCALE: 1/8" = 1'-0"



PLAN NORTH



BOWST ADDITION AND RENOVATIONS TO B-321
NIAGARA FALLS AIR RESERVE STATION, NY
P2#-508014

UPPER FIRST FLOOR PLAN

US ARMY CORPS OF ENGINEERS
LOUISVILLE DISTRICT
600 DR. MARTIN LUTHER KING, JR. PLACE
LOUISVILLE KY 40202-2239

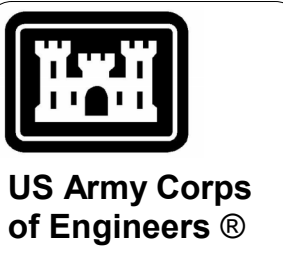


DESIGNED BY:
A. LUGNBILL
DRAWN BY:
A. MORRISON
CHECKED BY:
S. DUFFY
SUBMITTED BY:
SIZE:
ANSI D

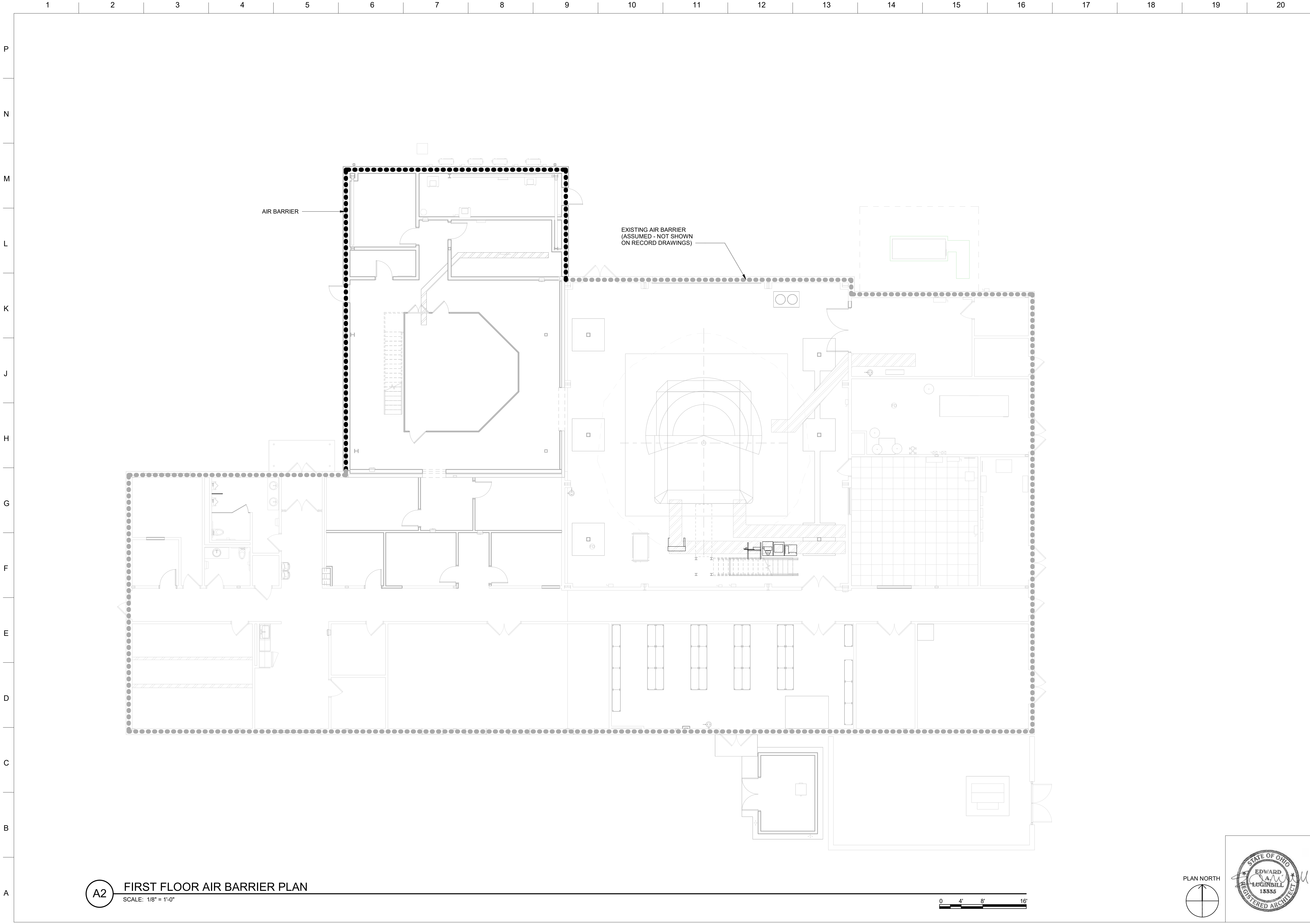
ISSUE DATE:
JUNE 26, 2024
SOLICITATION NO.:
CONTRACT NO.:

MARK	DESCRIPTION

DATE





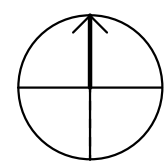


A2

FIRST FLOOR AIR BARRIER PLAN

SCALE: 1/8" = 1'-0"

PLAN NORTH



BOWST ADDITION AND RENOVATIONS TO B-321
NIAGARA FALLS AIR RESERVE STATION, NY
P2#- 508014

FIRST FLOOR AIR BARRIER PLAN

US ARMY CORPS OF ENGINEERS
LOUISVILLE DISTRICT
600 DR. MARTIN LUTHER KING, JR. PLACE
LOUISVILLE KY 40202-2239



DESIGNED BY:
A. LUGNBILL
DRAWN BY:
A. MORRISON
CHECKED BY:
S. DUFFY
SUBMITTED BY:
S. DUFFY
SIZE:
ANSI D

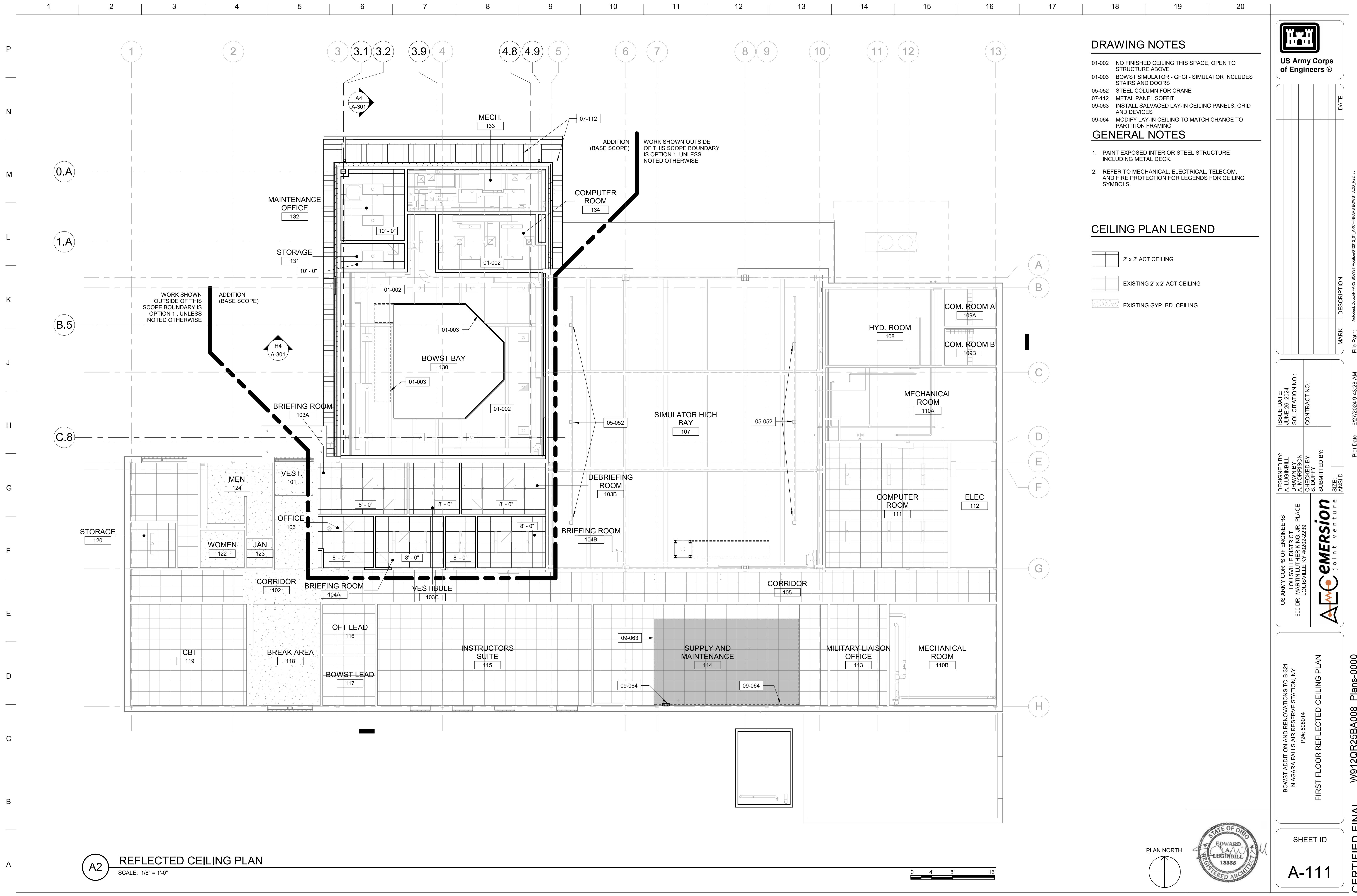
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JUNE 26, 2024
SOLICITATION NO.:
CONTRACT NO.:

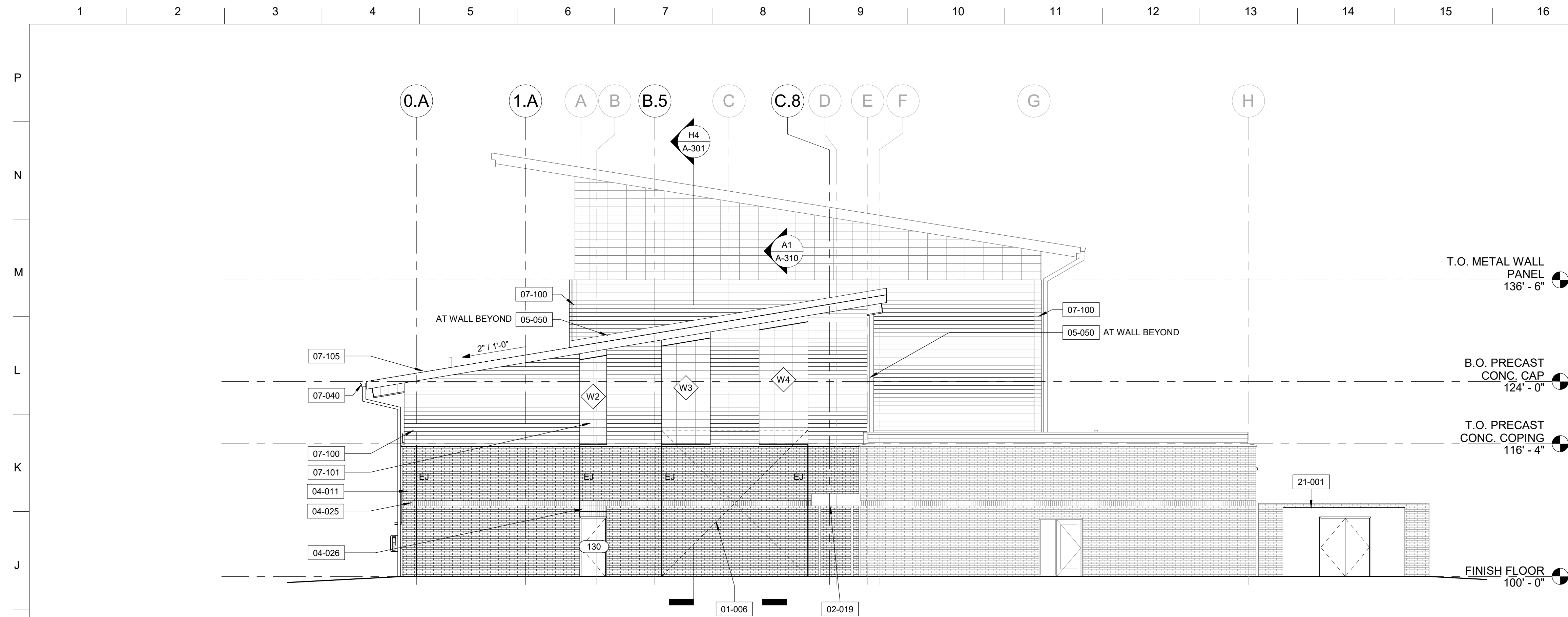
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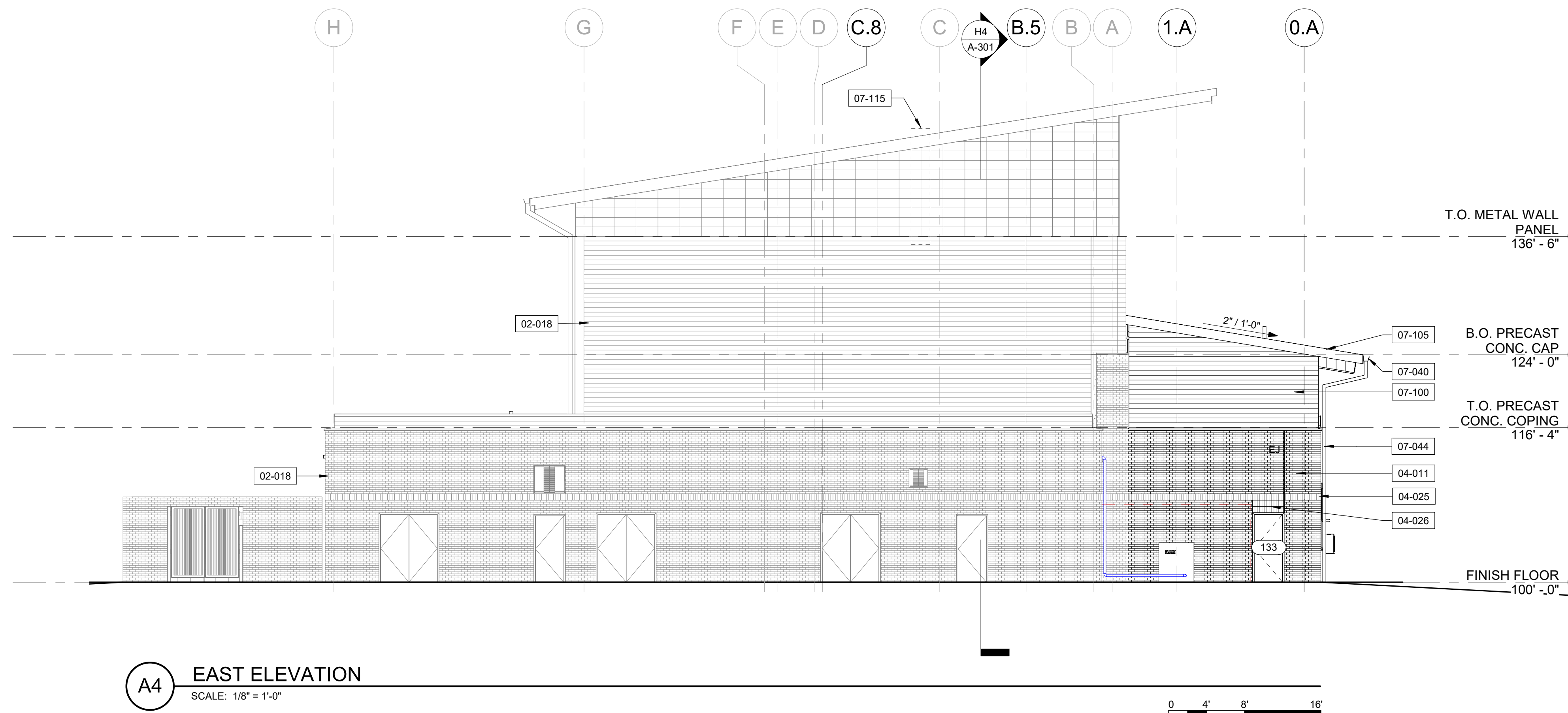
SHEET ID

A-104





H4 WEST ELEVATION
SCALE: 1/8" = 1'-0"



A4 EAST ELEVATION
SCALE: 1/8" = 1'-0"

EXTERIOR MATERIAL NOTES

1. EXTERIOR MATERIAL FINISHES MUST MATCH THE EXISTING BUILDING. FINISHES LISTED HERE PROVIDE REFERENCE INFORMATION ON THOSE MATERIALS, BUT DO NOT LIMIT THE CHOICES AVAILABLE FOR USE.
2. FINISHES LISTED HERE ARE FROM THE RECORD DRAWINGS AND MAY NOT BE THE ACTUAL PRODUCT INSTALLED.
3. CONTRACTING OFFICER WILL EVALUATE IF PROPOSED MATERIALS ARE AN ACCEPTABLE MATCH.
4. EXISTING FIELD BRICK (BR-1): CUNNINGHAM BRICK COMPANY (NO LONGER AVAILABLE); COLOR: WINESTONE VELDUR.
5. EXISTING MORTAR COLOR FOR BR-1: KOSMOS CEMENT; PRODUCT: KOSMORTAR; COLOR: RED.
6. EXISTING ACCENT BRICK (BR-2): BELDEN BRICK COMPANY; COLOR: BLACK.
7. EXISTING MORTAR COLOR FOR BR-2: KOSMOS CEMENT; PRODUCT: KOSMORTAR; COLOR: BLACK, 3 LBS.
8. METAL WALL AND SOFFIT PANELS: FIRESTONE UNO-CLAD, FINISH: KYNAR 5000/HYPAR 5000; COLOR: CHAMPAGNE METALLIC.
9. METAL TRIM, LOUVERS, GUTTERS, AND DOWNSPOUTS MUST MATCH THE METAL WALL PANEL FINISH.

DRAWING NOTES

01-006	LOCATION FOR FUTURE 18' X 18' OPENING
02-018	EXISTING BUILDING TO REMAIN
02-019	EXISTING ENTRANCE CANOPY TO REMAIN
04-011	BRICK (BR-1) - MATCH EXISTING
04-025	BRICK SOLDIER COURSE (BR-2)
04-026	DOUBLE BRICK SOLDIER COURSE (BR-2) OVER DOOR
05-050	EXPANSION JOINT
07-040	PRE-FINISHED METAL GUTTER, 5" WIDE X 8" DEEP WITH SUPPORT BRACKETS AT 32" O.C.
07-044	PRE-FINISHED METAL DOWNSPOUT, 4" X 3"
07-100	METAL SIDING - MATCH EXISTING
07-101	TRANSLUCENT ACRYLIC WALL PANEL
07-105	METAL FASCIA
07-115	FILL GAP IN EXISTING TRANSLUCENT PANELS WITH BACKER ROD AND SEALANT ON BOTH EXTERIOR AND INTERIOR FACES.
21-001	PACKAGED FIRE PUMP ENCLOSURE



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A. MORRISON	
CHECKED BY:	
S. DUFFY	
SUBMITTED BY:	
SIZE:	
ANSI D	

LOUISVILLE DISTRICT
600 DR. MARTIN LUTHER KING, JR. PLACE
LOUISVILLE KY 40202-2239



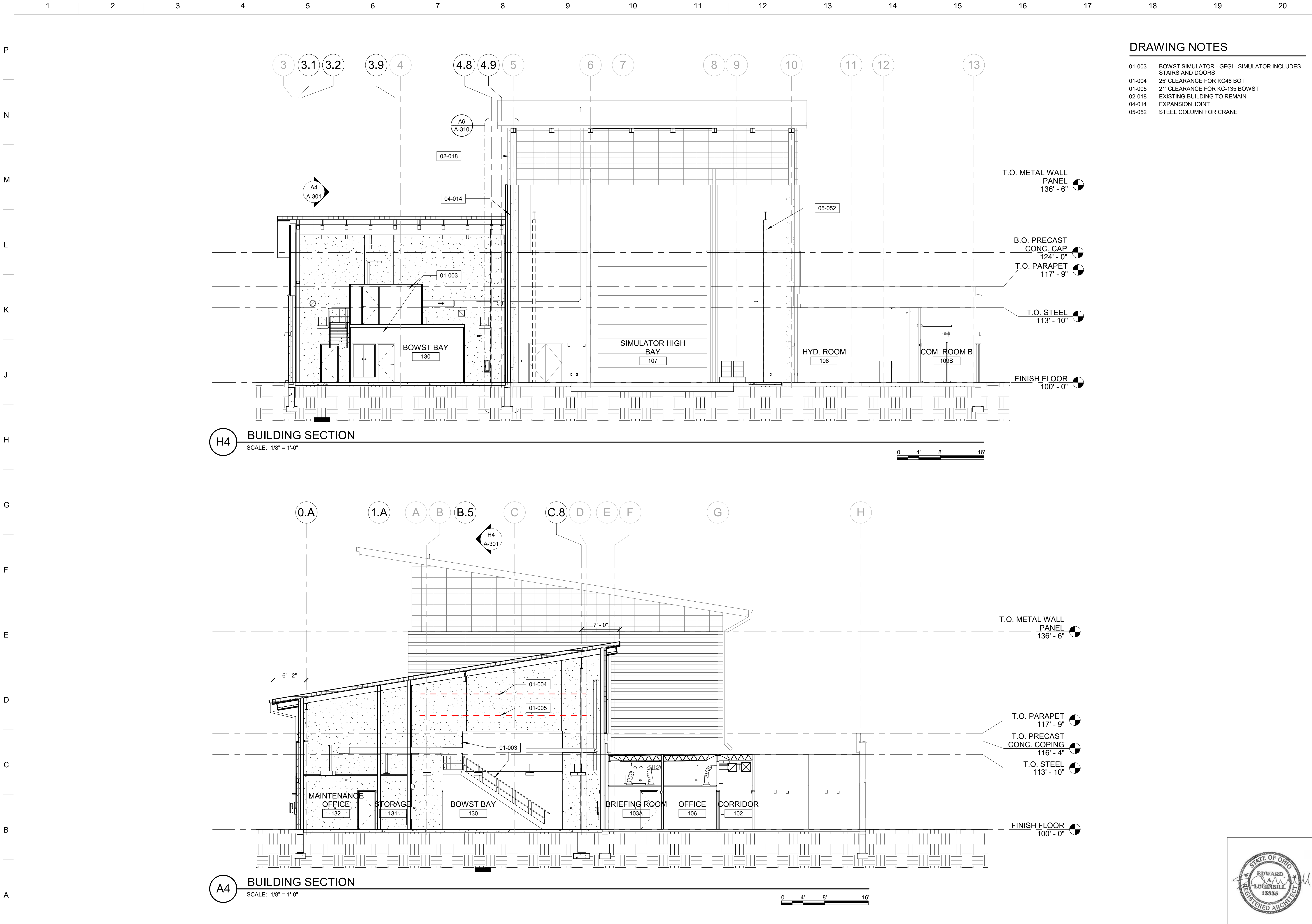
NIAGARA FALLS AIR RESERVE STATION, NY
P2#: 508014

EAST AND WEST ELEVATIONS

SHEET ID

A-202





DRAWING NOTES

- 01-003 BOWST SIMULATOR - GFGL - SIMULATOR INCLUDES STAIRS AND DOORS
- 01-004 25' CLEARANCE FOR KC46 BOT
- 01-005 21' CLEARANCE FOR KC-135 BOWST
- 02-018 EXISTING BUILDING TO REMAIN
- 04-014 EXPANSION JOINT
- 05-052 STEEL COLUMN FOR CRANE



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LOUISVILLE DISTRICT
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LOUISVILLE KY 40202-2239

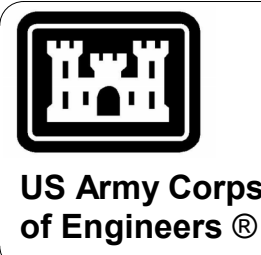
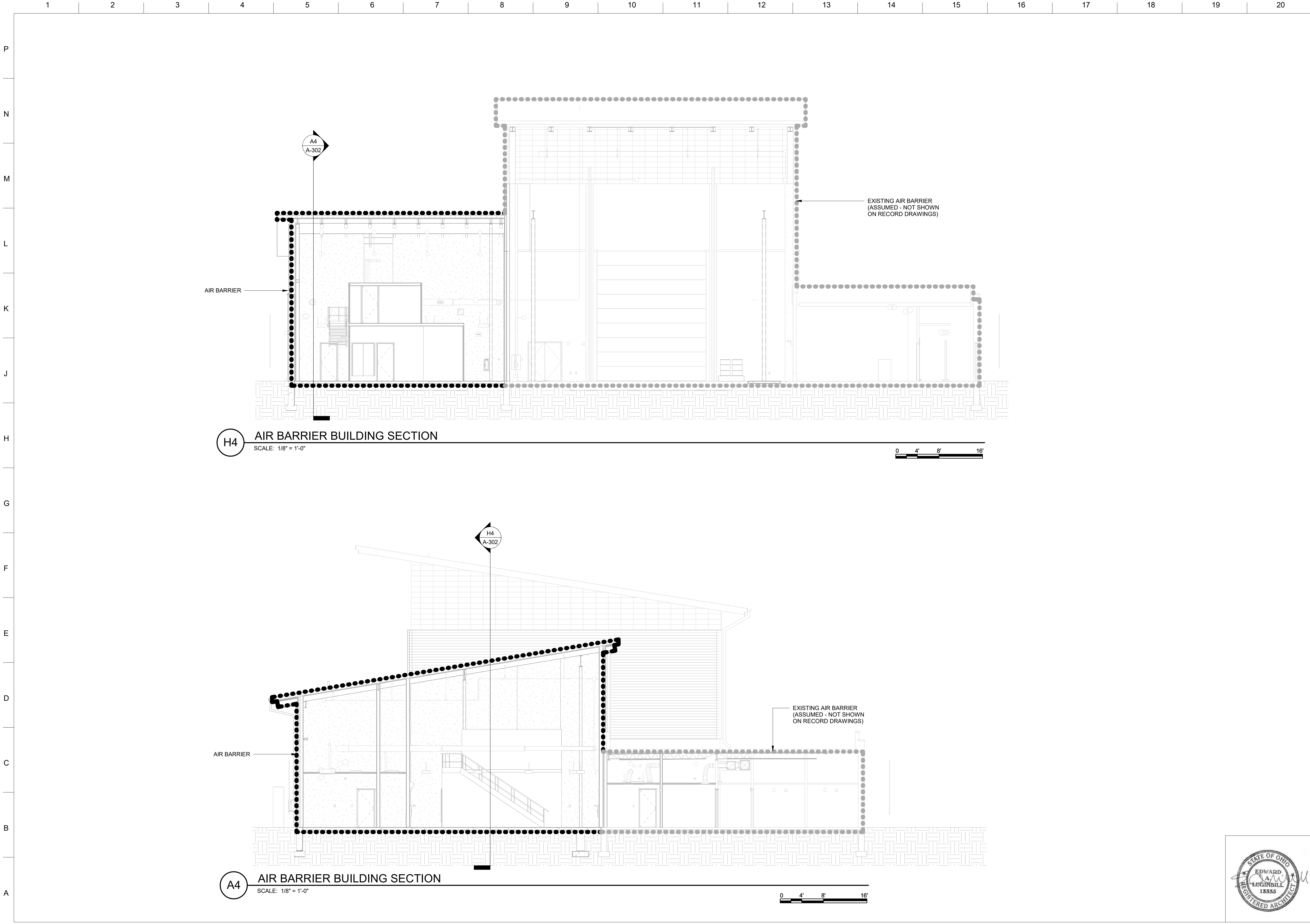
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BOWST ADDITION AND RENOVATIONS TO B-321
NIAGARA FALLS AIR RESERVE STATION, NY
P2#-508014

BUILDING SECTIONS

SHEET ID
A-301





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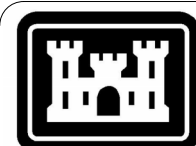
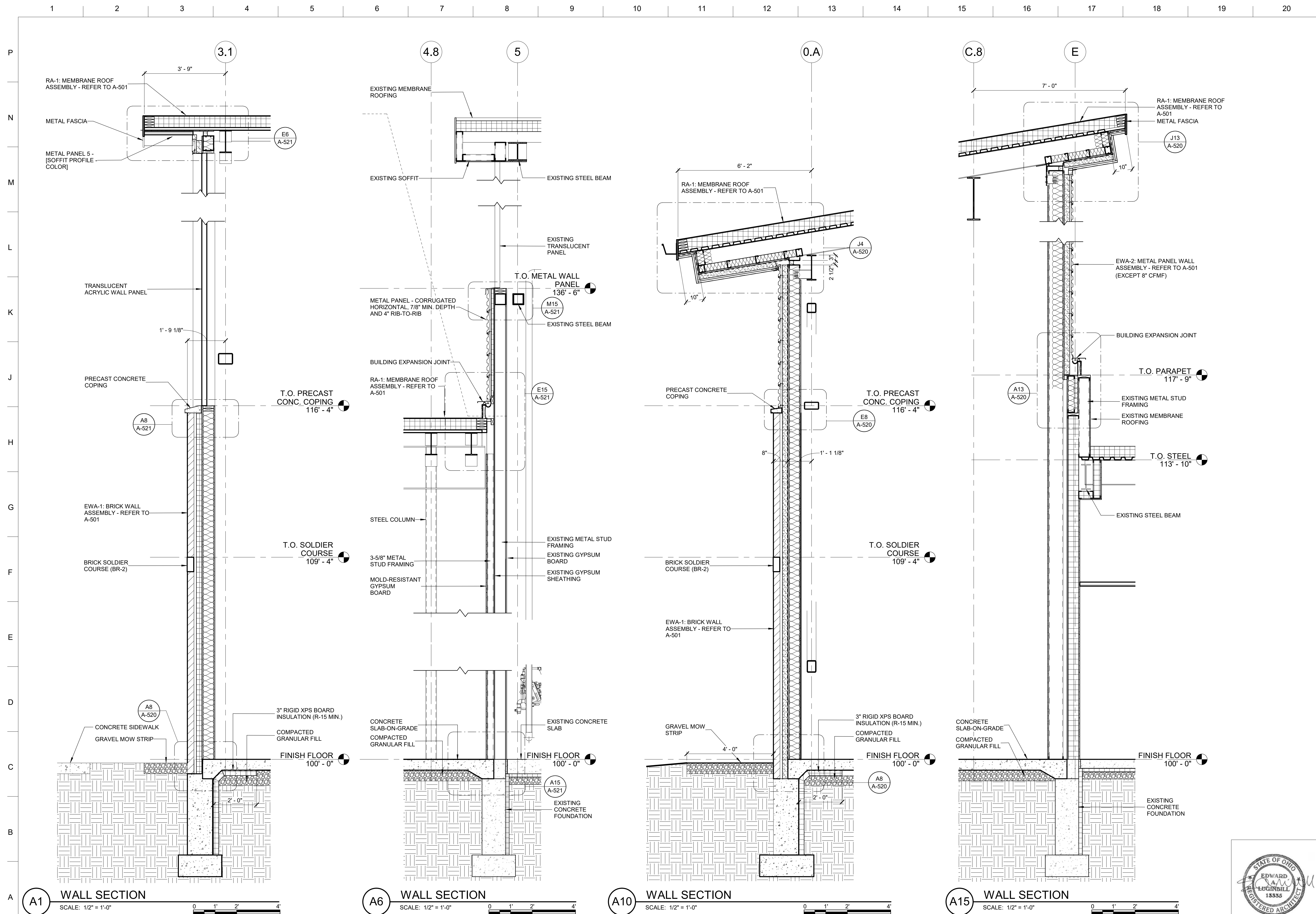
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DRAWN BY: A. MORRISON	SOLICITATION NO.:
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US ARMY CORPS OF ENGINEERS LOUISVILLE DISTRICT 600 DR. MARTIN LUTHER KING, JR. PLACE LOUISVILLE KY 40202-2239	AEC emersion joint venture
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BOWST ADDITION AND RENOVATIONS TO B-321 NIAGARA FALLS AIR RESERVE STATION, NY P2#-508014	AIR BARRIER BUILDING SECTIONS
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SHEET ID A-302





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DESIGNED BY: T. LUGNIBILL	ISSUE DATE: JUNE 26, 2024
DRAWN BY: M. MORRISON	SOLICITATION NO.:
CHECKED BY: S. DUFFY	CONTRACT NO.:
SUBMITTED BY:	
SIZE:	

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LOUISVILLE DISTRICT
600 DR. MARTIN LUTHER KING, JR. PLACE
LOUISVILLE KY 40202-2239

BOWST ADDITION AND RENOVATIONS TO B-321
NIAGARA FALLS AIR RESERVE STATION, NY
P2#: 508014

SHEET ID

A-310



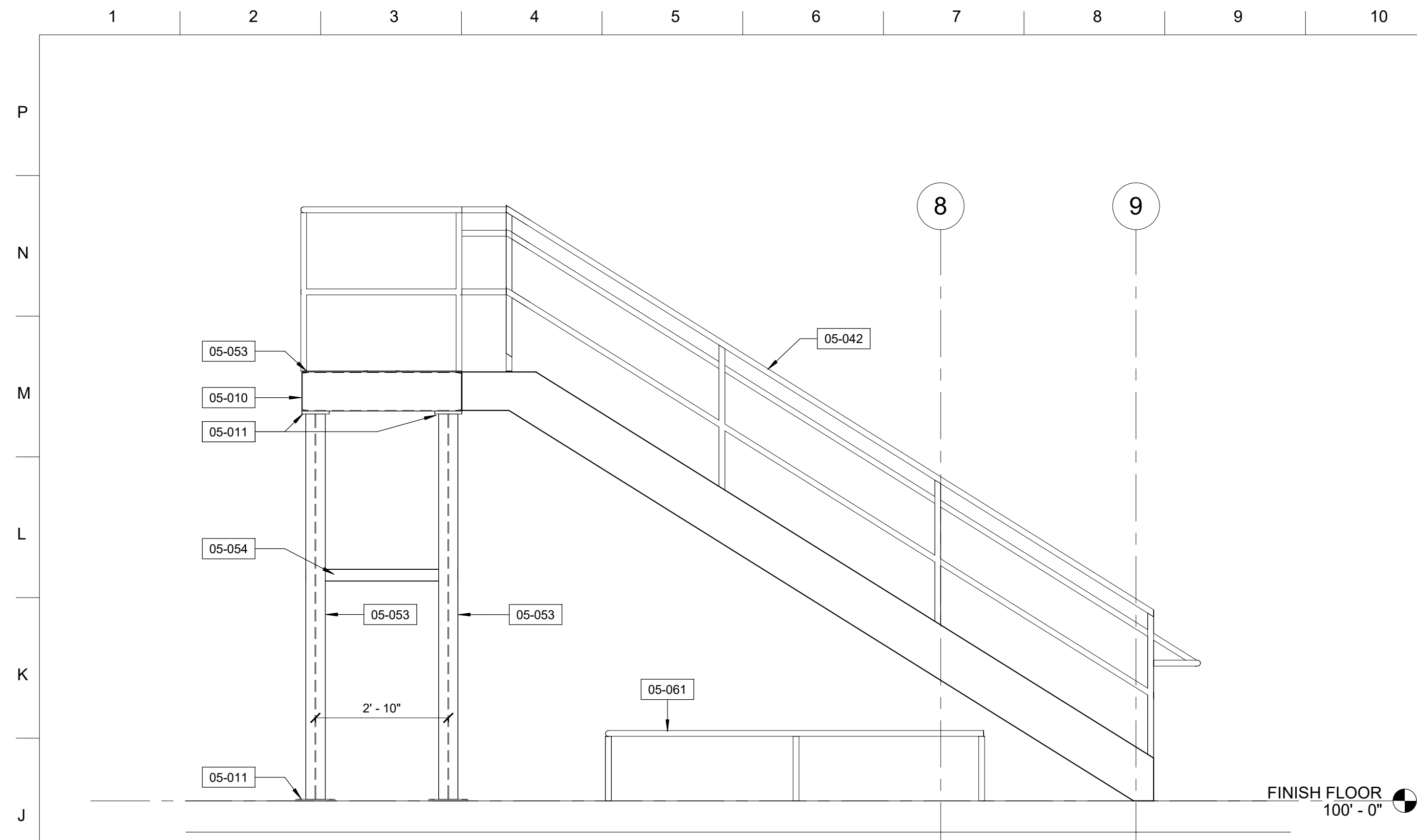
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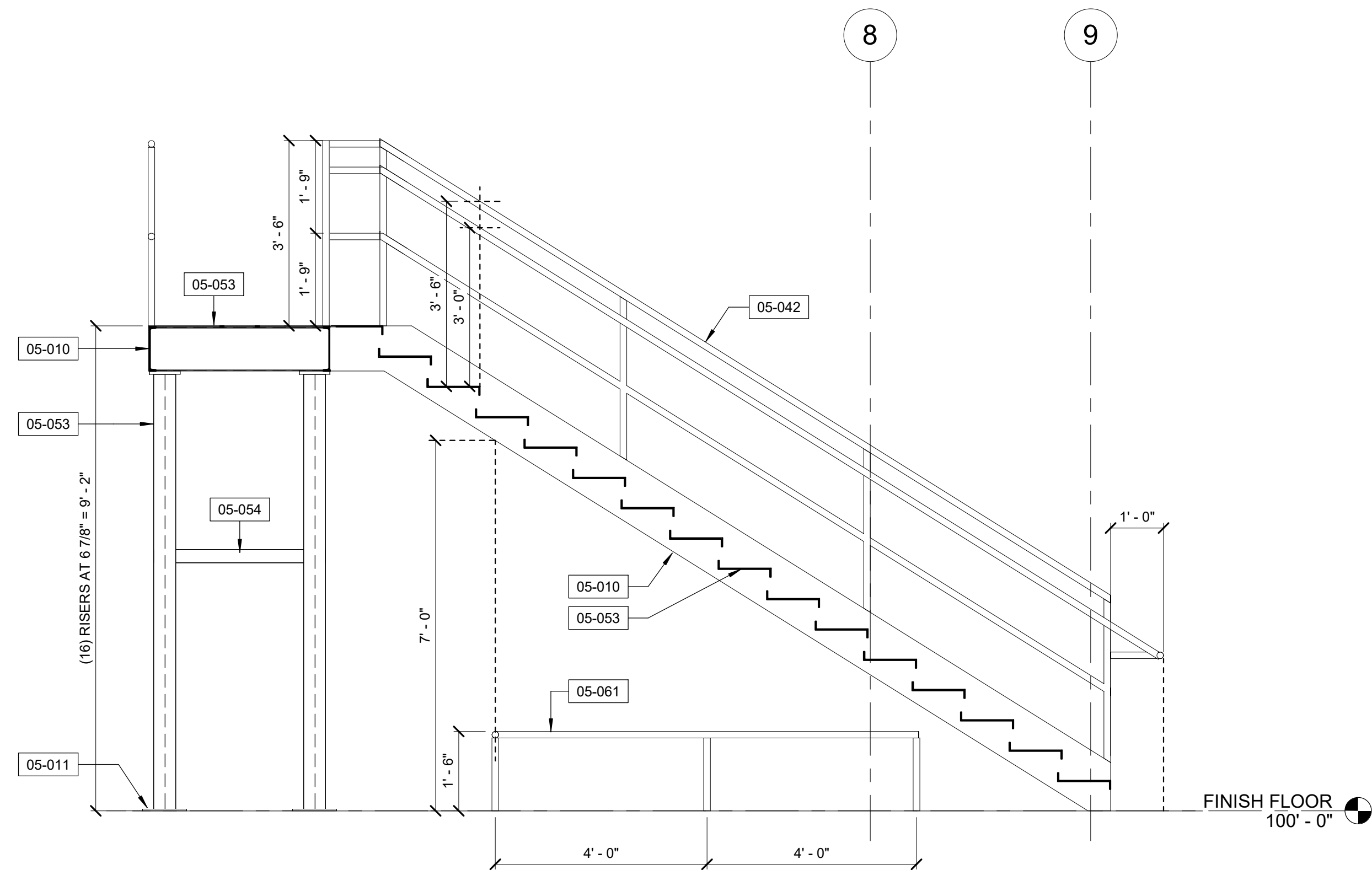
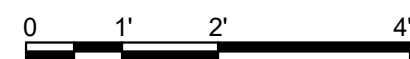
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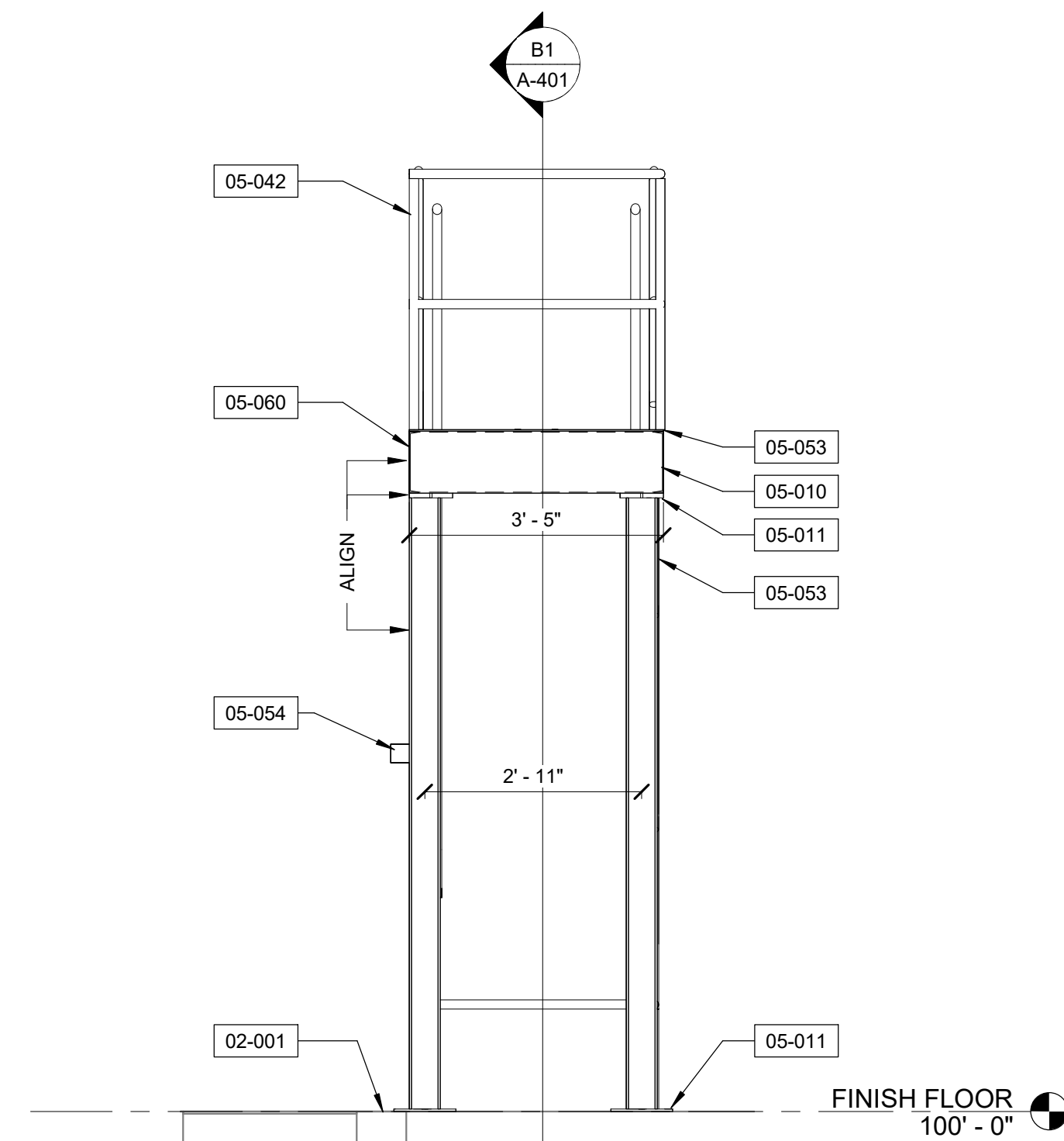
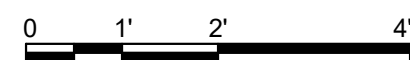
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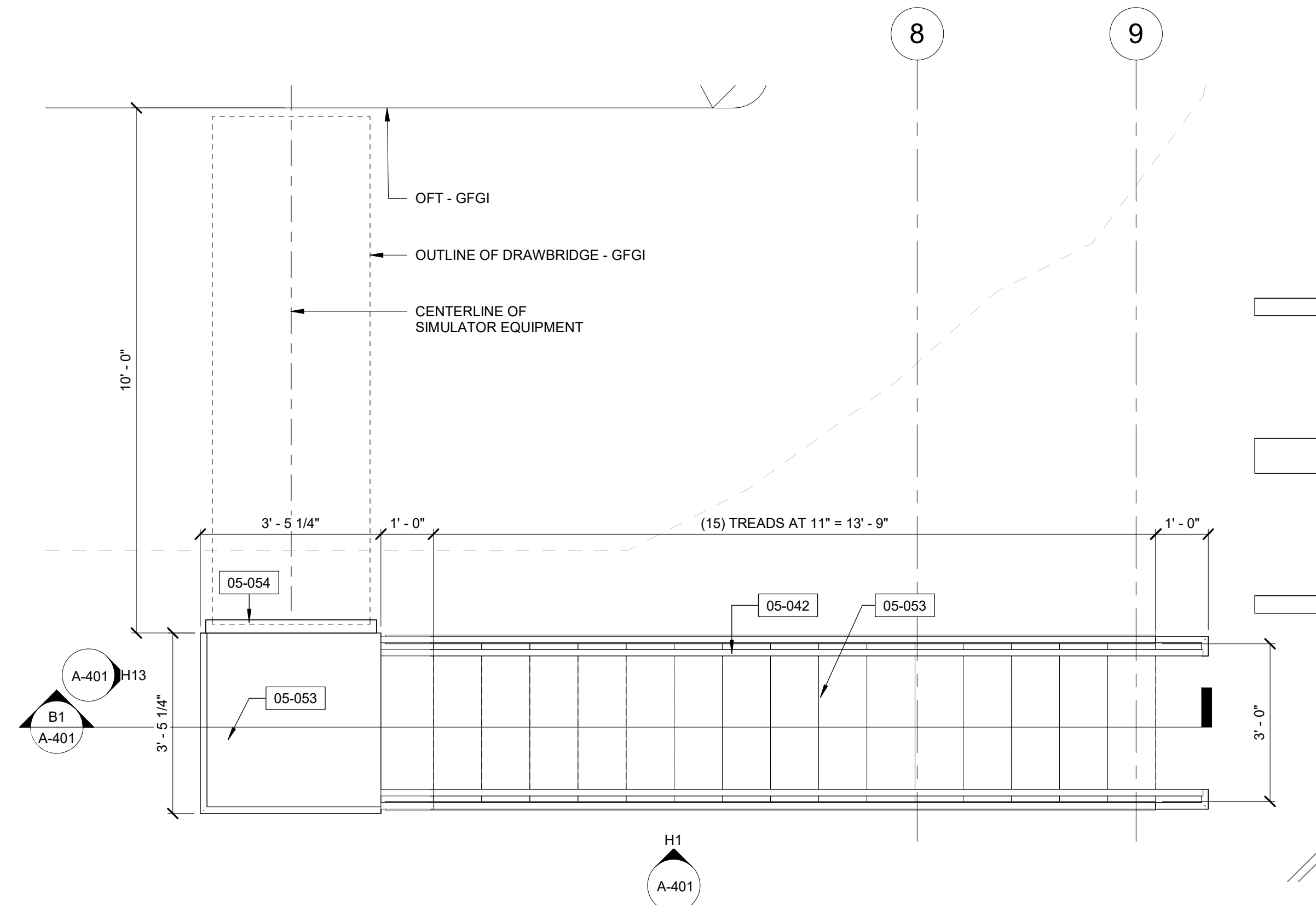
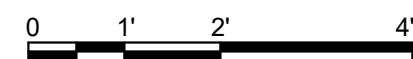
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SCALE: 1/2" = 1'-0"



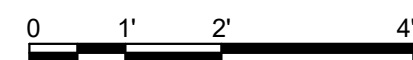
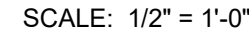
B1 STAIR SECTION
SCALE: 1/2" = 1'-0"



H13 STAIR ELEVATION
SCALE: 1/2" = 1'-0"



B9 ENLARGED STAIR PLAN
SCALE: 1/2" = 1'-0"



DRAWING NOTES

- | | |
|--------|--|
| 02-001 | EXISTING CONCRETE SLAB |
| 05-010 | STEEL CHANNEL |
| 05-011 | STEEL PLATE |
| 05-042 | STEEL GUARD AND HANDRAIL - 1 1/2" DIAMETER PIPE |
| 05-053 | 1/4" THICK DIAMOND STEEL PLATE |
| 05-054 | RAMP CYLINDER MOUNT - GFGI |
| 05-060 | SURFACE OF STEEL MEMBERS MUST BE ALIGNED TO FORM A FLAT SURFACE TO ALLOW FOR INSTALLATION OF GFGI DRAWBRIDGE |
| 05-061 | 1 1/2" DIAMETER STEEL TUBE PROTECTION UNDER STAIR - CORE DRILL CONCRETE SLAB FOR MOUNTING GUARD |



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600 DR. MARTIN LUTHER KING, JR. PLACE
LOUISVILLE KY 40202-2239



NIAGARA FALLS AIR RESERVE STATION, NY
P2#: 508014

SIMULATOR STAIR PLAN AND ELEVATIONS

SHEET ID

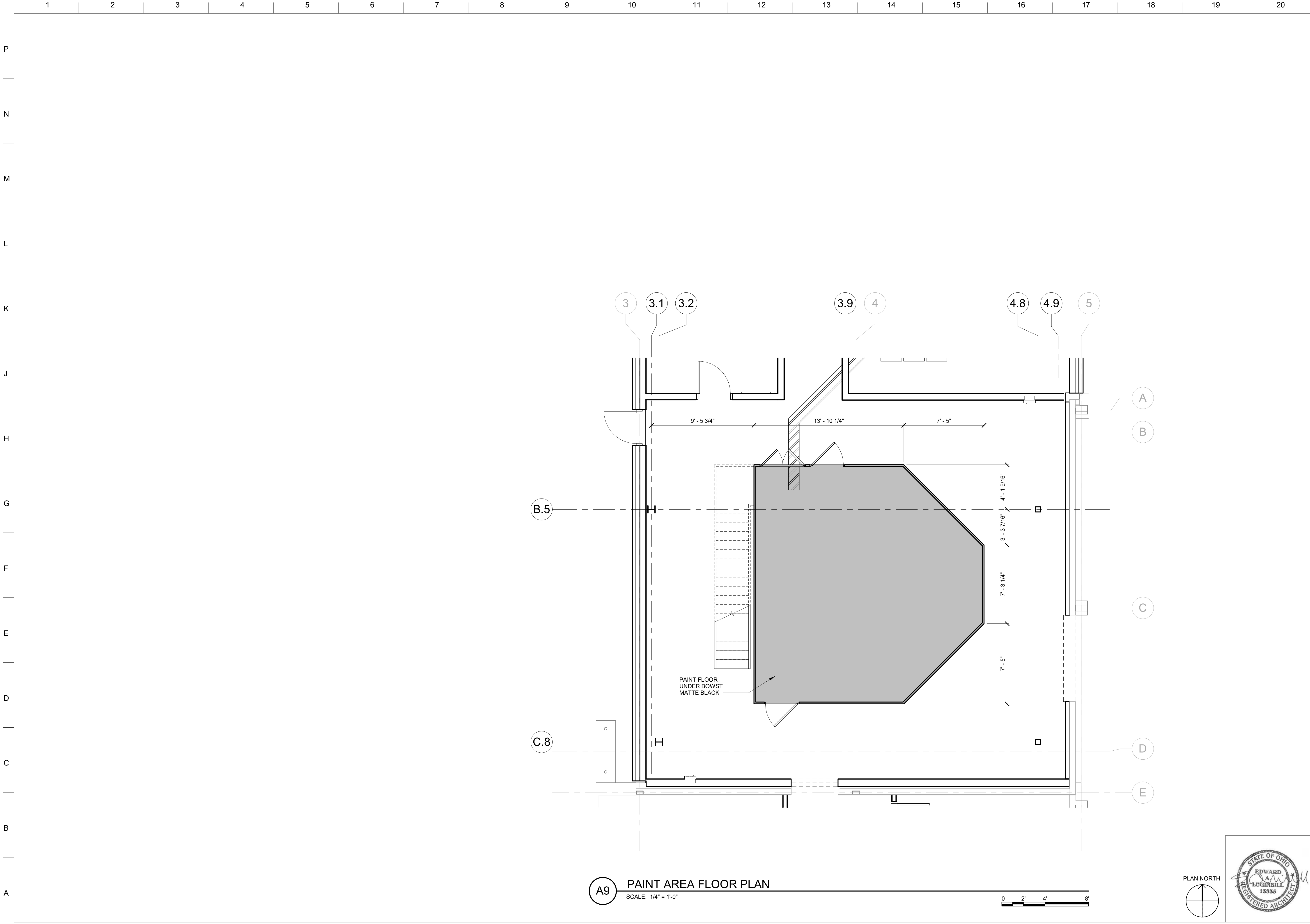
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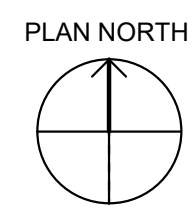
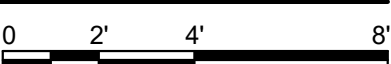
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A9 PAINT AREA FLOOR PLAN
SCALE: 1/4" = 1'-0"



BOWST ADDITION AND RENOVATIONS TO B-321
NIAGARA FALLS AIR RESERVE STATION, NY
P2#-508014

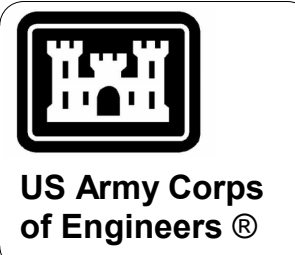
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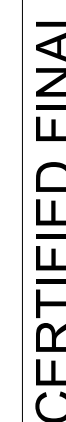
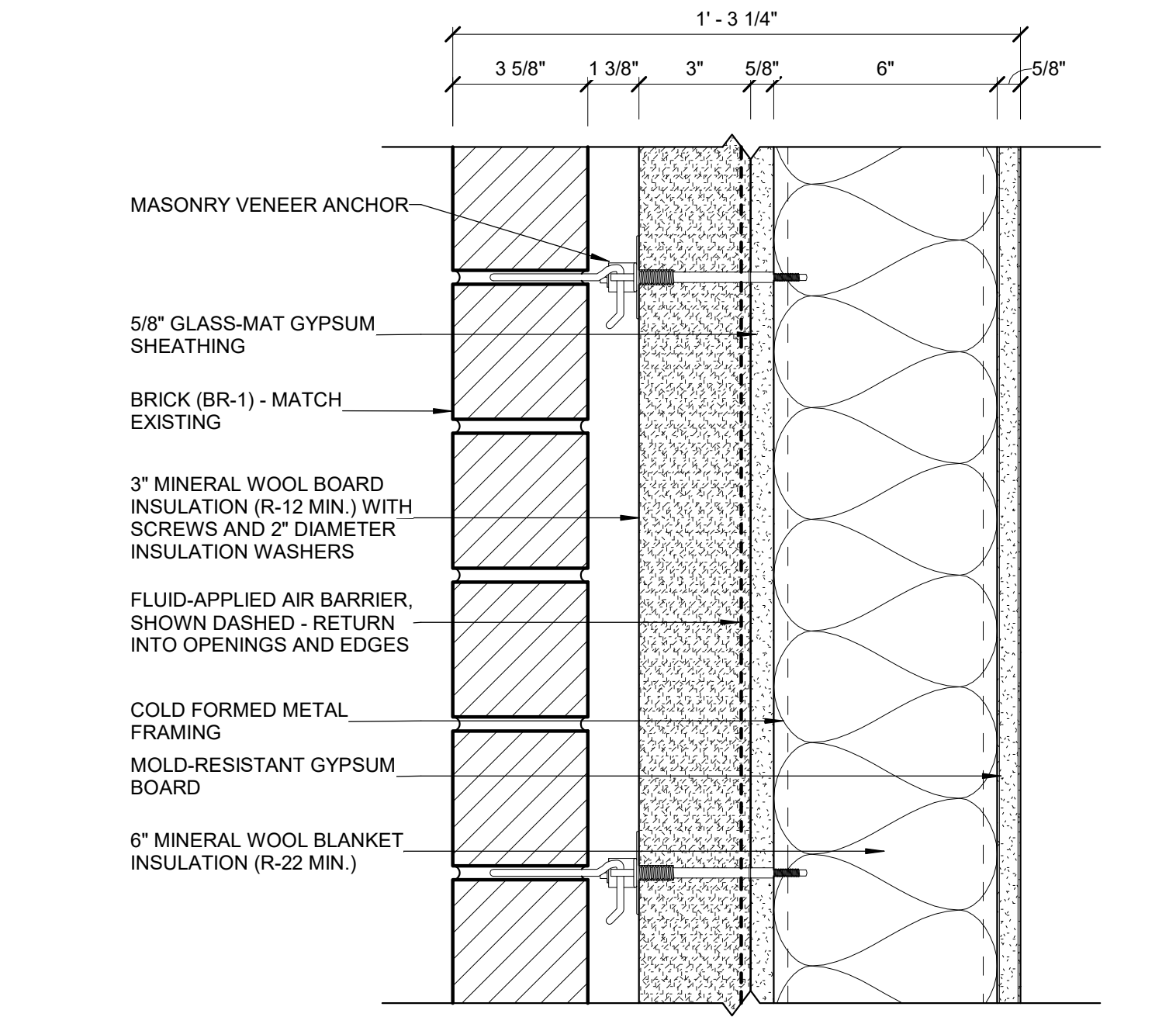
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SUBMITTED BY:	
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MARK	DESCRIPTION	DATE







PLAN DETAIL

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0 6" 1' 1 1/2"

PLAN DETAIL

SCALE: 1 1/2" = 1'-0"

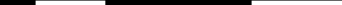
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A13 PLAN DETAIL
SCALE: 1 1/2" = 1'-0"



0 6" 1' 1 1/2'

E13 **PLAN DETAIL**
SCALE: 1 1/2" = 1'-0"



A13 PLAN DETAIL
SCALE: 1 1/2" = 1'-0"



0 6" 1' 1 1/2'

PLAN DETAIL

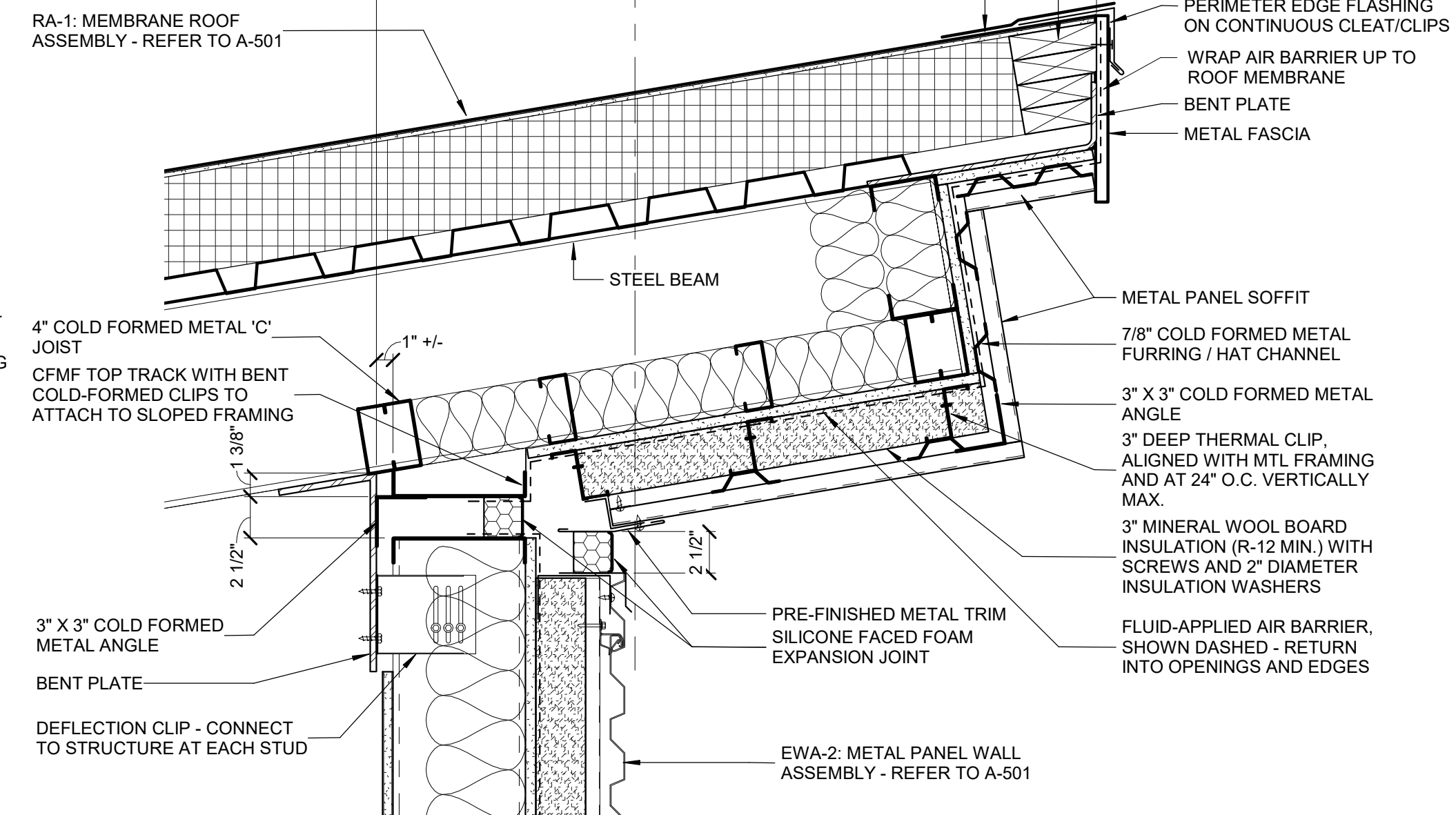
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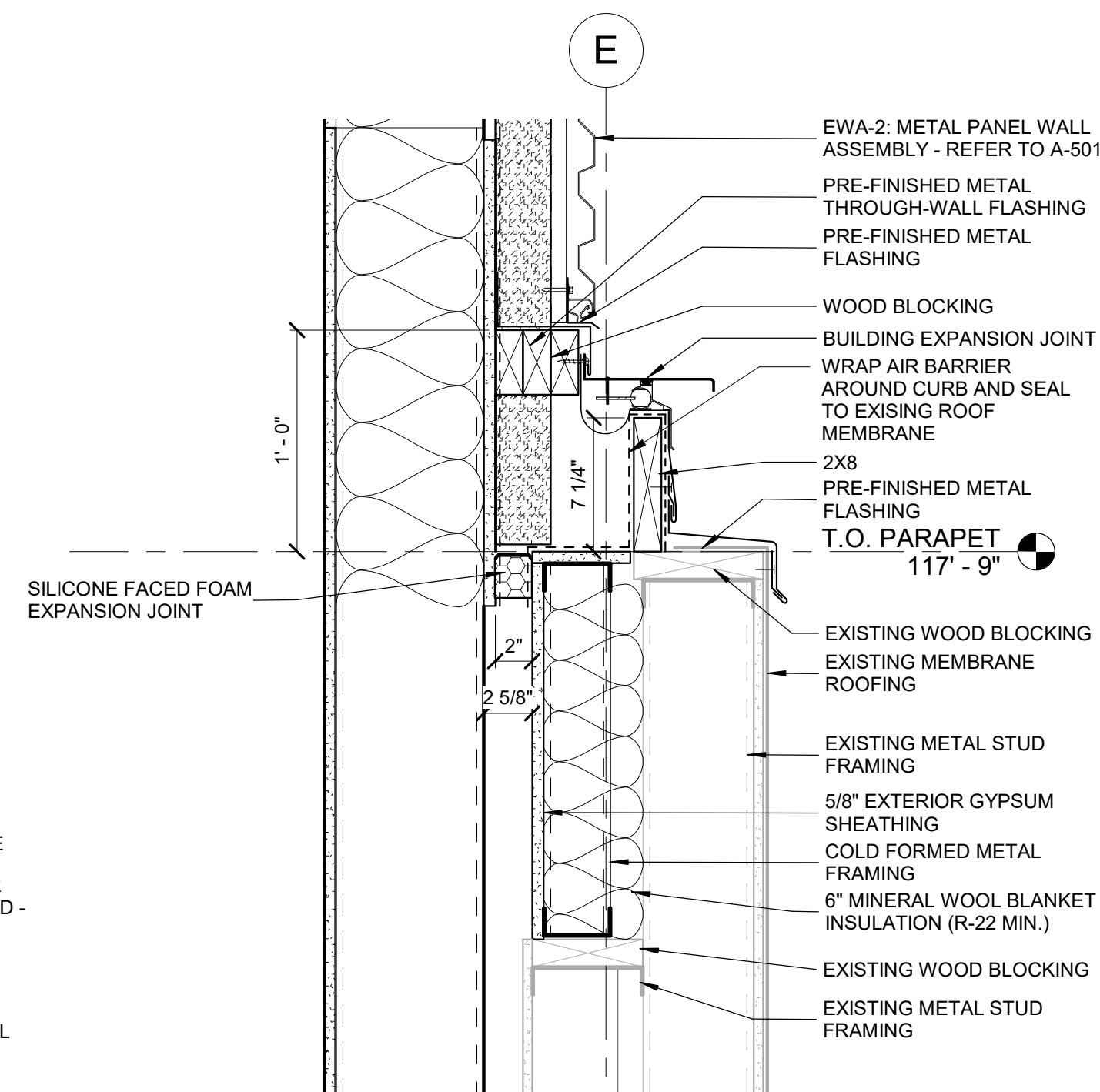
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0 6" 1' 1 1/2'

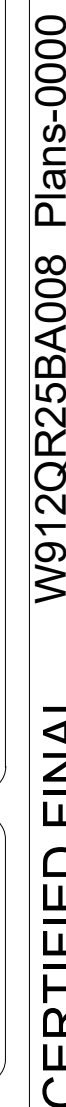
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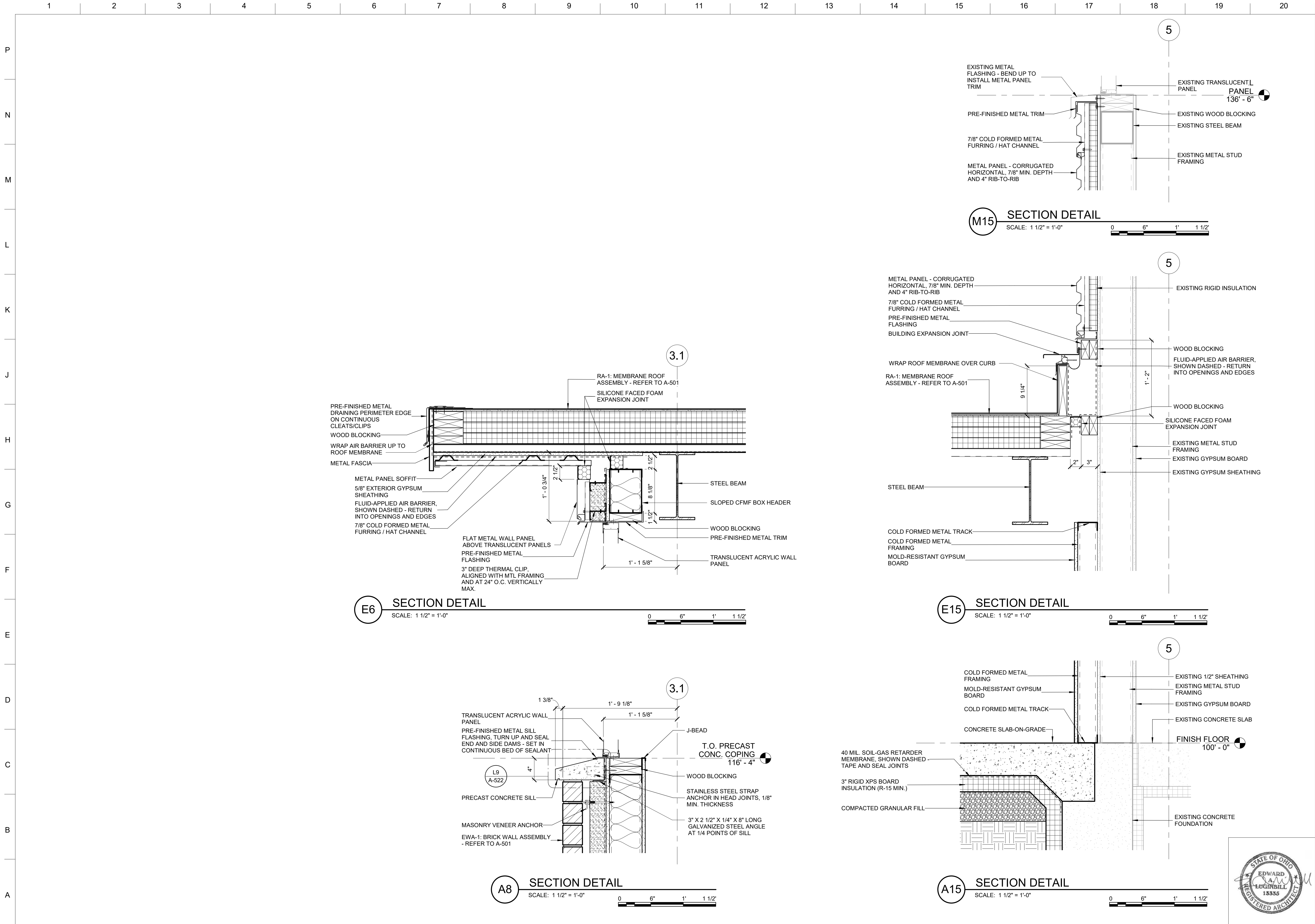


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0 6" 1' 1 1/2'

0 6" 1' 1 1/2'





MARK		DESCRIPTION	DATE

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LOUISVILLE DISTRICT
600 DR. MARTIN LUTHER KING, JR. PLACE
LOUISVILLE KY 40202-2239

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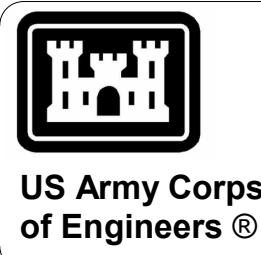
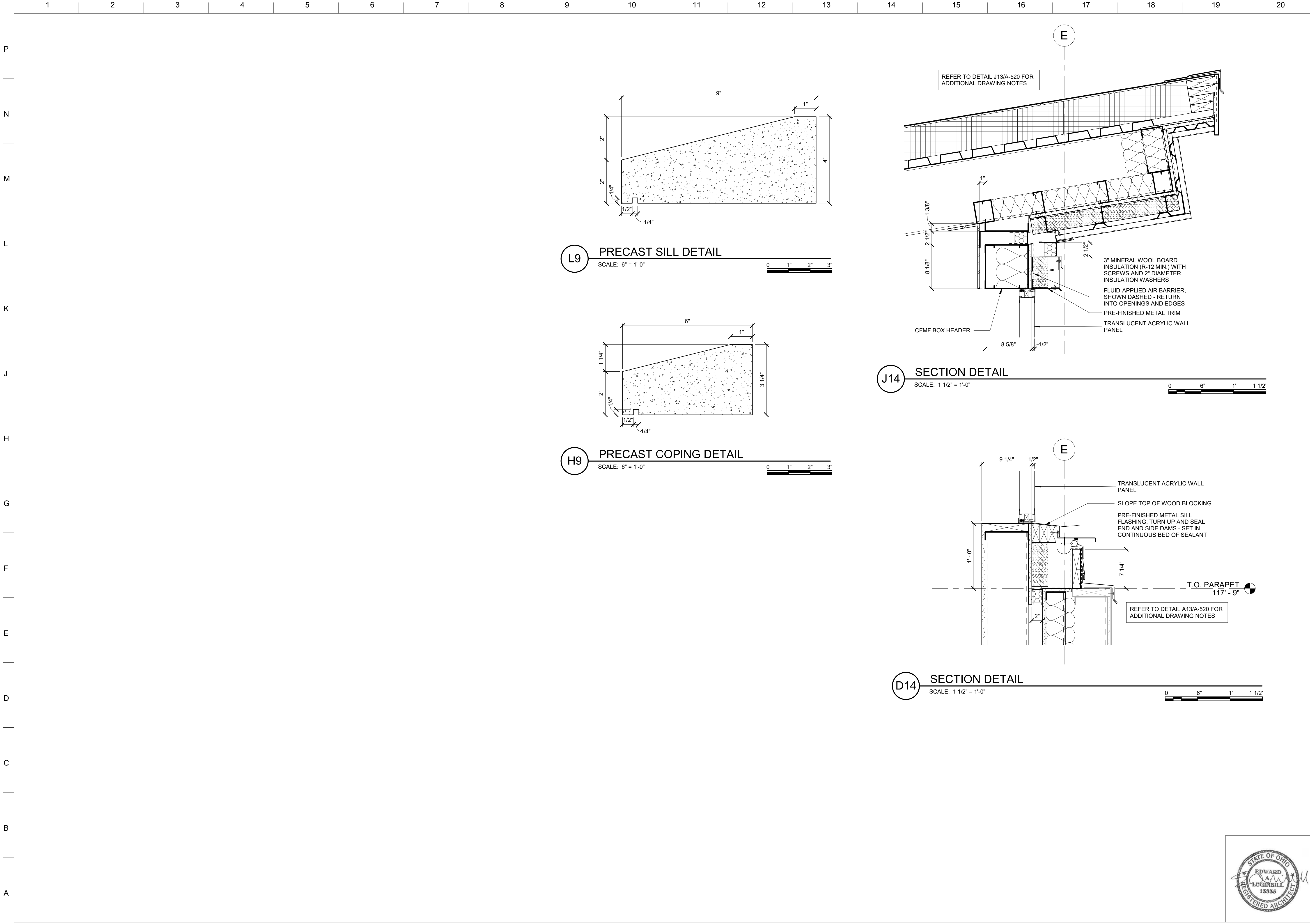
BOWST ADDITION AND RENOVATIONS TO B-321
NIAGARA FALLS AIR RESERVE STATION, NY
P2#-508014

SECTION DETAILS

SHEET ID

A-521





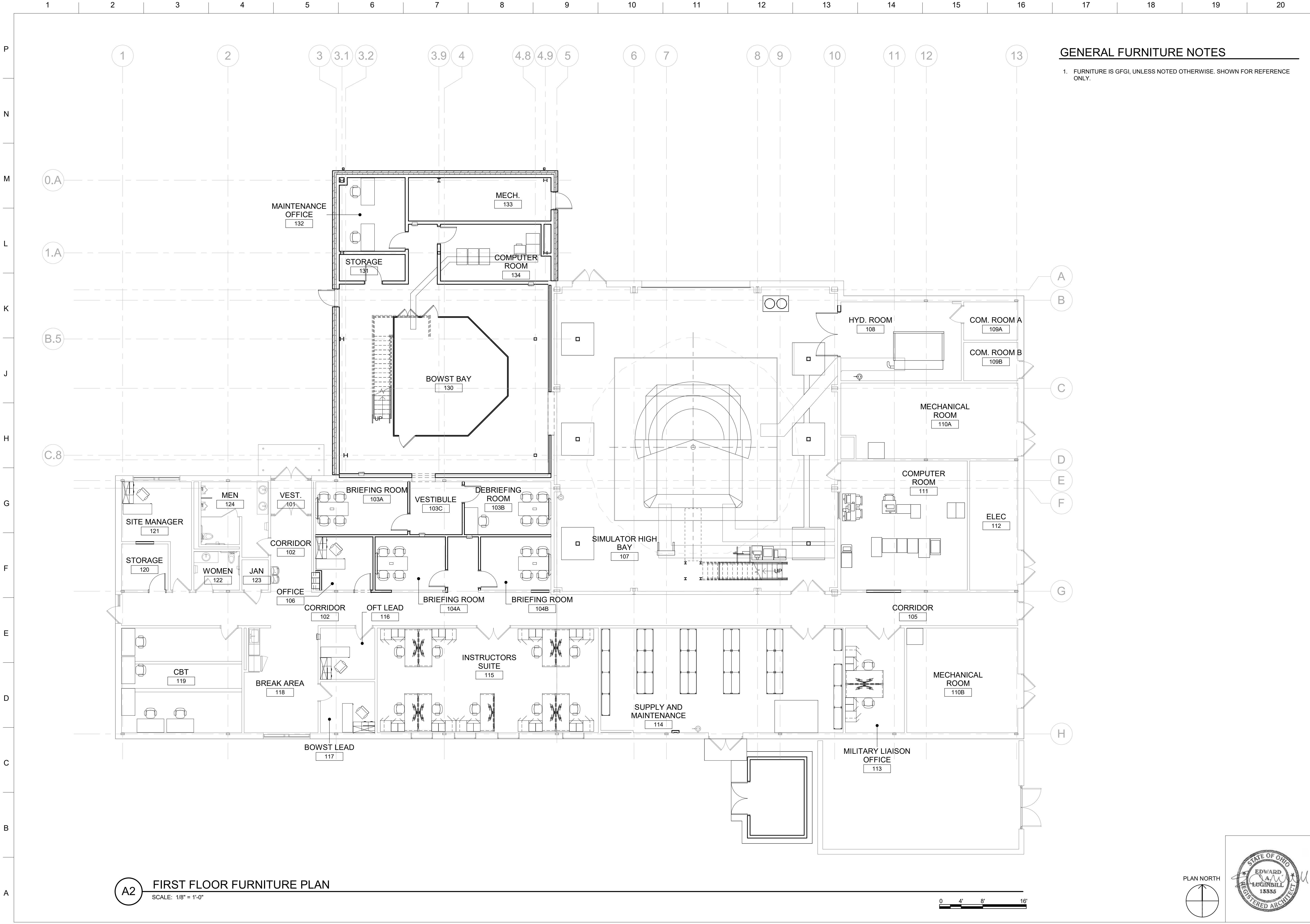
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US ARMY CORPS OF ENGINEERS LOUISVILLE DISTRICT 600 DR. MARTIN LUTHER KING, JR., PLACE LOUISVILLE KY 40202-2239	DESIGNED BY: A. LUGNIBILL	ISSUE DATE: JUNE 26, 2024	SOLICITATION NO.: A. MORRISON CONTRACT NO.:
	CHECKED BY: S. DUFFY		
	SUBMITTED BY:		
SIZE: ANSI D			

US ARMY CORPS OF ENGINEERS LOUISVILLE DISTRICT 600 DR. MARTIN LUTHER KING, JR. PLACE LOUISVILLE KY 40202-2239	
SECTION DETAILS	

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GENERAL FURNITURE NOTES

1. FURNITURE IS GFGI, UNLESS NOTED OTHERWISE. SHOWN FOR REFERENCE ONLY.



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LOUISVILLE DISTRICT
600 DR. MARTIN LUTHER KING, JR. PLACE
LOUISVILLE KY 40202-2239

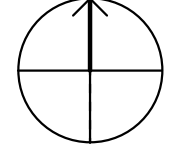
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BOWST ADDITION AND RENOVATIONS TO B-321
NIAGARA FALLS AIR RESERVE STATION, NY
P2#-508014

FIRST FLOOR FURNITURE PLAN

SHEET ID
I-101

PLAN NORTH

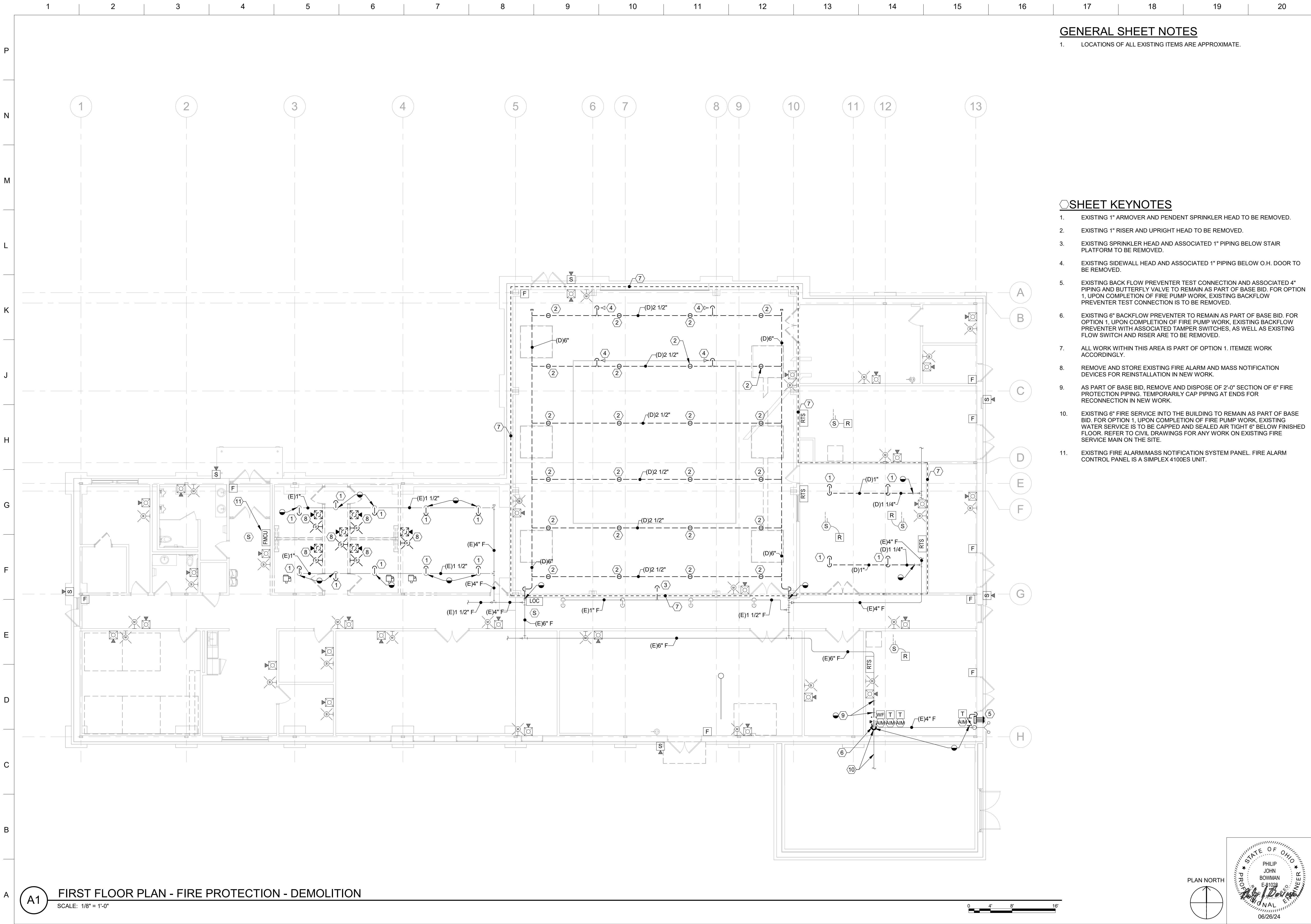


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GENERAL SHEET NOTES

1. LOCATIONS OF ALL EXISTING ITEMS ARE APPROXIMATE.

SHEET KEYNOTES

1. EXISTING 1" ARMOVER AND PENDENT SPRINKLER HEAD TO BE REMOVED.
2. EXISTING 1" RISER AND UPRIGHT HEAD TO BE REMOVED.
3. EXISTING SPRINKLER HEAD AND ASSOCIATED 1" PIPING BELOW STAIR PLATFORM TO BE REMOVED.
4. EXISTING SIDEWALL HEAD AND ASSOCIATED 1" PIPING BELOW O.H. DOOR TO BE REMOVED.
5. EXISTING BACK FLOW PREVENTER TEST CONNECTION AND ASSOCIATED 4" PIPING AND BUTTERFLY VALVE TO REMAIN AS PART OF BASE BID. FOR OPTION 1, UPON COMPLETION OF FIRE PUMP WORK, EXISTING BACKFLOW PREVENTER TEST CONNECTION IS TO BE REMOVED.
6. EXISTING 6" BACKFLOW PREVENTER TO REMAIN AS PART OF BASE BID. FOR OPTION 1, UPON COMPLETION OF FIRE PUMP WORK, EXISTING BACKFLOW PREVENTER WITH ASSOCIATED TAMPER SWITCHES, AS WELL AS EXISTING FLOW SWITCH AND RISER ARE TO BE REMOVED.
7. ALL WORK WITHIN THIS AREA IS PART OF OPTION 1. ITEMIZE WORK ACCORDINGLY.
8. REMOVE AND STORE EXISTING FIRE ALARM AND MASS NOTIFICATION DEVICES FOR REINSTALLATION IN NEW WORK.
9. AS PART OF BASE BID, REMOVE AND DISPOSE OF 2'-0" SECTION OF 6" FIRE PROTECTION PIPING. TEMPORARILY CAP PIPING AT ENDS FOR RECONNECTION IN NEW WORK.
10. EXISTING 6" FIRE SERVICE INTO THE BUILDING TO REMAIN AS PART OF BASE BID. FOR OPTION 1, UPON COMPLETION OF FIRE PUMP WORK, EXISTING WATER SERVICE IS TO BE CAPPED AND SEALED AIR TIGHT 6" BELOW FINISHED FLOOR. REFER TO CIVIL DRAWINGS FOR ANY WORK ON EXISTING FIRE SERVICE MAIN ON THE SITE.
11. EXISTING FIRE ALARM/MASS NOTIFICATION SYSTEM PANEL. FIRE ALARM CONTROL PANEL IS A SIMPLEX 4100ES UNIT.



DATE	DESCRIPTION	MARK

DESIGNED BY: P. BOWMAN	ISSUE DATE: JUNE 26, 2024
DRAWN BY: P. BOWMAN	SOLICITATION NO.:
CHECKED BY: B. JOHNSON	CONTRACT NO.:
SUBMITTED BY:	
SIZE: ANSI D	

US ARMY CORPS OF ENGINEERS
LOUISVILLE DISTRICT
600 DR. MARTIN LUTHER KING, JR. PLACE
LOUISVILLE KY 40202-2239

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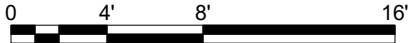
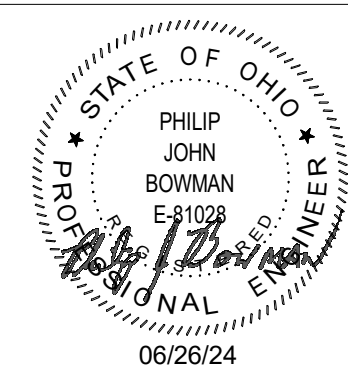
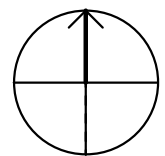
BOWST ADDITION AND RENOVATIONS TO B-321
NIAGARA FALLS AIR RESERVE STATION, NY
P2# 506014

FIRST FLOOR PLAN - FIRE PROTECTION -
DEMOLITION

SHEET ID

FD101

PLAN NORTH



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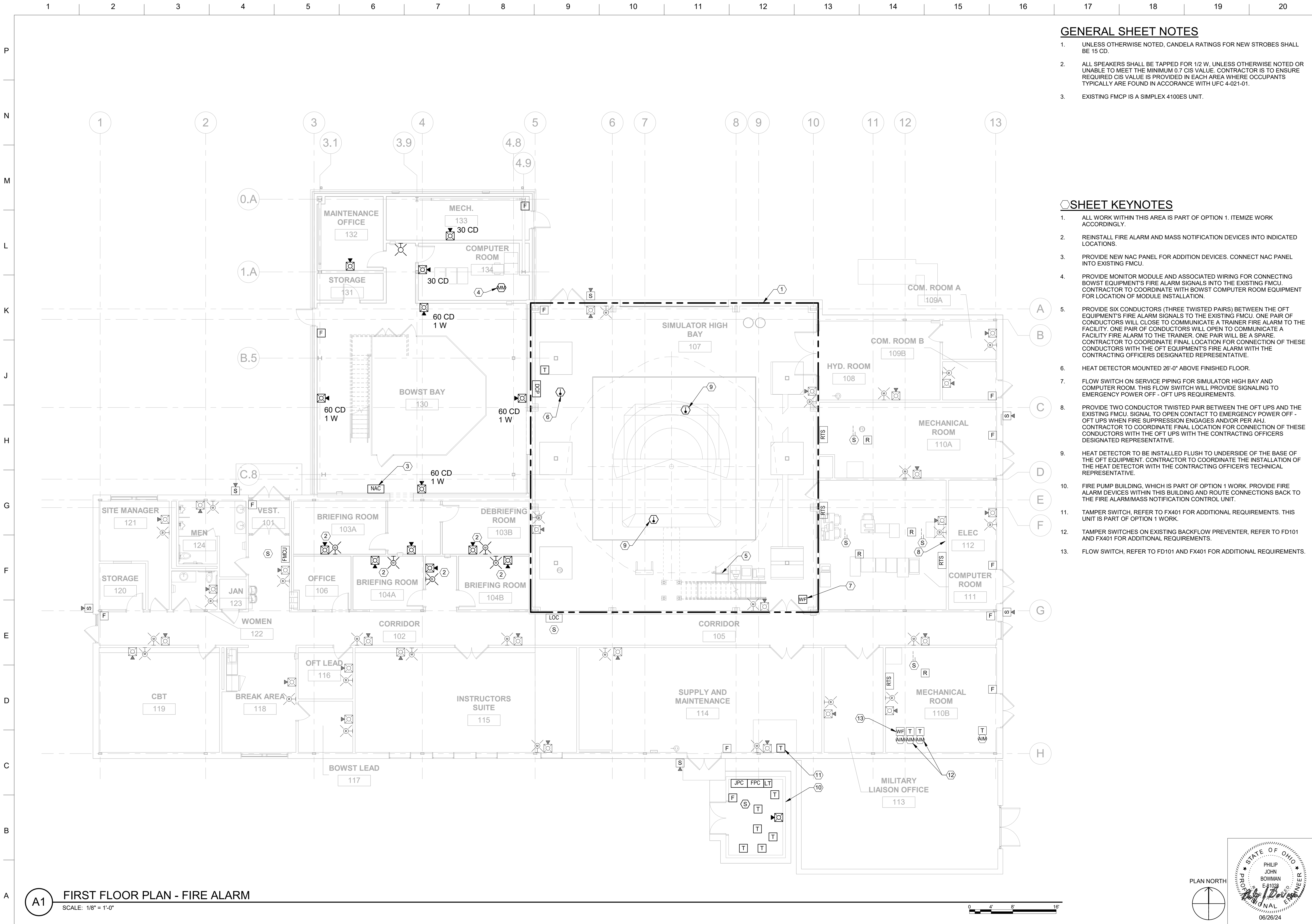
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GENERAL SHEET NOTES

- 1. UNLESS OTHERWISE NOTED, CANDELA RATINGS FOR NEW STROBES SHALL BE 15 CD.
- 2. ALL SPEAKERS SHALL BE TAPPED FOR 1/2 W. UNLESS OTHERWISE NOTED OR UNABLE TO MEET THE MINIMUM 0.7 CIS VALUE. CONTRACTOR IS TO ENSURE REQUIRED CIS VALUE IS PROVIDED IN EACH AREA WHERE OCCUPANTS TYPICALLY ARE FOUND IN ACCORDANCE WITH UFC 4-021-01.
- 3. EXISTING FMCP IS A SIMPLEX 4100ES UNIT.

SHEET KEYNOTES

- 1. ALL WORK WITHIN THIS AREA IS PART OF OPTION 1. ITEMIZE WORK ACCORDINGLY.
- 2. REINSTALL FIRE ALARM AND MASS NOTIFICATION DEVICES INTO INDICATED LOCATIONS.
- 3. PROVIDE NEW NAC PANEL FOR ADDITION DEVICES. CONNECT NAC PANEL INTO EXISTING FMCU.
- 4. PROVIDE MONITOR MODULE AND ASSOCIATED WIRING FOR CONNECTING BOWST EQUIPMENT'S FIRE ALARM SIGNALS INTO THE EXISTING FMCU. CONTRACTOR TO COORDINATE WITH BOWST COMPUTER ROOM EQUIPMENT FOR LOCATION OF MODULE INSTALLATION.
- 5. PROVIDE SIX CONDUCTORS (THREE TWISTED PAIRS) BETWEEN THE OFT EQUIPMENT'S FIRE ALARM SIGNALS TO THE EXISTING FMCU. ONE PAIR OF CONDUCTORS WILL CLOSE TO COMMUNICATE A TRAINER FIRE ALARM TO THE FACILITY. ONE PAIR OF CONDUCTORS WILL OPEN TO COMMUNICATE A FACILITY FIRE ALARM TO THE TRAINER. ONE PAIR WILL BE A SPARE. CONTRACTOR TO COORDINATE FINAL LOCATION FOR CONNECTION OF THESE CONDUCTORS WITH THE OFT EQUIPMENT'S FIRE ALARM WITH THE CONTRACTING OFFICERS DESIGNATED REPRESENTATIVE.
- 6. HEAT DETECTOR MOUNTED 26'-0" ABOVE FINISHED FLOOR.
- 7. FLOW SWITCH ON SERVICE PIPING FOR SIMULATOR HIGH BAY AND COMPUTER ROOM. THIS FLOW SWITCH WILL PROVIDE SIGNALING TO EMERGENCY POWER OFF - OFT UPS REQUIREMENTS.
- 8. PROVIDE TWO CONDUCTOR TWISTED PAIR BETWEEN THE OFT UPS AND THE EXISTING FMCU. SIGNAL TO OPEN CONTACT TO EMERGENCY POWER OFF - OFT UPS WHEN FIRE SUPPRESSION ENGAGES AND/OR PER AHJ. CONTRACTOR TO COORDINATE FINAL LOCATION FOR CONNECTION OF THESE CONDUCTORS WITH THE OFT UPS WITH THE CONTRACTING OFFICERS DESIGNATED REPRESENTATIVE.
- 9. HEAT DETECTOR TO BE INSTALLED FLUSH TO UNDERSIDE OF THE BASE OF THE OFT EQUIPMENT. CONTRACTOR TO COORDINATE THE INSTALLATION OF THE HEAT DETECTOR WITH THE CONTRACTING OFFICER'S TECHNICAL REPRESENTATIVE.
- 10. FIRE PUMP BUILDING, WHICH IS PART OF OPTION 1 WORK. PROVIDE FIRE ALARM DEVICES WITHIN THIS BUILDING AND ROUTE CONNECTIONS BACK TO THE FIRE ALARM/MASS NOTIFICATION CONTROL UNIT.
- 11. TAMPER SWITCH, REFER TO FX401 FOR ADDITIONAL REQUIREMENTS. THIS UNIT IS PART OF OPTION 1 WORK.
- 12. TAMPER SWITCHES ON EXISTING BACKFLOW PREVENTER, REFER TO FD101 AND FX401 FOR ADDITIONAL REQUIREMENTS.
- 13. FLOW SWITCH, REFER TO FD101 AND FX401 FOR ADDITIONAL REQUIREMENTS.



US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: P. BOWMAN	ISSUE DATE: JUNE 26, 2024
DRAWN BY: K. HALLER	SOLICITATION NO.:
CHECKED BY: B. JOHNSON	CONTRACT NO.:
SUBMITTED BY:	
SIZE: ANSI D	

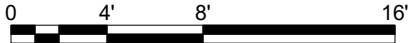
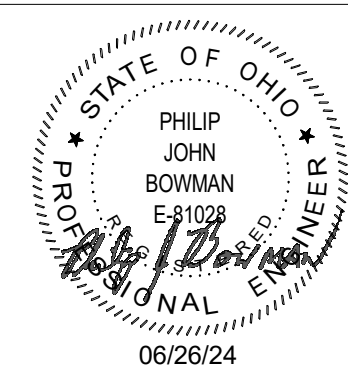
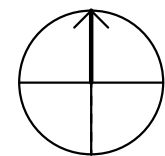
US ARMY CORPS OF ENGINEERS
LOUISVILLE DISTRICT
600 DR. MARTIN LUTHER KING, JR. PLACE
LOUISVILLE KY 40202-2239

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BOWST ADDITION AND RENOVATIONS TO B-321 NIAGARA FALLS AIR RESERVE STATION, NY P2# 508014	FIRST FLOOR PLAN - FIRE ALARM
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SHEET ID FA101

PLAN NORTH



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FIRE ALARM INPUT ¹

MANUAL PULL STATION	•	•	•	•		•		•		
AUTOMATIC SPRINKLER FLOW SWITCH	•	•	•	•		•			•	
AUTOMATIC SPRINKLER SYSTEM TAMPER SWITCH					•		•			
SMOKE DETECTOR	•	•	•	•		•				
HEAT DETECTORS (DELUGE SYSTEM, BELOW OFT)	•	•	•	•		•				•
HEAT DETECTOR (OVERHEAD)	•	•	•	•		•				
OTHER TROUBLE CONDITION					•		•			
FIRE PUMP RUN					•			•		
FIRE PUMP CONTROLLER OFF NORMAL					•			•		
FIRE PUMP/PUMP CONTROLLER TROUBLE					•			•		
FIRE PUMP PRIMARY POWER FAILURE					•			•		
PHASE REVERSAL ON FIRE PUMP					•			•		
LOSS OF PHASE ON FIRE PUMP					•			•		

MASS NOTIFICATION INPUT ¹

OPERATION 1

ACTIVATE VISUAL * ALERT STROBES

ACTIVATE NOTIFICATION DEVICE ON EXTERIOR OF BUILDING

ALARM AUDIBLE FIRE ALARM EVACUATION

ALARM CONDITION AT FACU

SUPERVISORY CONDITION AT FACU

TROUBLE CONDITION AT FACU

TRANSMIT TROUBLE SIGNAL TO FIRE DEPT. VIA FIRE ALARM LOOP

DEACTIVATE ACCESS CONTROL DOORS ²

EMERGENCY POWER OFF -OFT UPS

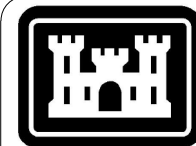
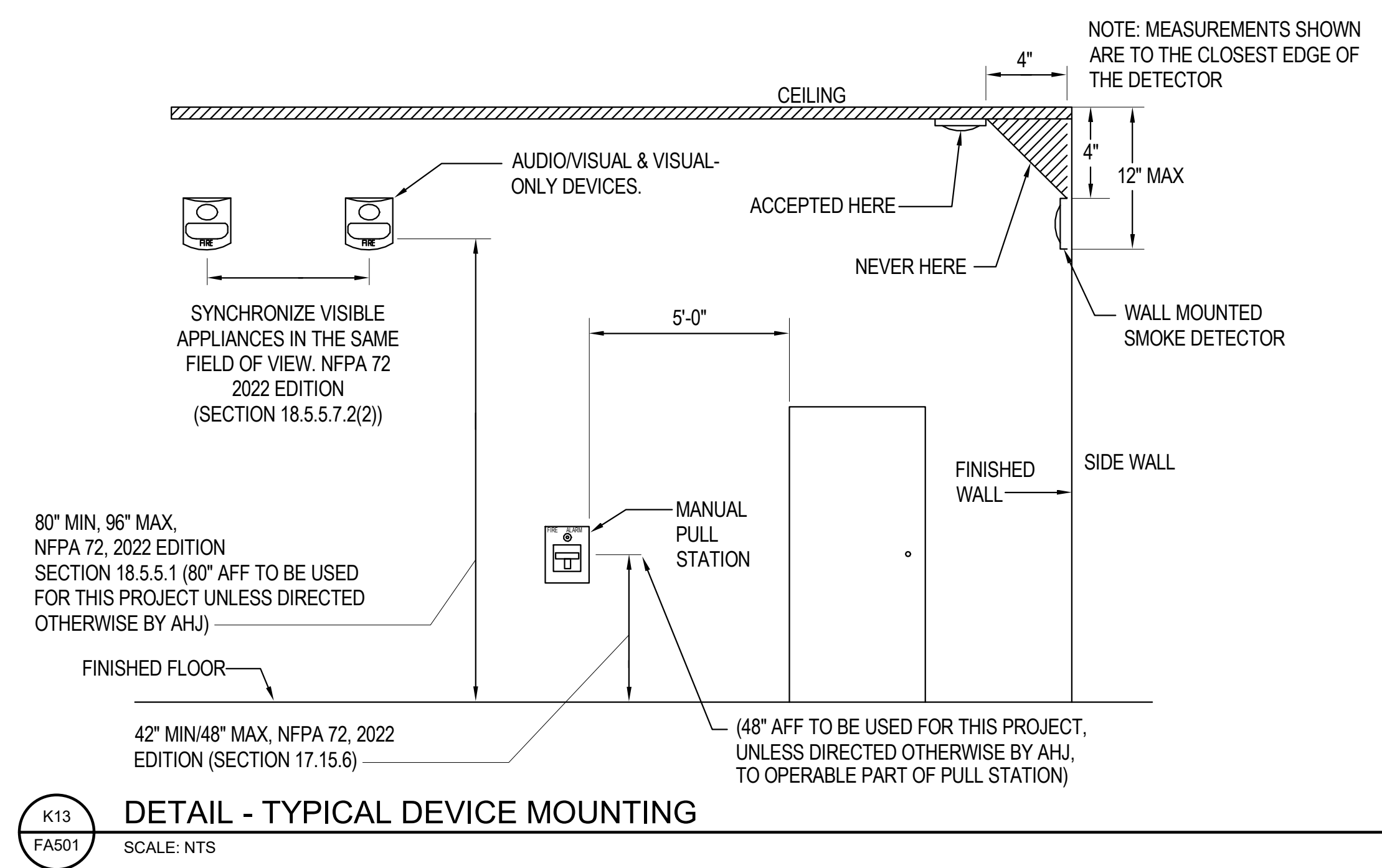
ACTIVATE DELUGE SYSTEM VIA DELUGE CONTROL PANEL

NOTES:

1. EXISTING PANEL HAS FIRE ALARM/MASS NOTIFICATION PROGRAMMING, THIS MATRIX PROVIDES PROGRAMMING FOR NEW INPUT REQUIREMENTS. CONTRACTOR TO CONFIRM THAT EXISTING PROGRAMMING AND DEVICES AND ALL NEW DEVICES AND PROGRAMMING ARE IN ACCORDANCE WITH UFC 3-600-01 AND 4-021-01.
2. FLOW SWITCH TO BE UTILIZED FOR OPERATION MUST BE ASSOCIATED WITH SIMULATOR HIGH BAY AND OFT COMPUTER ROOM SPRINKLER SYSTEM.

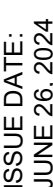
FIRE ALARM / MASS NOTIFICATION SYSTEM SEQUENCE OF OPERATIONS

SCALE: NTS



**US Army Corps
of Engineers ®**

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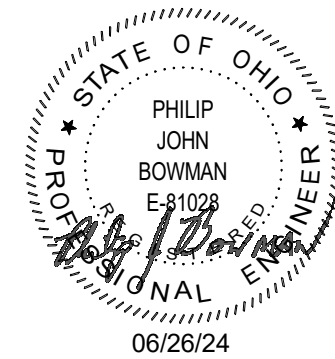
	US ARMY CORPS OF ENGINEERS LOUISVILLE DISTRICT 600 DR. MARTIN LUTHER KING, JR. PLACE LOUISVILLE KY 40202-2239		DESIGNED BY: P. BOWMAN	ISSUE DATE: JUNE 26, 2024
			DRAWN BY: K. HALLER CHECKED BY: J. FEE	SOLICITATION NO.: CONTRACT NO.:
		SUBMITTED BY:		
		SIZE: ANSI D		

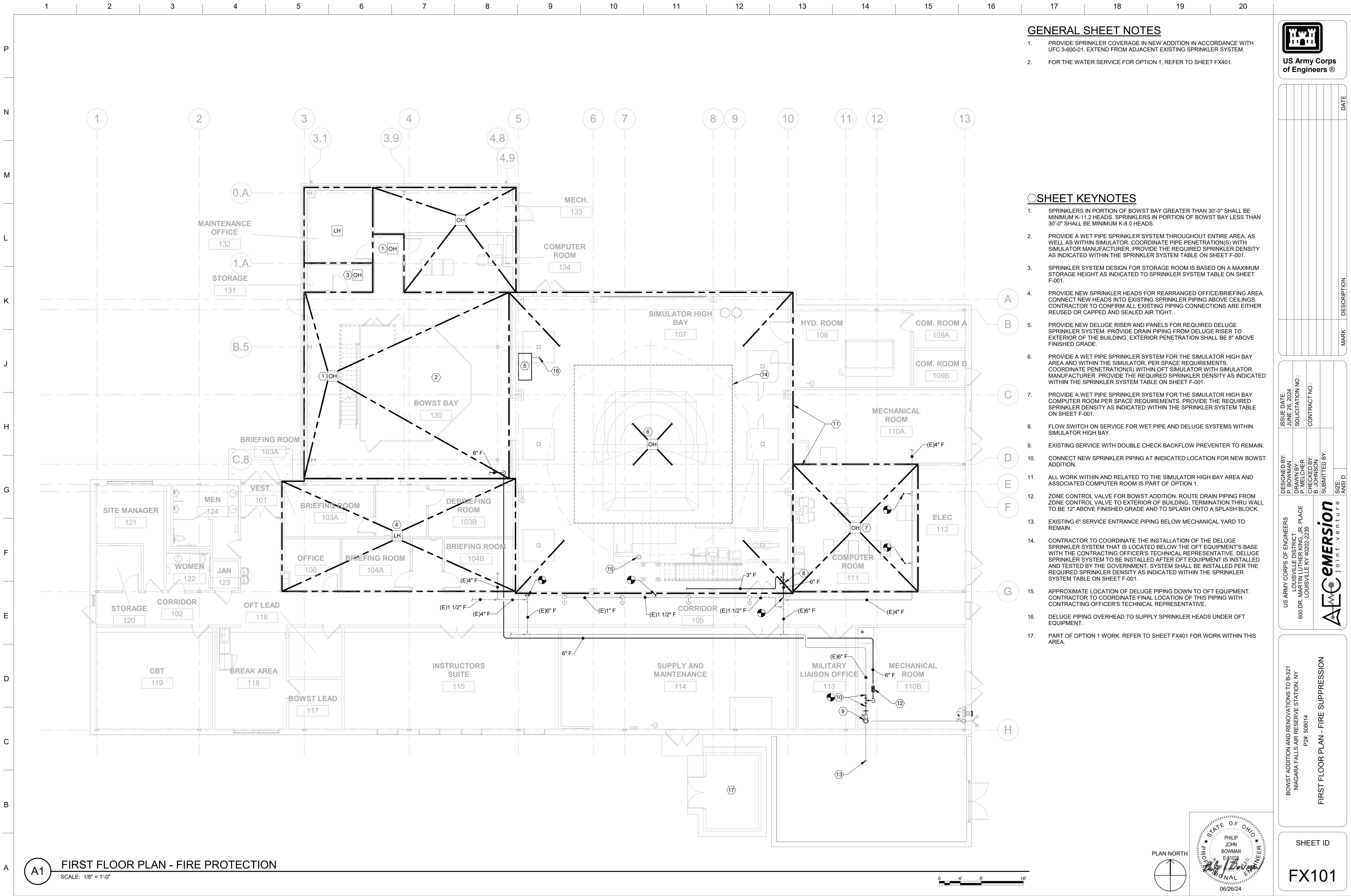
BOWST ADDITION AND RENOVATIONS TO B-321
 NIAGARA FALLS AIR RESERVE STATION, NY

DETAILS - FIRE ALARM

SHEET ID

FA501

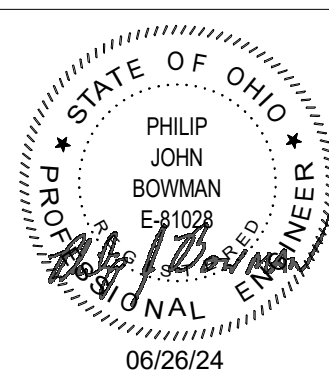
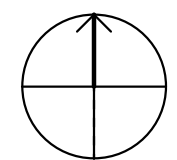






1. PROVIDE SPRINKLER COVERAGE IN NEW ADDITION IN ACCORDANCE WITH UFC 3-600-01. EXTEND FROM ADJACENT EXISTING SPRINKLER SYSTEM.
2. THIS SHEET REPRESENTS WORK FOR OPTION 1 ONLY.
3. PUMP BUILDING WILL AN UNSPRINKLERED STRUCTURE.

1. 8" FIRE SERVICE FROM SITE, REFER TO CIVIL DRAWINGS FOR CONTINUATION.
2. 8" DOUBLE CHECK BACKFLOW PREVENTER, REFER TO DETAIL C15 ON SHEET FX501 FOR ADDITIONAL INFORMATION.
3. BACKFLOW PREVENTER FORWARD FLOW TEST CONNECTION. TEST HEADER TO BE PROVIDED WITH FOUR 5" STORZ CONNECTIONS TO SUPPORT FULL DEMAND OF SPRINKLER SYSTEM.
4. FIRE PUMP TEST HEADER. TEST HEADER TO BE PROVIDED WITH FOUR 5" STORZ CONNECTIONS TO SUPPORT FULL DEMAND OF FIRE PUMP.
5. 6" PIPING DOWN THROUGH FLOOR OF FIRE PUMP BUILDING AND TO MAIN BUILDING. PIPING TO BE ROUTED A MINIMUM OF 5'-0" BELOW FINISHED GRADE BETWEEN BUILDINGS.
6. 6" PIPING UP THROUGH FLOOR WITHIN MAIN BUILDING. PROVIDE SUPERVISED SHUTOFF VALVE 2'-0" ABOVE FINISHED FLOOR AT ENTRY INTO MAIN BUILDING. ROUTE PIPING UP WALL AND BELOW EXISTING STRUCTURE TO MECHANICAL ROOM 110B.
7. PROVIDE NEW CONNECTION TO THE EXISTING PIPING FROM THE FIRE DEPARTMENT CONNECTION.
8. CONNECT INTO THE EXISTING SPRINKLER SYSTEMS AT THE INDICATED LOCATION. MAKE CONNECTION ONLY UPON COMPLETION OF FIRE PUMP INSTALLATION AND TESTING OF SYSTEM IS ACCEPTED.
9. FIRE PUMP BUILDING, WHICH SHALL BE A 2-HOUR FIRE-RATED CONSTRUCTION PRE-MANUFACTURED BUILDING PROVIDED BY THE FIRE PUMP MANUFACTURER. THE BUILDING IS TO BE PROVIDED WITH LIGHTING, AN EXHAUST FAN WITH GRAVITY LOUVER, A VENTILATION LOUVER WITH MOTORIZED DAMPER, AN UNIT HEATER, FLOOR DRAIN CONNECTED INTO SANITARY PIPING BELOW GRADE, RECEPTACLES, AND ALL EQUIPMENT INDICATED ON THIS PLAN.
10. FIRE PUMP CONTROLLER.
11. JOCKEY PUMP CONTROLLER.
12. MINI POWER ZONE.
13. ELECTRIC UNIT HEATER DISCONNECT SWITCH.
14. CLEARANCE AREA IN FRONT OF ELECTRICAL EQUIPMENT, AS REQUIRED PER THE NATIONAL ELECTRIC CODE.
15. REFER TO DETAIL F5 ON SHEET FX501 FOR ARRANGEMENT OF PIPING FOR THE FIRE PUMP BUILDING.
16. FLOW SWITCH FOR BUILDING SPRINKLER SYSTEM.

PLAN NORTH

DESIGNED BY:	ISSUE DATE:
P. BOWMAN	JUNE 26, 2024
DRAWN BY:	SOLICITATION NO.:
P. MELCHER	
CHECKED BY:	CONTRACT NO.:
B. JOHNSON	
SUBMITTED BY:	
SIZE:	
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US ARMY CORPS OF ENGINEERS
LOUISVILLE DISTRICT
600 DR. MARTIN LUTHER KING, JR. PLACE
LOUISVILLE KY 40202-2239

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BOWST ADDITION AND RENOVATIONS TO B-321
 NIAGARA FALLS AIR RESERVE STATION, NY
 P2#: 508014

SHEET ID

FX401

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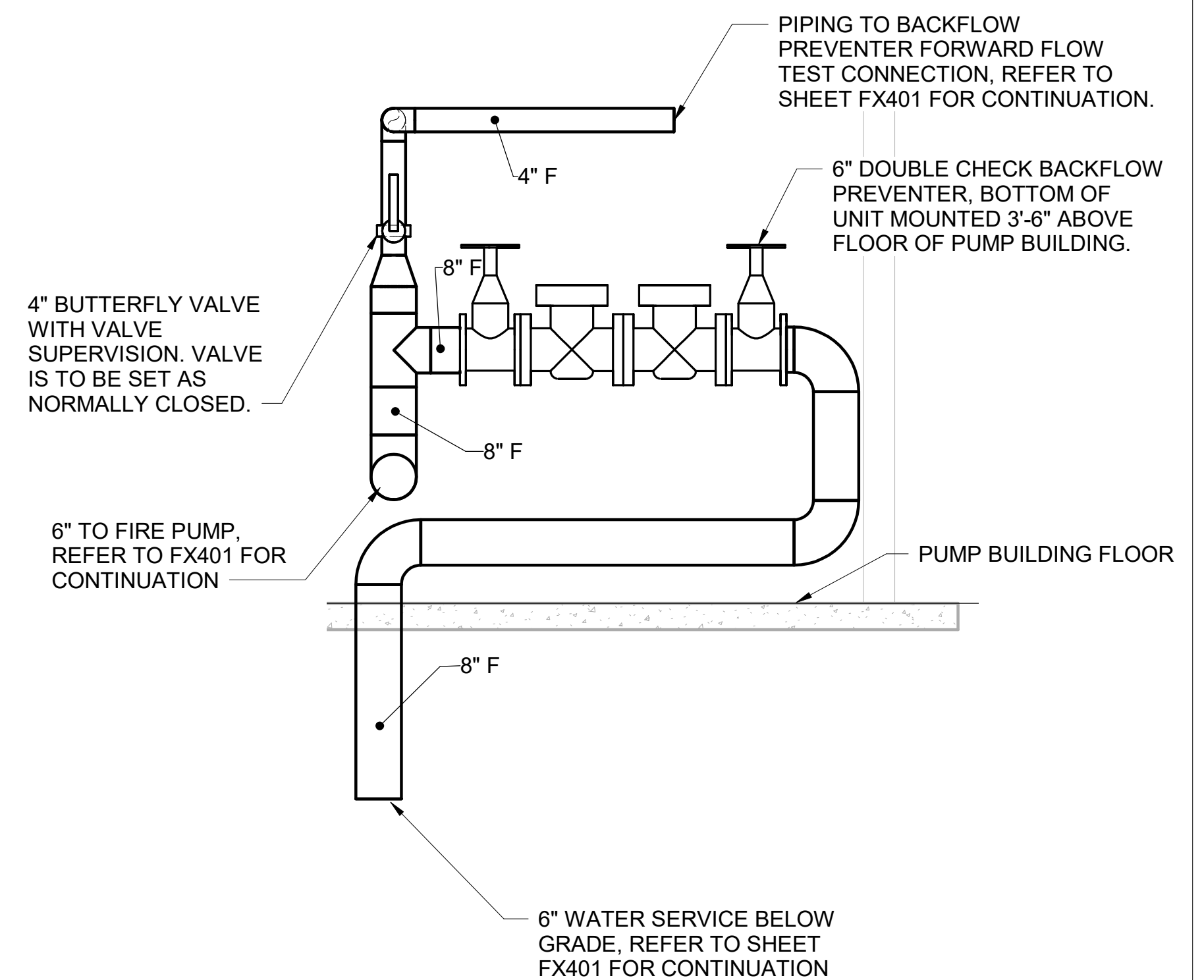
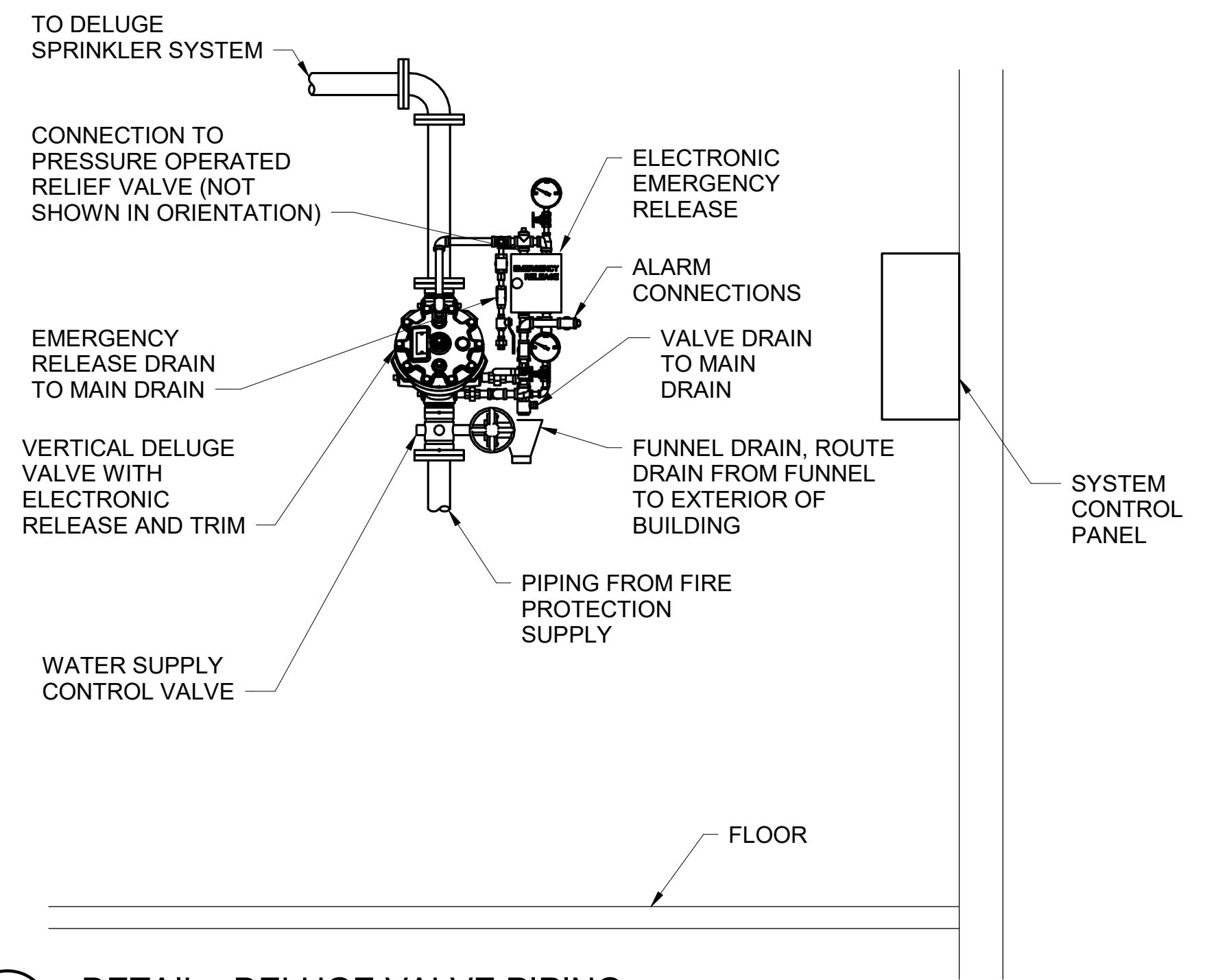
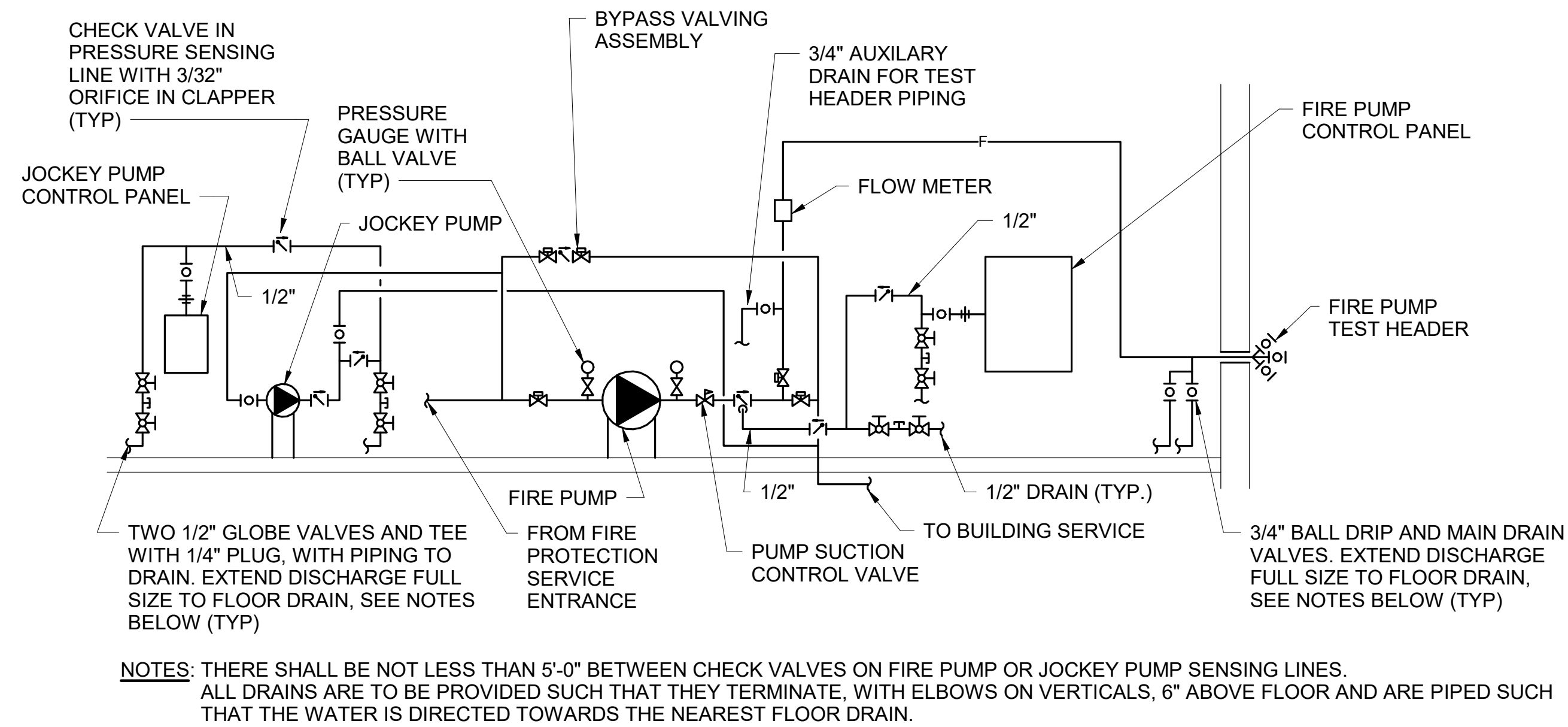
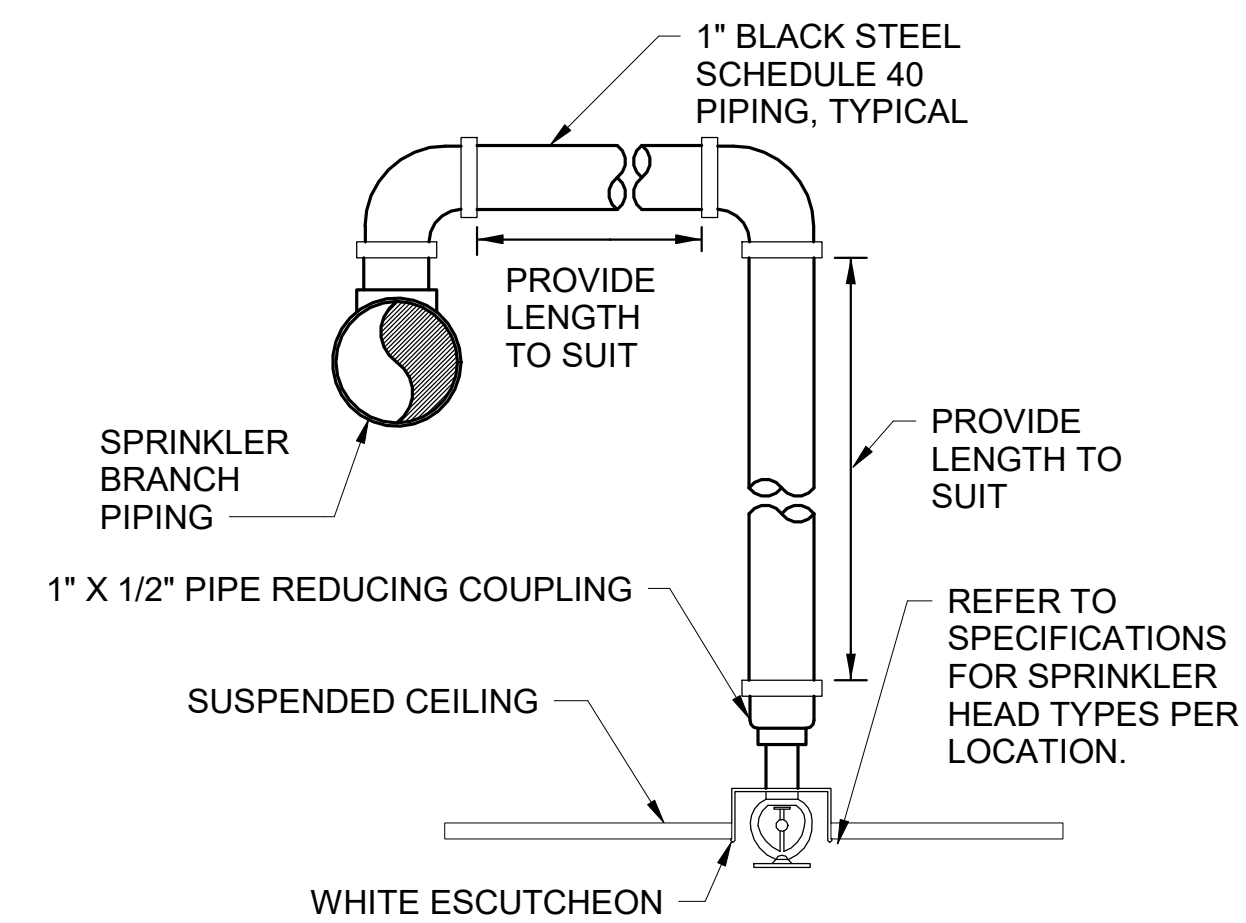
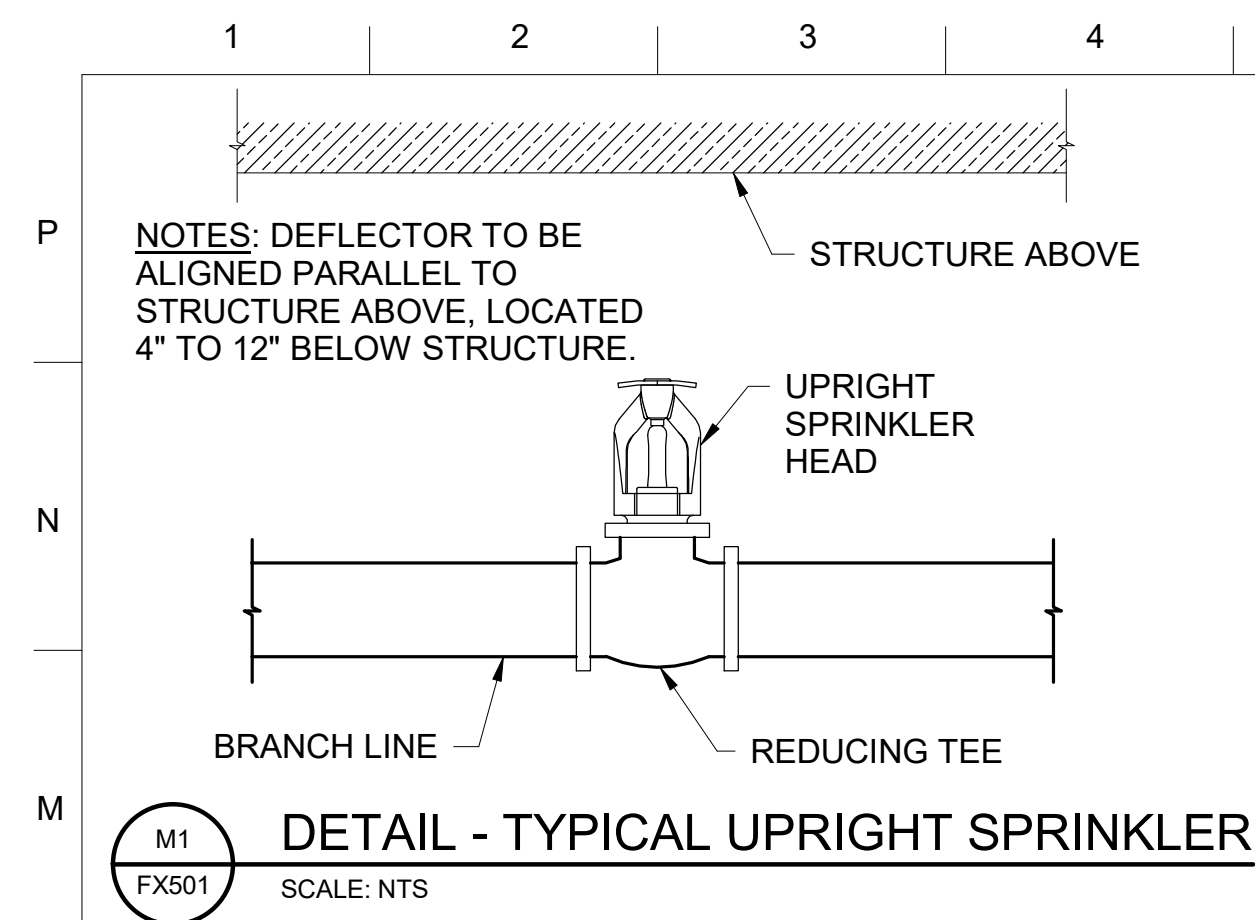
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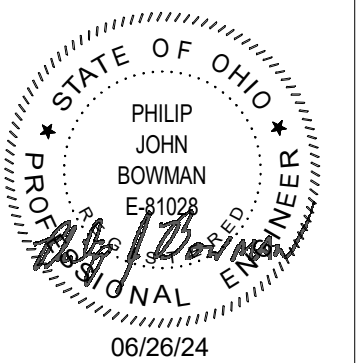
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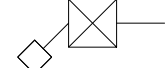
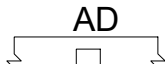
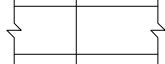
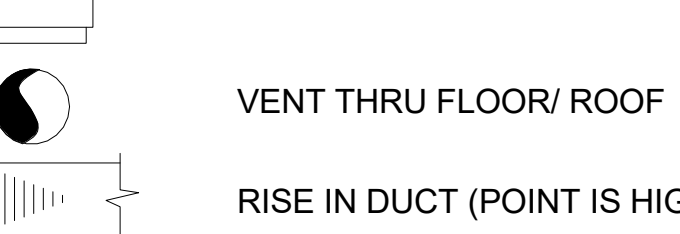
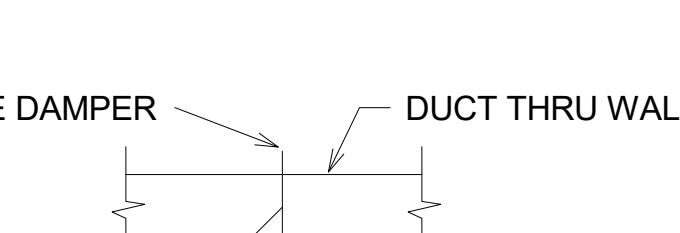
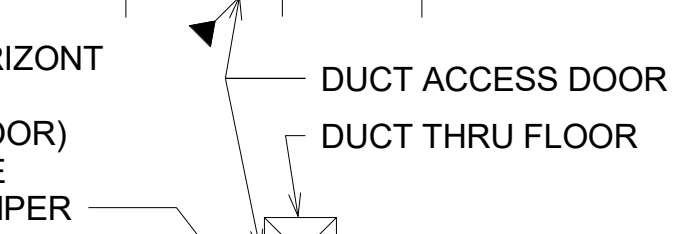
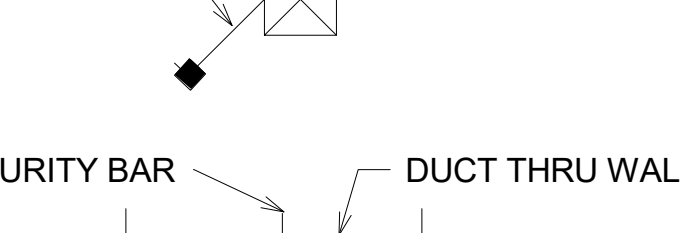
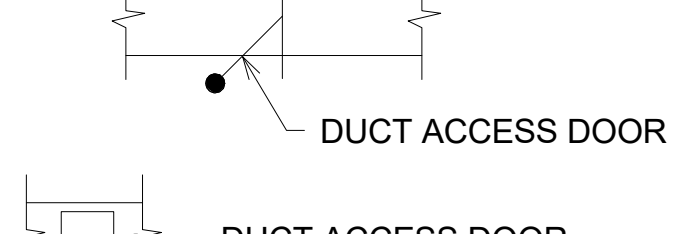
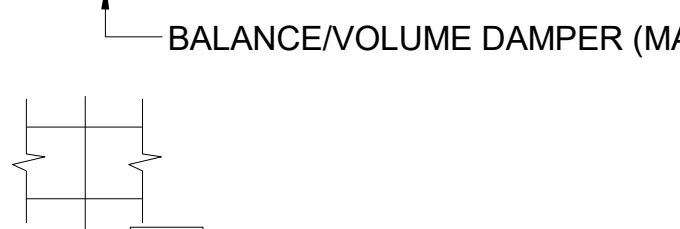
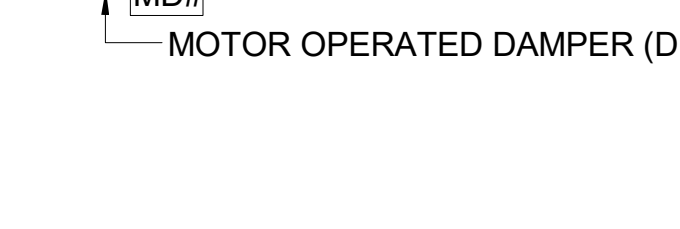
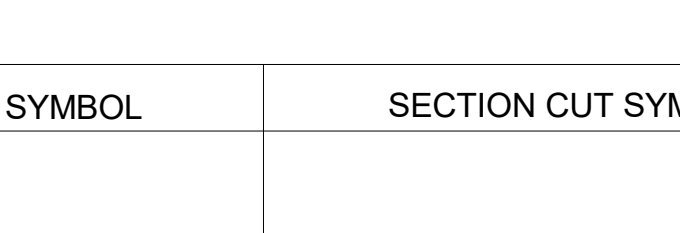
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P. BOWMAN		P. WELCHER		CONTRACT NO.:	
CHECKED BY:		B. JOHNSON			

DETAILS - FIRE SUPPRESSION

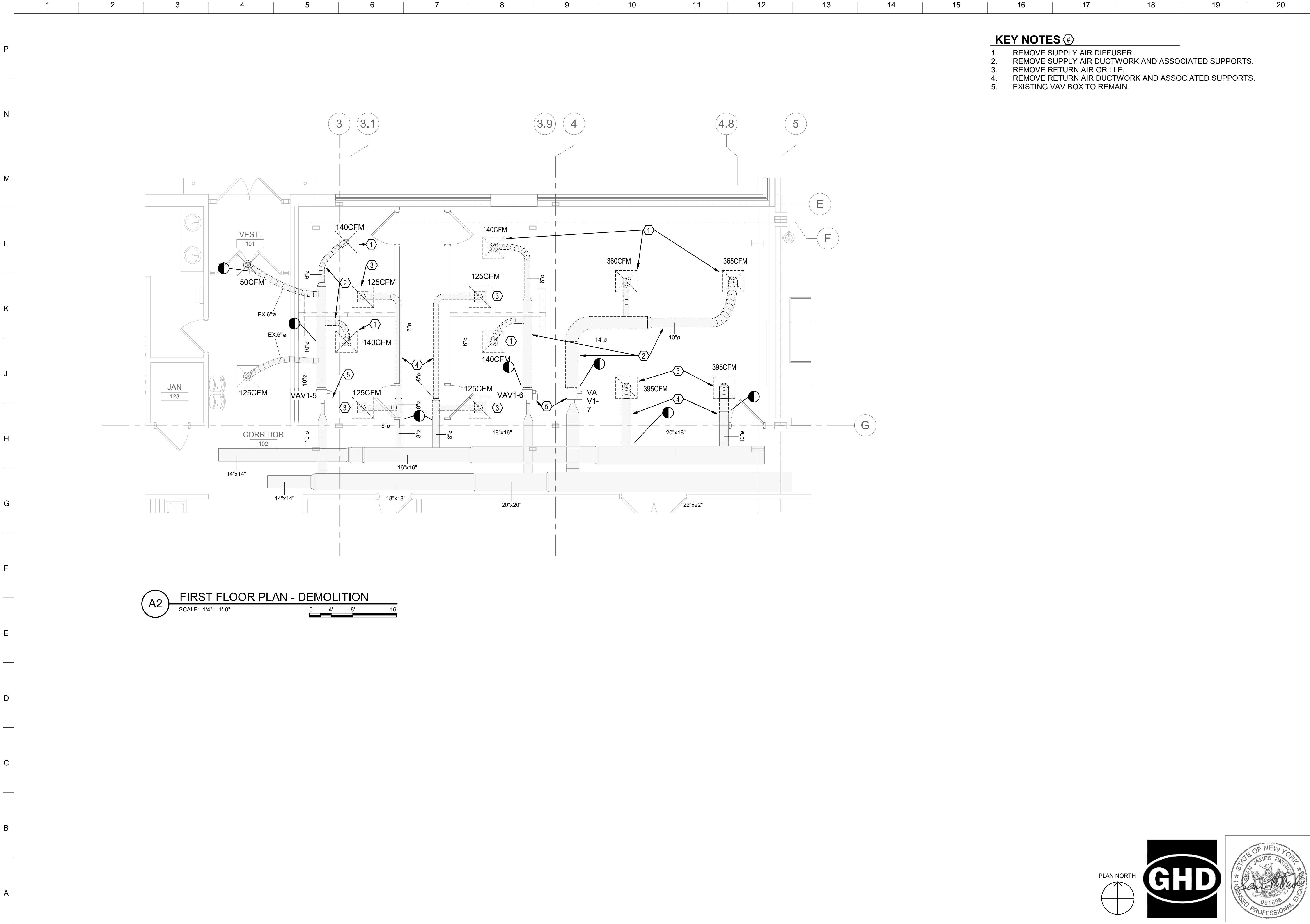
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FX501



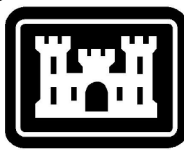
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P	MECHANICAL LEGEND										MECHANICAL PIPING LEGEND					MECHANICAL ABBREVIATIONS					GENERAL MECHANICAL NOTES								
N		THERMOSTAT OR TEMPERATURE SENSOR		VERTICAL (WALL) FIRE DAMPER							CWR	CONDENSER WATER RETURN PIPING	DWGS DRAWINGS					1. THESE GENERAL MECHANICAL NOTES APPLY TO ALL MECHANICAL DRAWINGS IN THIS SET. 2. SHEET SPECIFIC NOTES ARE INDICATED ON EACH SHEET UNDER GENERAL SHEET NOTES. 3. PROVIDE ALL MATERIALS AND EQUIPMENT AND PERFORM ALL LABOR REQUIRED TO INSTALL COMPLETE AND OPERABLE MECHANICAL SYSTEMS AS INDICATED ON THE DRAWINGS, AS SPECIFIED, AND AS REQUIRED BY CODE. 4. INSTALL ALL MECHANICAL EQUIPMENT AND APPURTENANCES IN ACCORDANCE WITH MANUFACTURER'S RECOMMENDATIONS, CONTRACT DOCUMENTS, APPLICABLE CODES, AND REGULATIONS. 5. EXISTING DUCT AND PIPING SIZES INDICATED ARE APPROXIMATE. FIELD VERIFY ALL EXISTING CONDITIONS. PATCH ALL HOLES FROM REMOVAL OF PIPING, DUCTWORK, OR EQUIPMENT TO MATCH ADJACENT CONSTRUCTION. 6. CONCEAL ALL DUCTWORK AND PIPING ABOVE SUSPENDED CEILINGS AND IN WALLS, UNLESS OTHERWISE NOTED. PROVIDE VIBRATION ISOLATORS FOR ALL MECHANICAL EQUIPMENT AND PIPING TO PREVENT TRANSMISSION OF VIBRATION TO BUILDING STRUCTURE. 7. PROVIDE ACCESS PANELS IN WALLS AND CEILINGS, WHERE REQUIRED, TO SERVE FIRE DAMPERS, VALVES, SMOKE DETECTORS, AND OTHER CONCEALED MECHANICAL EQUIPMENT THAT REQUIRES ROUTINE MAINTENANCE. ACCESS PANELS SHALL BE SIZED AS REQUIRED TO ALLOW WORKING CLEARANCE FOR DUCT ACCESS DOORS. 8. ALL 90 DEGREE RECTANGULAR MITERED DUCT ELBOWS IN SUPPLY AND RETURN DUCTWORK SHALL HAVE TURNING VANES. PROVIDE AN ACCESS DOOR UPSTREAM OF ELBOW FOR TURNING VANE INSPECTION AND CLEANING. 9. PROVIDE PIPE SLEEVES FOR ALL PIPING PENETRATIONS THRU WALLS AND FLOORS. EACH PIPE SHALL BE SEPARATELY SLEEVED. 10. PROVIDE CONTINUOUS SEALANT JOINT AROUND PIPE SLEEVE, BOTH SIDES OF WALL, WATERTIGHT. ALL OPENINGS IN FIRE WALLS DUE TO DUCTWORK, PIPING, CONDUIT, ETC., SHALL BE FIRE STOPPED. 11. PROVIDE P-TRAPS IN CONDENSATE DRAIN PIPING AT EACH AC UNIT UNLESS THE RTU IS EQUIPPED WITH AN INTERNAL TRAP. 12. CONDENSATE DRAIN PIPING CLEANOUTS SHALL BE INSTALLED AT EACH CHANGE IN DIRECTION GREATER THAN 45 DEGREES. WHERE MORE THAN ONE CHANGE OF DIRECTION OCCURS IN A RUN OF PIPING, ONLY ONE CLEANOUT SHALL BE REQUIRED FOR EACH 40 FEET OF DEVELOPED LENGTH. COMBINATION WYE AND 1/8 BENDS SHALL BE USED IN LIEU OF TEES, UNLESS OTHERWISE INDICATED ALL CONDENSATE LINES SHALL SLOPE DOWNWARD AND BE GRAVITY FED. 13. MOUNT THERMOSTATS 48 INCHES ABOVE THE FINISHED FLOOR AND RECESSED, FLUSH WITH WALL SURFACE, WHERE POSSIBLE UNLESS OTHERWISE NOTED. SEE ELECTRICAL DRAWINGS FOR COORDINATION WITH LIGHT SWITCHES. 14. TESTING AND BALANCING (TAB): TAB ALL EQUIPMENT AND DUCTWORK; 2-SEASONS.											
M		HUMIDITY SENSOR		HORIZONTAL (CEILING) FIRE DAMPER							- - -CWS-	CONDENSER WATER SUPPLY PIPING	DX DIRECT EXPANSION																
L		AIR TERMINAL TAG WITH AIR FLOW (CFM)		ACCESS DOOR (AD), ACCESS PANEL (AP)							- - -CHWR-	CHILLED WATER RETURN PIPING	EA EXHAUST AIR																
K		EMERGENCY HVAC SYSTEM SHUTOFF SWITCH		VOLUME DAMPER							- - -CHWS-	CHILLED WATER SUPPLY PIPING	EC ELECTRICALLY COMMUTATED (MOTOR) / ELECTRIC HEAT COIL																
J		SPACE CO2 SENSOR / DDC POINT TAG		EXHAUST AIR (NEG.) DUCT SECTION							CND	CONDENSATE DRAIN PIPING	EER ENERGY EFFICIENCY RATIO																
H		RECTANGULAR DUCT DIMENSIONS (INCHES)		CEILING CASSETTE							- - -DCW-	DOMESTIC COLD WATER PIPING	EF EXHAUST FAN																
G		ROUND DUCT DIMENSION		CEILING CASSETTE DUCTED							DHW	DOMESTIC HOT WATER PIPING	EFF EFFECTIVENESS / EFFICIENCY																
F		FLEXIBLE ROUND DUCT (5'-0" MAX LENGTH)		DUCTED FAN COIL UNIT							DTR	DUAL TEMPERATURE RETURN PIPING	EH ELECTRIC HEAT																
E		DOWNWARD DIRECTION OF SLOPED PIPING		VENT THRU FLOOR/ ROOF							HWR	HOT WATER RETURN PIPING	ERV ENERGY RECOVERY VENTILATOR																
D		SUPPLY DUCT, SUPPLY DUCT DOWN/UP		RISE IN DUCT (POINT IS HIGH)							- - -HWS-	HOT WATER SUPPLY PIPING	ESP EXTERNAL STATIC PRESSURE																
C		RETURN DUCT, RETURN DUCT DOWN/UP		FIRE DAMPER							LPC	LOW PRESSURE CONDENSATE PIPING	ET EXPANSION TANK																
B		2-WAY VALVE		HORIZONTAL (FLOOR) FIRE DAMPER							- - -RL-	REFRIGERANT LIQUID PIPING	ETR EXISTING TO REMAIN																
A		3-WAY VALVE		SECURITY BAR							RS	REFRIGERANT SUCTION PIPING	EWT ENTERING WATER TEMPERATURE																
		AUTOMATIC BALANCING VALVE		DUCT THRU WALL							VAC	VACUUM (AIR)	E EXISTING																
		BALL VALVE		DUCT THRU FLOOR									F FAHRENHEIT																
		BUTTERFLY VALVE		DUCT ACCESS DOOR									FCU FAN COIL UNIT																
		CHECK VALVE		DUCT THRU WALL									FD FIRE DAMPER																
		DRAIN VALVE		DUCT ACCESS DOOR									FT FEET																
		GATE VALVE		BALANCE/VOLUME DAMPER (MANUAL)									FTWG FEET WATER GAUGE																
		GLOBE VALVE		MOTOR OPERATED DAMPER (DDC)									GP GLYCOL PUMP																
		MANUAL FLOW CONTROL BALANCING VALVE											GPM GALLONS PER MINUTE																
		PRESSURE REDUCING VALVE											GRH GRAVITY ROOF HOOD																
		RELIEF VALVE											H HUMIDIFIER (ELECTRIC)																
		SOLENOID VALVE											HC HEAT COIL (DX)																
		SQAURE HEAD BALANCING VALVE											HGRC HOT GAS REHEAT COIL (DX)																
		TEMPERATURE PRESSURE RELIEF VALVE											HGRH HOT GAS REHEAT																
		BACKFLOW PREVENTER											HHWS HOT HEATING WATER SYSTEM																
		AUTOMATIC AIR VENT											HP HEAT PUMP, HORSEPOWER																
		ECCENTRIC REDUCER											HP HEAT PUMP, HORSEPOWER																
		FLANGE											HPU HYDRAULIC POWER UNIT																
		MANUAL AIR VENT											HT HEIGHT																
		PIPE CAP											HX HEAT EXCHANGER																
		PIPE DOWN											IC IN CEILING																
		SUCTION REDUCER											IDU INDOOR UNIT (VRF)																
		PIPE UP											IN. INCH																
		PRESSURE GAUGE WITH SHUTOFF COCK											IWG INCHES WATER GAUGE																
		STEAM TRAP											KW KILOWATT																
		STRAINER WITH BLOWDOWN VALVE											LBS POUNDS																
		THERMOMETER											LPS LOW PRESSURE STEAM																
		UNION											LV LOUVER																
		VACUUM BREAKER											LWT LEAVING WATER TEMPERATURE																
		WYE STRAINER											MAU MAKE-UP AIR UNIT																
		DIFFERENTIAL PRESSURE SENSOR (DUCTWORK)											MAX MAXIMUM																
		NEW WORK NOTE TAG											MBH 1,000 BTUH																

																				</	



KEY NOTES

- 1. REMOVE SUPPLY AIR DIFFUSER.
- 2. REMOVE SUPPLY AIR DUCTWORK AND ASSOCIATED SUPPORTS.
- 3. REMOVE RETURN AIR GRILLE.
- 4. REMOVE RETURN AIR DUCTWORK AND ASSOCIATED SUPPORTS.
- 5. EXISTING VAV BOX TO REMAIN.



US Army Corps
of Engineers

MARK	DESCRIPTION	DATE

DESIGNED BY: S. PATRICK	ISSUE DATE: JUNE 26, 2024
DRAWN BY: H. NAYBI	SOLICITATION NO.:
CHECKED BY: D. DESIMONE	CONTRACT NO.:
SUBMITTED BY:	
SIZE: ANSI D	

US ARMY CORPS OF ENGINEERS
LOUISVILLE DISTRICT
600 DR. MARTIN LUTHER KING, JR. PLACE
LOUISVILLE KY 40202-2239

AECemersion
joint venture

BOWST ADDITION AND RENOVATIONS TO B-321
NIAGARA FALLS AIR RESERVE STATION, NY
P2#- 508014

FIRST FLOOR PLAN DEMOLITION PLAN
MECHANICAL HVAC

SHEET ID

M-101





**US Army Corps
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KEYNOTES:
1.) PROVIDE RADON EXHAUST SYSTEM. EXTEND 4" SOLID WALL PVC BELOW SLAB. PROVIDE REF-3 EXHAUST FAN IN VERTICAL PIPE. EXTEND 4" PVC PIPE DISCHARGE OUT THROUGH ROOF.

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Autodesk Docs://NFARS BOWST Addition1012012 01 MECH-NFARS BOWST ADD R22.IM

File Path:

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Plot Date: 6/20/2024 10:29:45 AM

Plot Date:

US ARMY CORPS OF ENGINEERS
LOUISVILLE DISTRICT
6000 DR. MARTIN LUTHER KING, JR. PLACE
LOUISVILLE KY 40202-2239



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joint venture

BOWST ADDITION AND RENOVATIONS TO B-321
NIAGARA FALLS AIR RESERVE STATION, NY

P2#: 508014

FIRST FLOOR PLAN NEW WORK PLAN
MECHANICAL HVAC

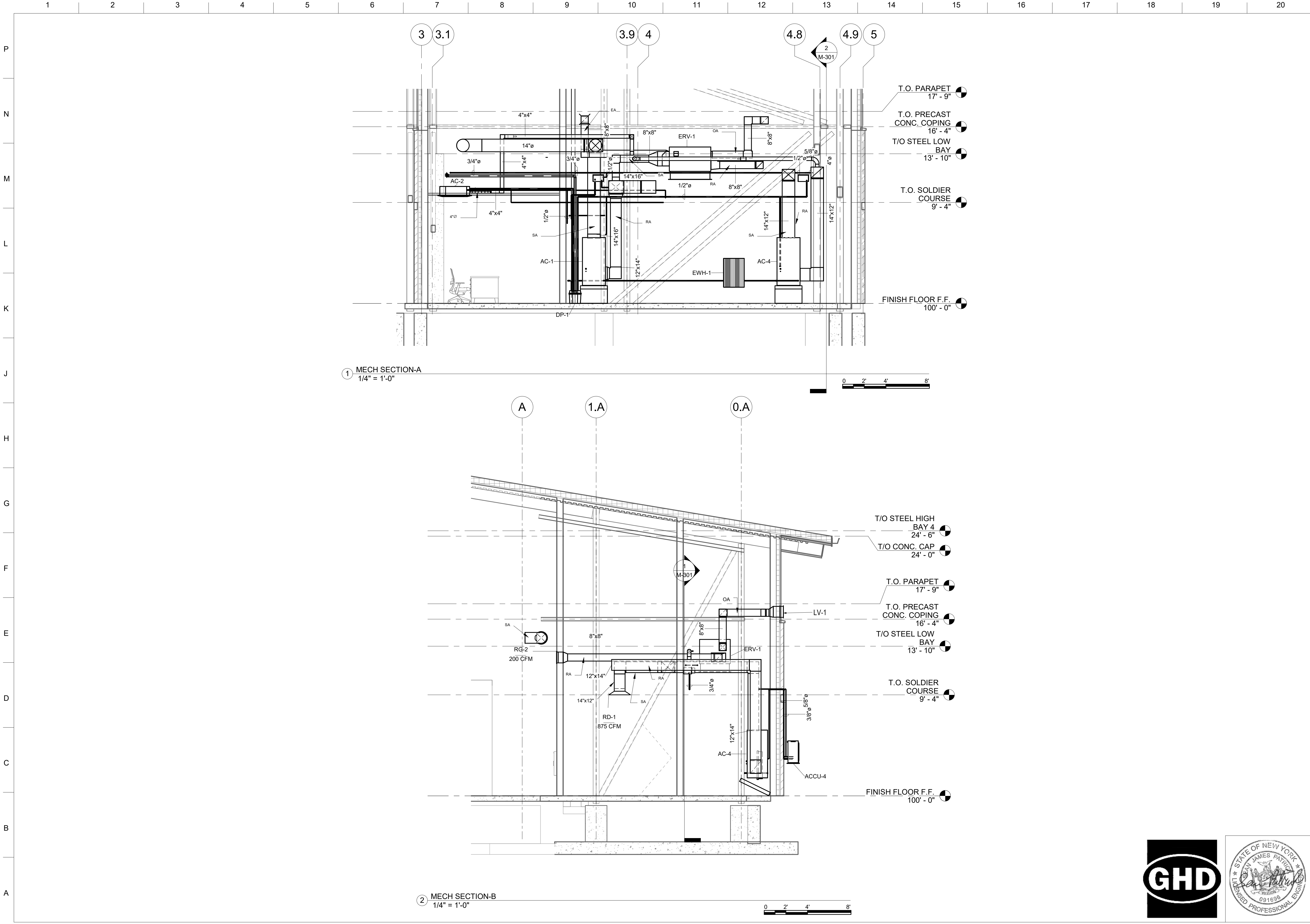
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SHEET ID

M-111





1 MECH SECTION-A
1/4" = 1'-0"

2 MECH SECTION-B
1/4" = 1'-0"



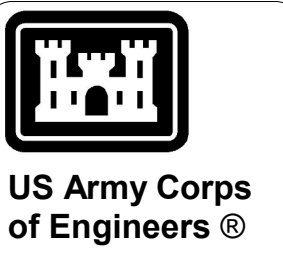
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NIAGARA FALLS AIR RESERVE STATION, NY
P2#-508014
MECHANICAL HVAC - SECTIONS

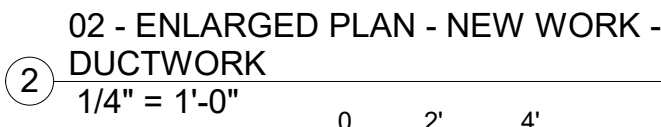
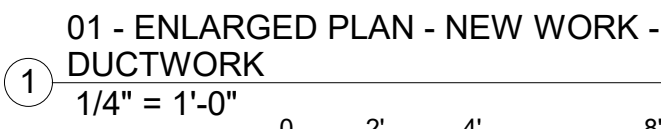
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LOUISVILLE DISTRICT
600 DR. MARTIN LUTHER KING, JR. PLACE
LOUISVILLE KY 40202-2239



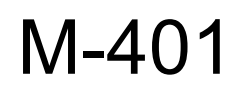
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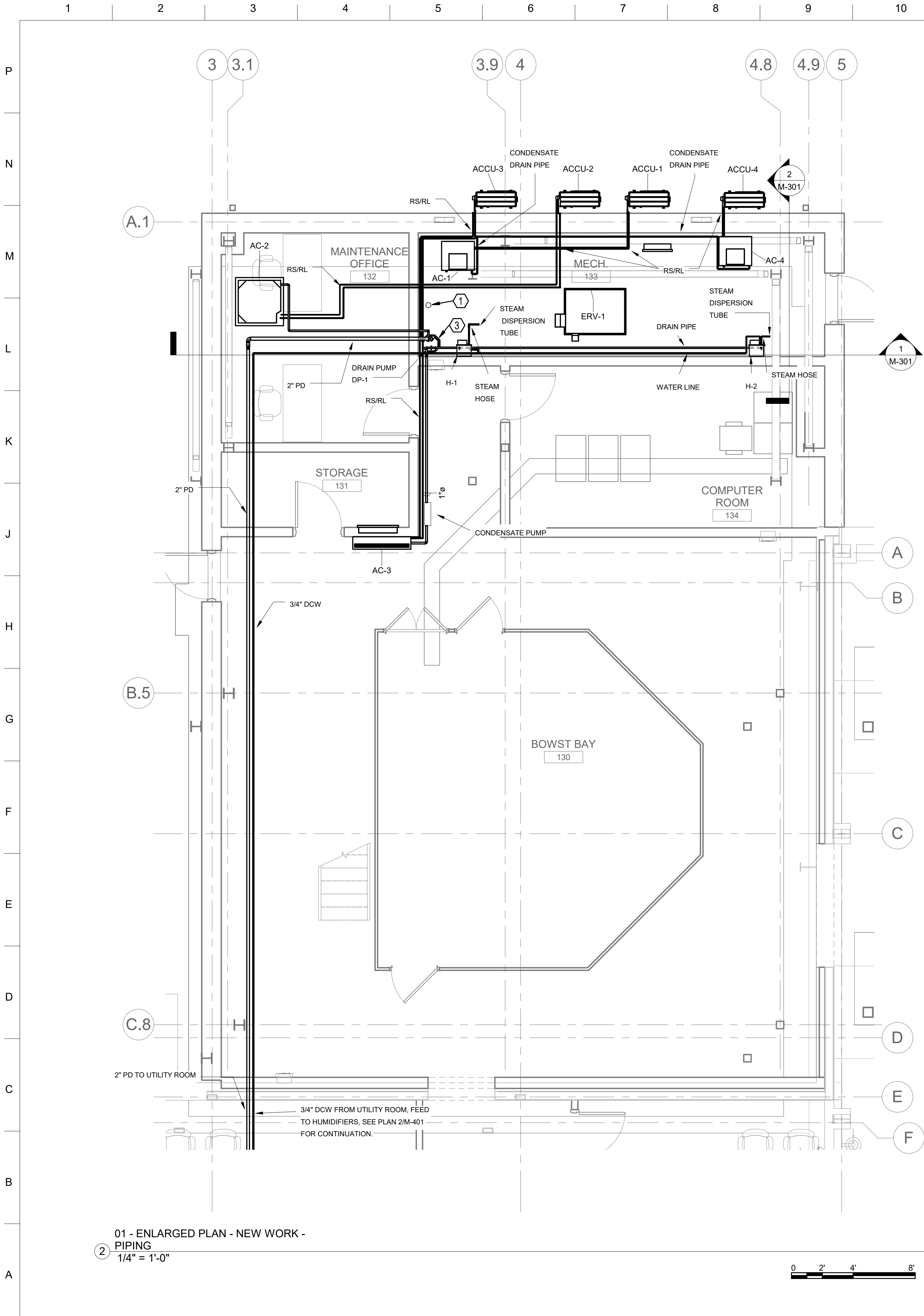
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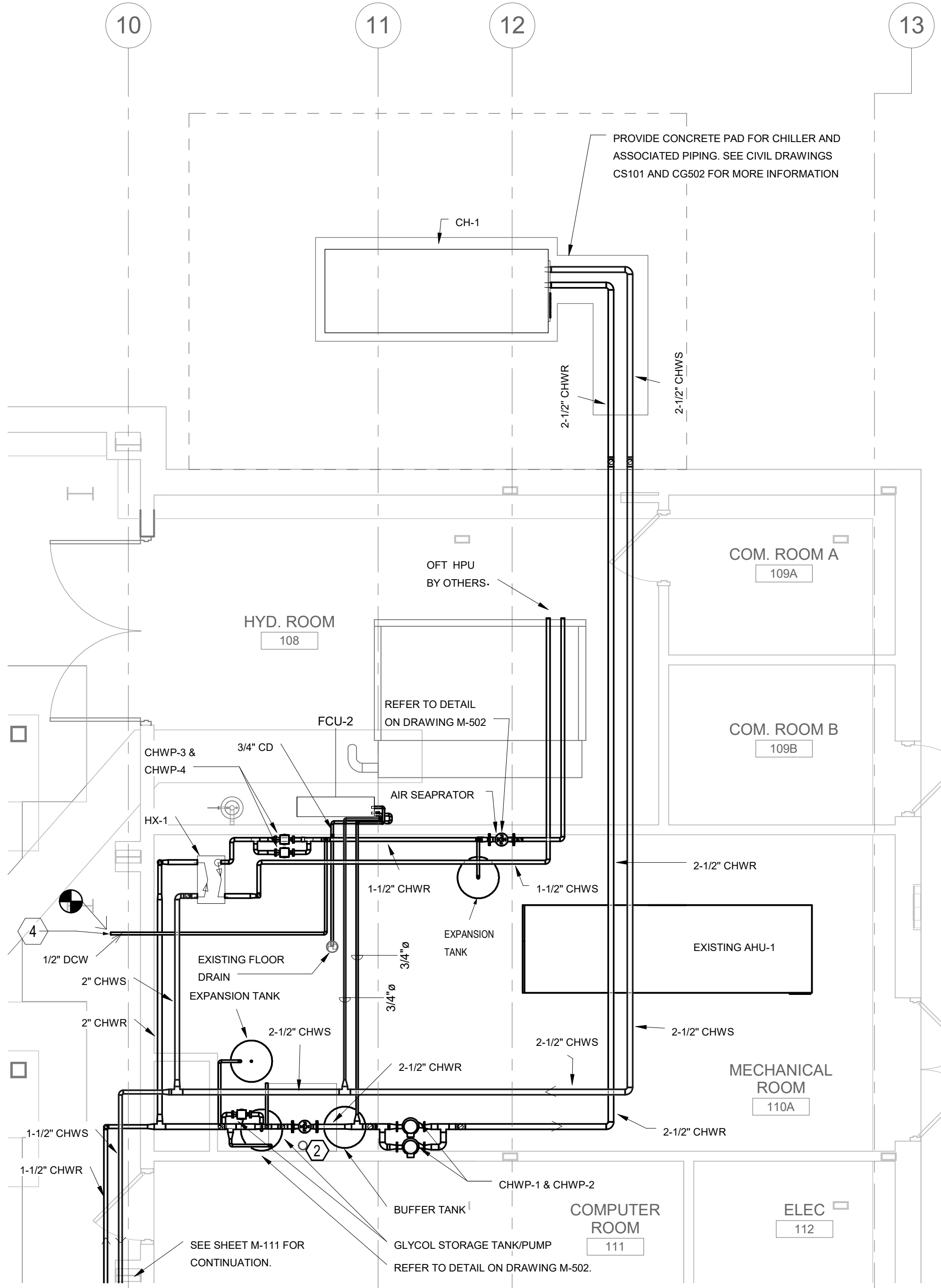


PLAN NORTH





02 - ENLARGED PLAN - NEW WORK-PIPING
1/4" = 1'-0"



KEY NOTES

- RADON EXHAUST SYSTEM, REF-1, PROVIDE 4" PVC SOLID WALL PIPE TO UNDERSLAB PERFORATED PIPE. INSTALL RADON EXHAUST FAN IN VERTICAL PIPE. EXTEND 4" PVC THROUGH ROOF.
- PROVIDE RADON EXHAUST SYSTEM. EXTEND 4" SOLID WALL PVC BELOW SLAB. PROVIDE REF-2 EXHAUST FAN IN VERTICAL PIPE. EXTEND 4" PVC PIPE DISCHARGE OUT THROUGH ROOF.
- EXTEND DRAIN LINES FROM AC-1, AC-2, AC-3, AC-4, H-1, AND H-2 TO INLET OF DRAIN PUMP DP-1.
- CONNECT TO EXISTING DCW. EXTEND TO CHWP-3/4 SYSTEM FOR MAKE-UP WATER FILL.

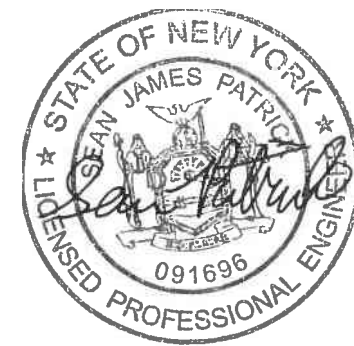


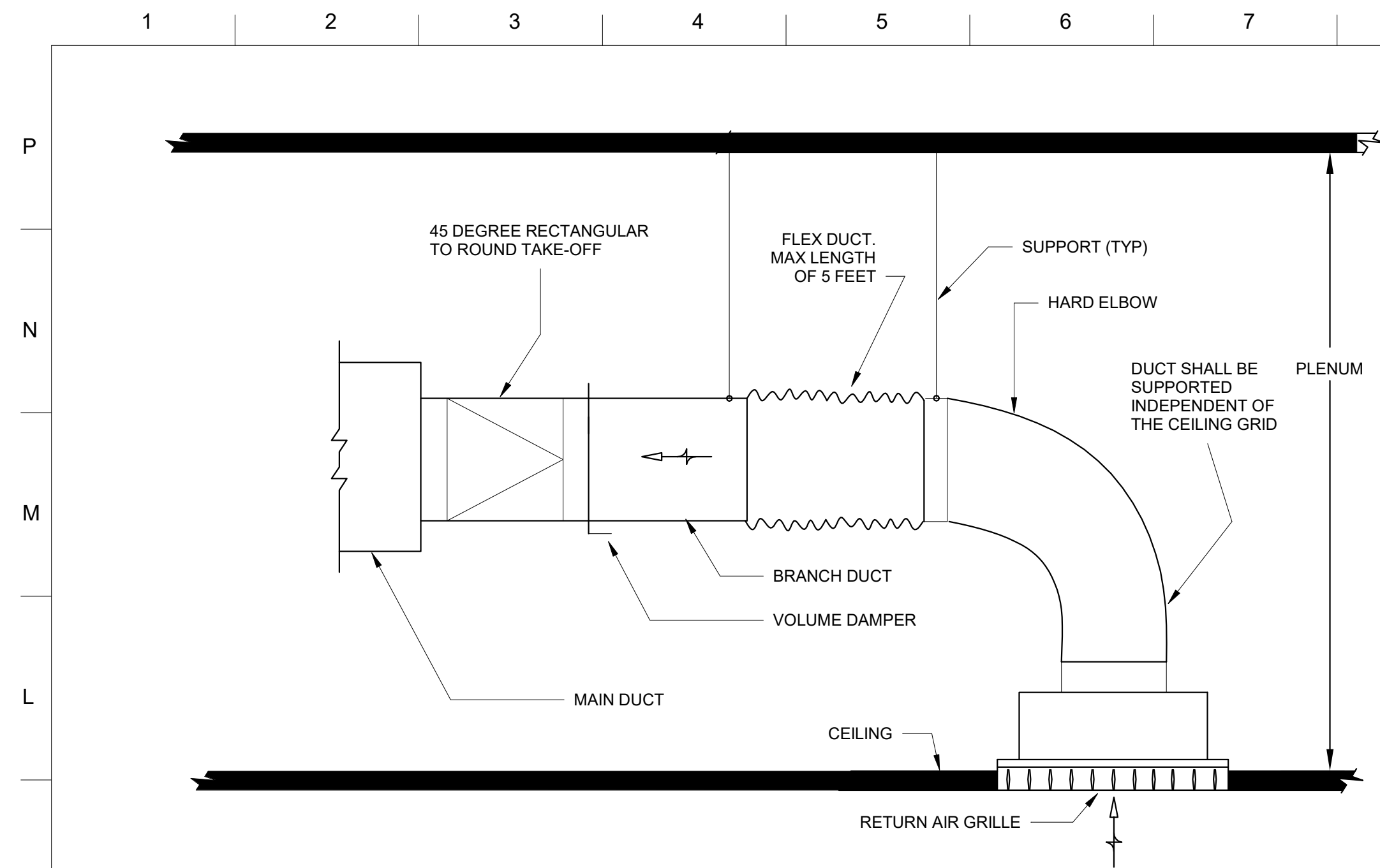
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US ARMY CORPS OF ENGINEERS LOUISVILLE DISTRICT 600 DR. MARTIN LUTHER KING, JR. PLACE LOUISVILLE KY 40202-2239	MECHANICAL HVAC - ENLARGED PLAN
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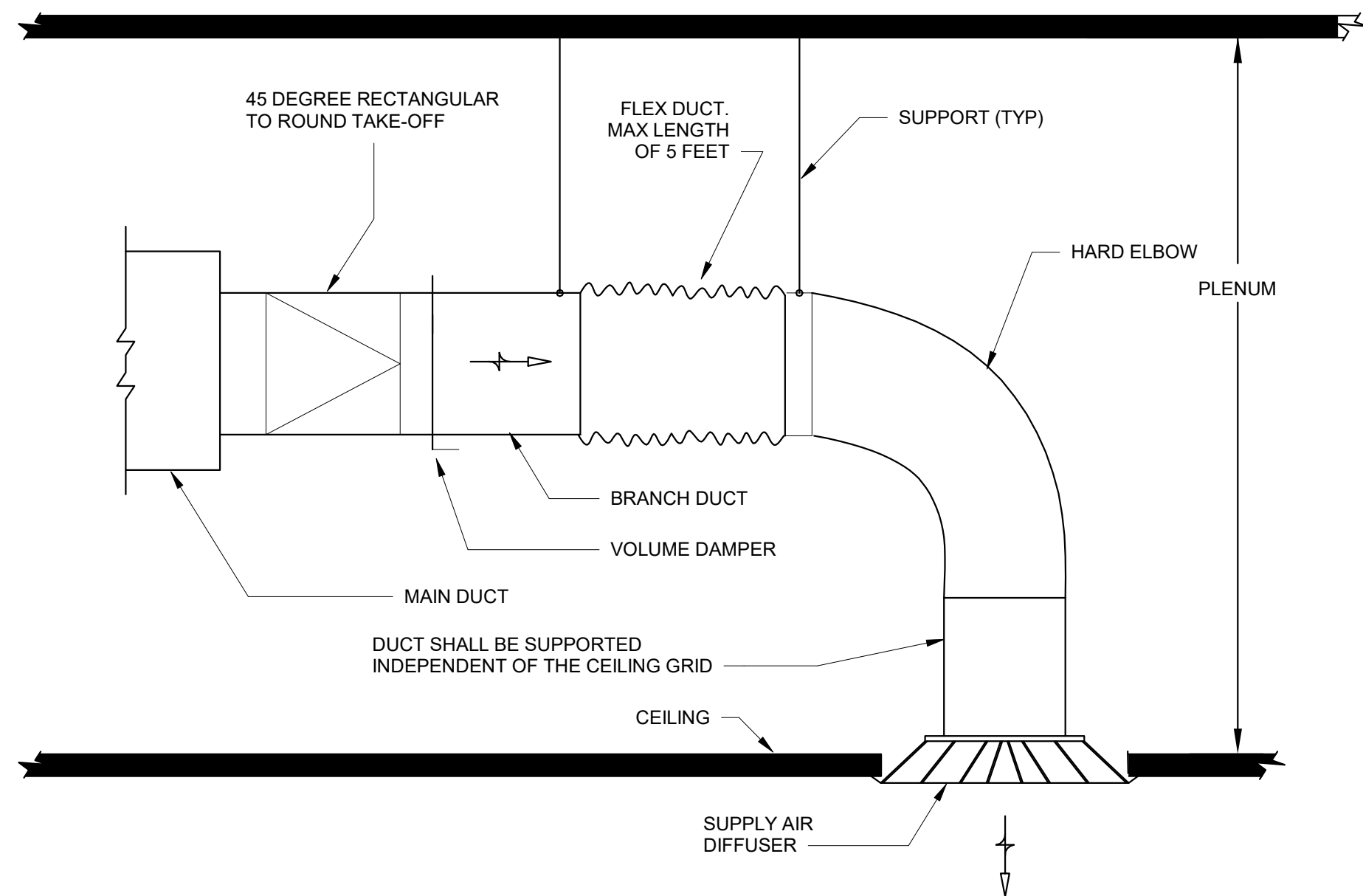
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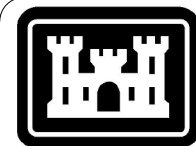


RETURN AIR GRILLE DETAIL

NTS



SUPPLY AIR DIFFUSER DETAIL



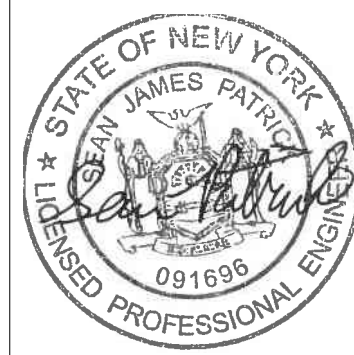
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 AEC <i>emersion</i> joint venture	DESIGNED BY: S. PATRICK	ISSUE DATE: JUNE 26, 2024
	DRAWN BY: H. NAYBI	SOLICITATION NO.: CONTRACT NO.:
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BOWST ADDITION AND RENOVATIONS TO B-321
 NIAGARA FALLS AIR RESERVE STATION, NY

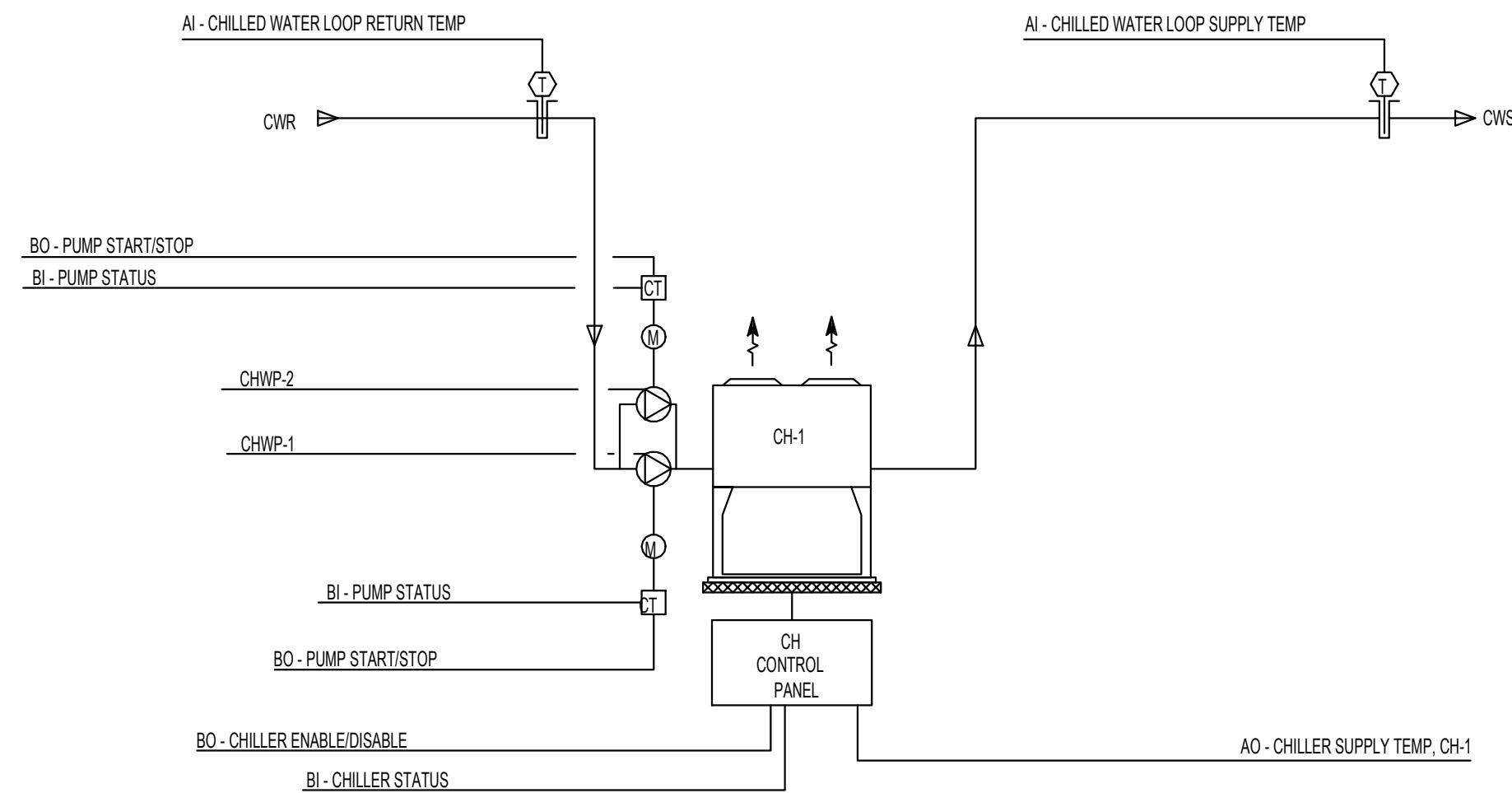
MECHANICAL HVAC - DETAILS



SHEET ID

M-503





CHILLED WATER LOOP

CHILLER SYSTEM AND PUMP - RUN CONDITIONS:

THE CHILLER SYSTEM MUST RUN SUBJECT TO ITS OWN INTERNAL SAFETIES AND CONTROLS. THE CHILLER SYSTEM MUST BE ENABLED TO RUN WHENEVER

- A CHILLED WATER DISTRIBUTION PUMP IS COMMANDED TO RUN

CHILLER CIRCULATION PUMP DUTY/STANDBY:

CHILLER DUTY PUMP MUST BE RUN WHENEVER THE ASSOCIATED CHILLER IS ENABLED, UNLESS SHUTDOWN BY SAFETIES. THE OPERATING PUMP MUST STOP AFTER A PREDETERMINED MINIMUM RUN TIME AFTER THE ASSOCIATED CHILLER HAS SHUTDOWN. ON FAILURE OF THE DUTY PUMP, THE STANDBY PUMP MUST RUN AND THE FAILED PUMP MUST TURN OFF.

CHILLED WATER SUPPLY TEMPERATURE CONTROL:

THE CHILLER CONTROLLER MUST MEASURE ITS RESPECTIVE SUPPLY WATER TEMPERATURE AND MODULATE RESPECTIVE CHILLER TO MAINTAIN SETPOINT. THE CHILLER SYSTEM MUST RUN SUBJECT TO ITS OWN INTERNAL SAFETIES AND CONTROLS. THE FOLLOWING SETPOINTS ARE RECOMMENDED VALUES. ALL SETPOINTS MUST BE FIELD ADJUSTED DURING THE COMMISSIONING PERIOD TO MEET THE REQUIREMENTS OF ACTUAL FIELD CONDITIONS.

THE CHILLER AND DUTY CIRCULATION PUMP MUST BE ENERGIZED TO MAINTAIN SETPOINT AS FOLLOWS

EACH CHILLER MUST MODULATE TO MAINTAIN ITS SUPPLY WATER TEMPERATURE AT 45°F (ADJ.).

THE CHILLER MUST DE-ENERGIZE WHEN LOOP RETURN WATER TEMPERATURE FALLS TO WITHIN 2°F (ADJ.) OF SUPPLY WATER TEMPERATURE SETPOINT

TO PREVENT SHORT CYCLING, THERE MUST BE A USER DEFINABLE (ADJ.) DELAY BETWEEN CHILLER ENERGIZING/DE-ENERGIZING AND MUST HAVE A USER DEFINABLE (ADJ.) MINIMUM RUNTIME.

ALARMS MUST BE PROVIDED AS FOLLOWS:

- FAILURE: COMMANDED ON, BUT THE STATUS IS OFF.
- RUNNING IN HAND: COMMANDED OFF, BUT THE STATUS IS ON.
- RUNTIME EXCEEDED: RUNTIME EXCEEDS A USER DEFINABLE LIMIT.
- HIGH SUPPLY TEMPERATURE: IF THE CHILLER SUPPLY TEMPERATURE IS MORE THAN 55°F (ADJ.).
- HIGH RETURN TEMPERATURE: IF THE LOOP RETURN WATER TEMPERATURE IS MORE THAN 57°F (ADJ.) FOR LONGER THAN 15 MINUTES (ADJ.).
- LOW SUPPLY TEMPERATURE: IF THE CHILLER SUPPLY WATER TEMPERATURE IS LESS THAN 45°F (ADJ.) FOR LONGER THAN 15 MINUTES (ADJ.).
- LOW FLOW: IF THE CHILLER SUPPLY WATER FLOW RATE IS LESS THAN THE SCHEDULED PUMP FLOW RATE (ADJ.) FOR LONGER THAN 15 MINUTES (ADJ.).

- CHILLER PUMP CHWP-1,2 (TYP. OF 2):
- FAILURE: COMMANDED ON, BUT THE STATUS IS OFF.
 - RUNNING IN HAND: COMMANDED OFF, BUT THE STATUS IS ON.
 - RUNTIME EXCEEDED: STATUS RUNTIME EXCEEDS A USER DEFINABLE LIMIT.
 - RUNNING IN HAND: COMMANDED OFF, BUT THE STATUS IS ON.

CHILLER (CH-1) AND CIRCULATION PUMP (CHWP-1,2) CONTROL DIAGRAM - MECHANICAL



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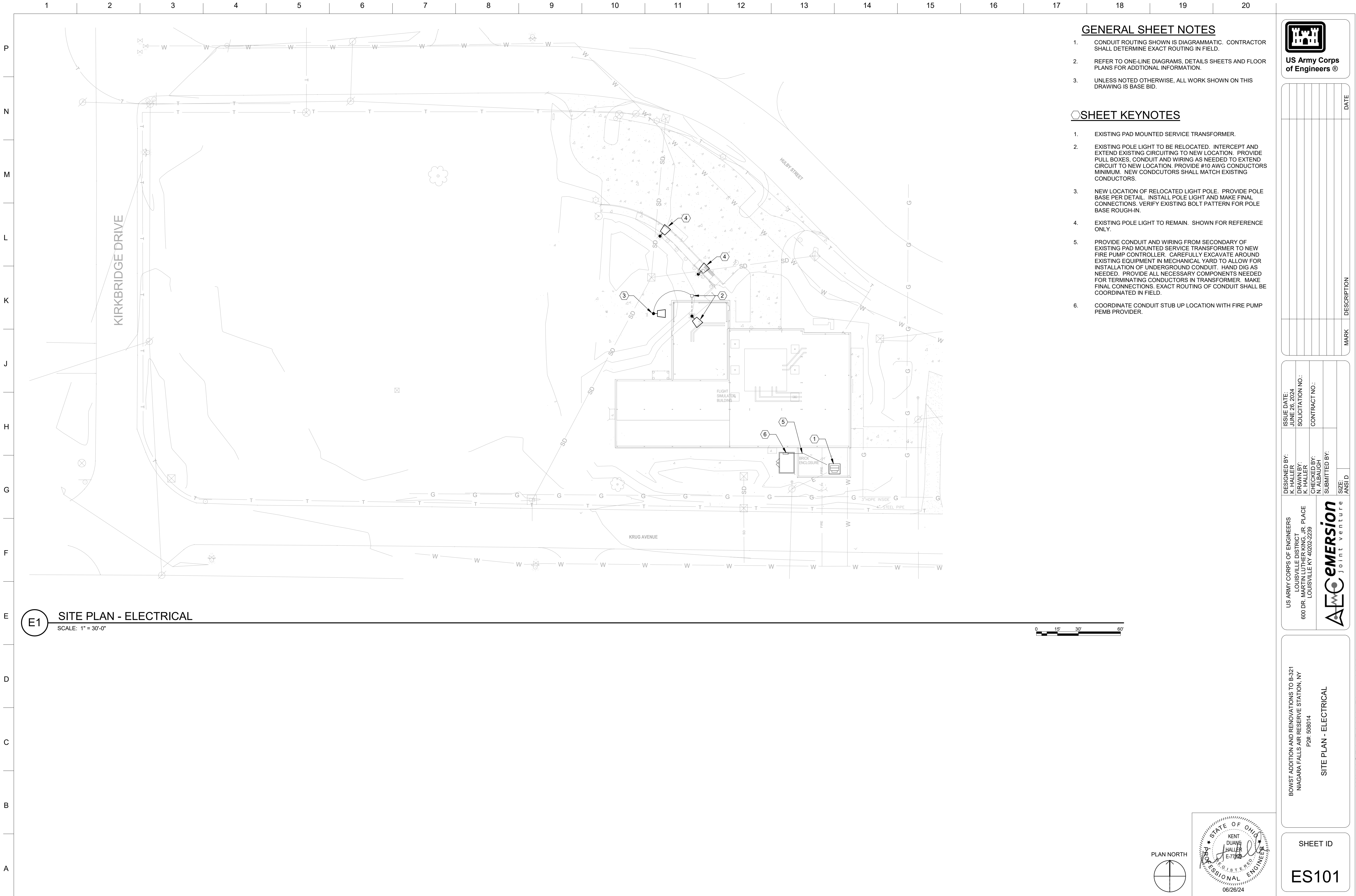
B30WST ADDITION AND RENOVATIONS TO B-321
NIAGARA FALLS AIR RESERVE STATION, NY

MECHANICAL HVAC - CONTROLS 2

SHEET ID

M-605



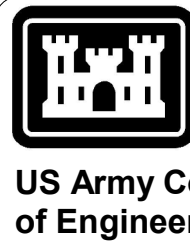


GENERAL SHEET NOTES

- 1. CONDUIT ROUTING SHOWN IS DIAGRAMMATIC. CONTRACTOR SHALL DETERMINE EXACT ROUTING IN FIELD.
- 2. REFER TO ONE-LINE DIAGRAMS, DETAILS SHEETS AND FLOOR PLANS FOR ADDITIONAL INFORMATION.
- 3. UNLESS NOTED OTHERWISE, ALL WORK SHOWN ON THIS DRAWING IS BASE BID.

SHEET KEYNOTES

- 1. EXISTING PAD MOUNTED SERVICE TRANSFORMER.
- 2. EXISTING POLE LIGHT TO BE RELOCATED. INTERCEPT AND EXTEND EXISTING CIRCUITING TO NEW LOCATION. PROVIDE PULL BOXES, CONDUIT AND WIRING AS NEEDED TO EXTEND CIRCUIT TO NEW LOCATION. PROVIDE #10 AWG CONDUCTORS MINIMUM. NEW CONDCUTORS SHALL MATCH EXISTING CONDUCTORS.
- 3. NEW LOCATION OF RELOCATED LIGHT POLE. PROVIDE POLE BASE PER DETAIL. INSTALL POLE LIGHT AND MAKE FINAL CONNECTIONS. VERIFY EXISTING BOLT PATTERN FOR POLE BASE ROUGH-IN.
- 4. EXISTING POLE LIGHT TO REMAIN. SHOWN FOR REFERENCE ONLY.
- 5. PROVIDE CONDUIT AND WIRING FROM SECONDARY OF EXISTING PAD MOUNTED SERVICE TRANSFORMER TO NEW FIRE PUMP CONTROLLER. CAREFULLY EXCAVATE AROUND EXISTING EQUIPMENT IN MECHANICAL YARD TO ALLOW FOR INSTALLATION OF UNDERGROUND CONDUIT. HAND DIG AS NEEDED. PROVIDE ALL NECESSARY COMPONENTS NEEDED FOR TERMINATING CONDUCTORS IN TRANSFORMER. MAKE FINAL CONNECTIONS. EXACT ROUTING OF CONDUIT SHALL BE COORDINATED IN FIELD.
- 6. COORDINATE CONDUIT STUB UP LOCATION WITH FIRE PUMP PEMB PROVIDER.



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LOUISVILLE DISTRICT
600 DR. MARTIN LUTHER KING, JR. PLACE
LOUISVILLE KY 40202-2239

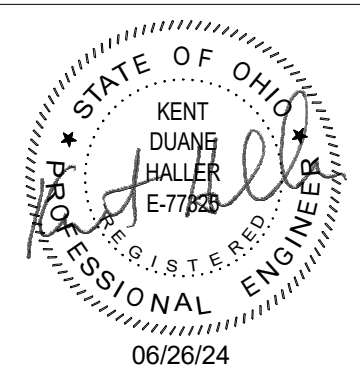
emersion
joint venture

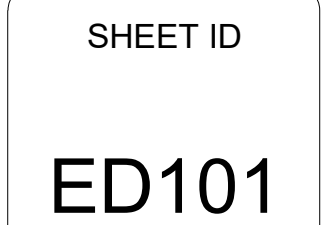
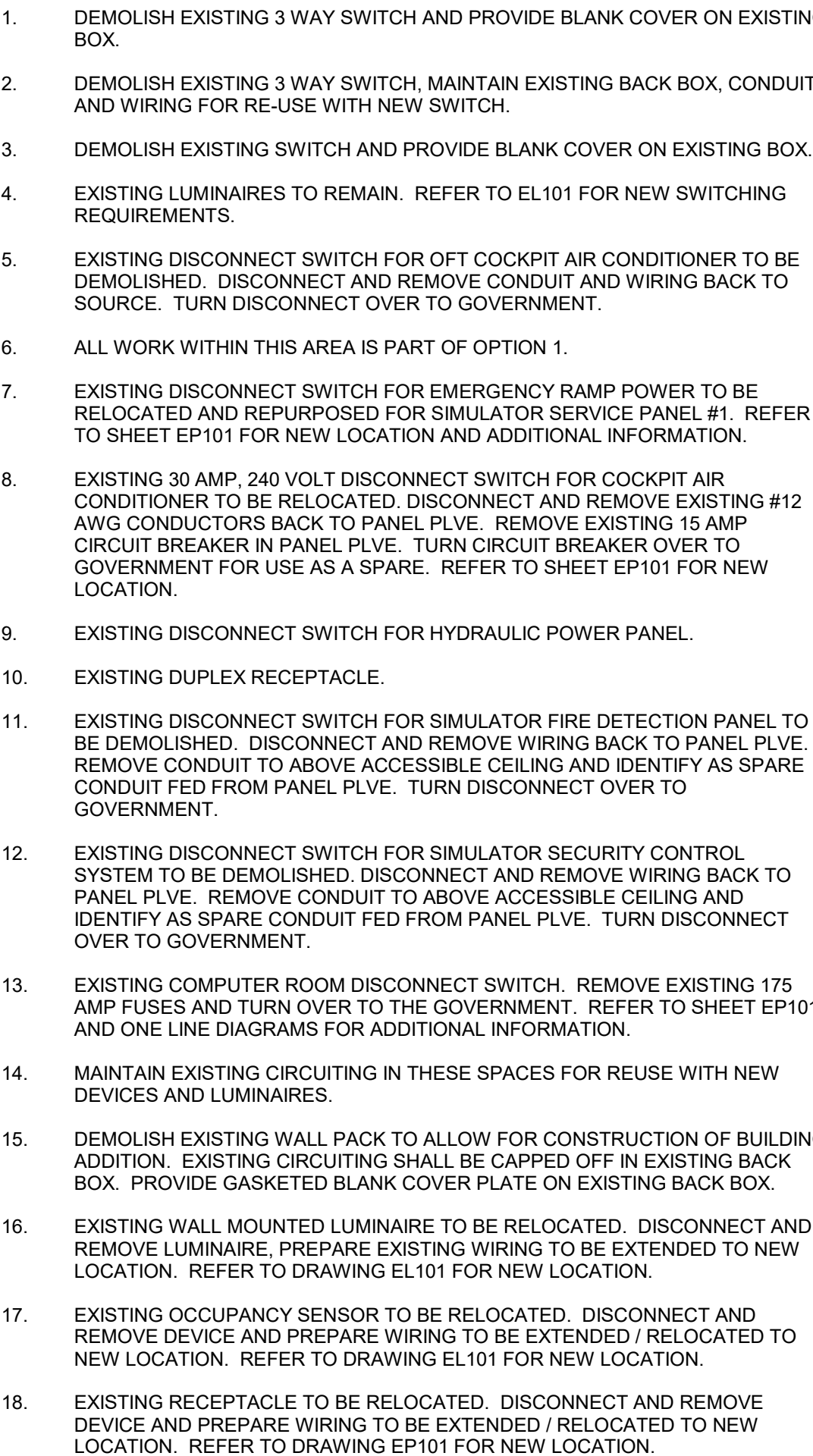
BOWST ADDITION AND RENOVATIONS TO B-321
NIAGARA FALLS AIR RESERVE STATION, NY
P2# 508014

SITE PLAN - ELECTRICAL

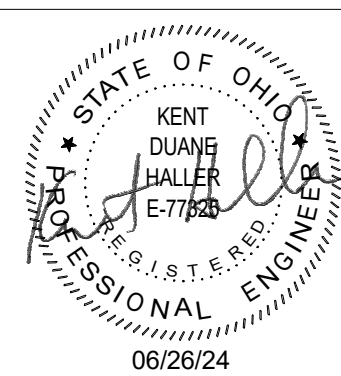
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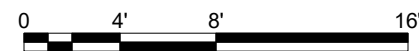
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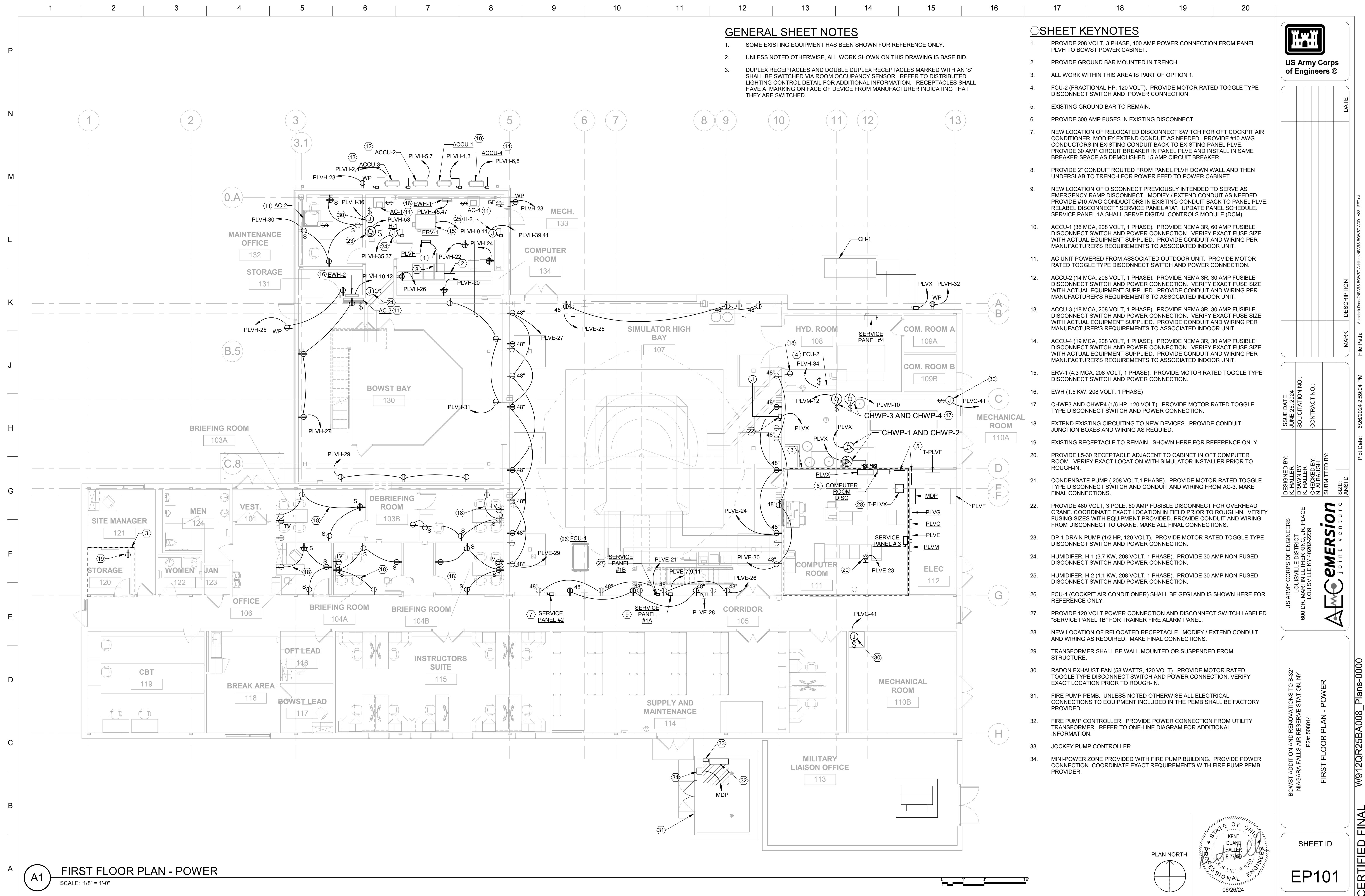


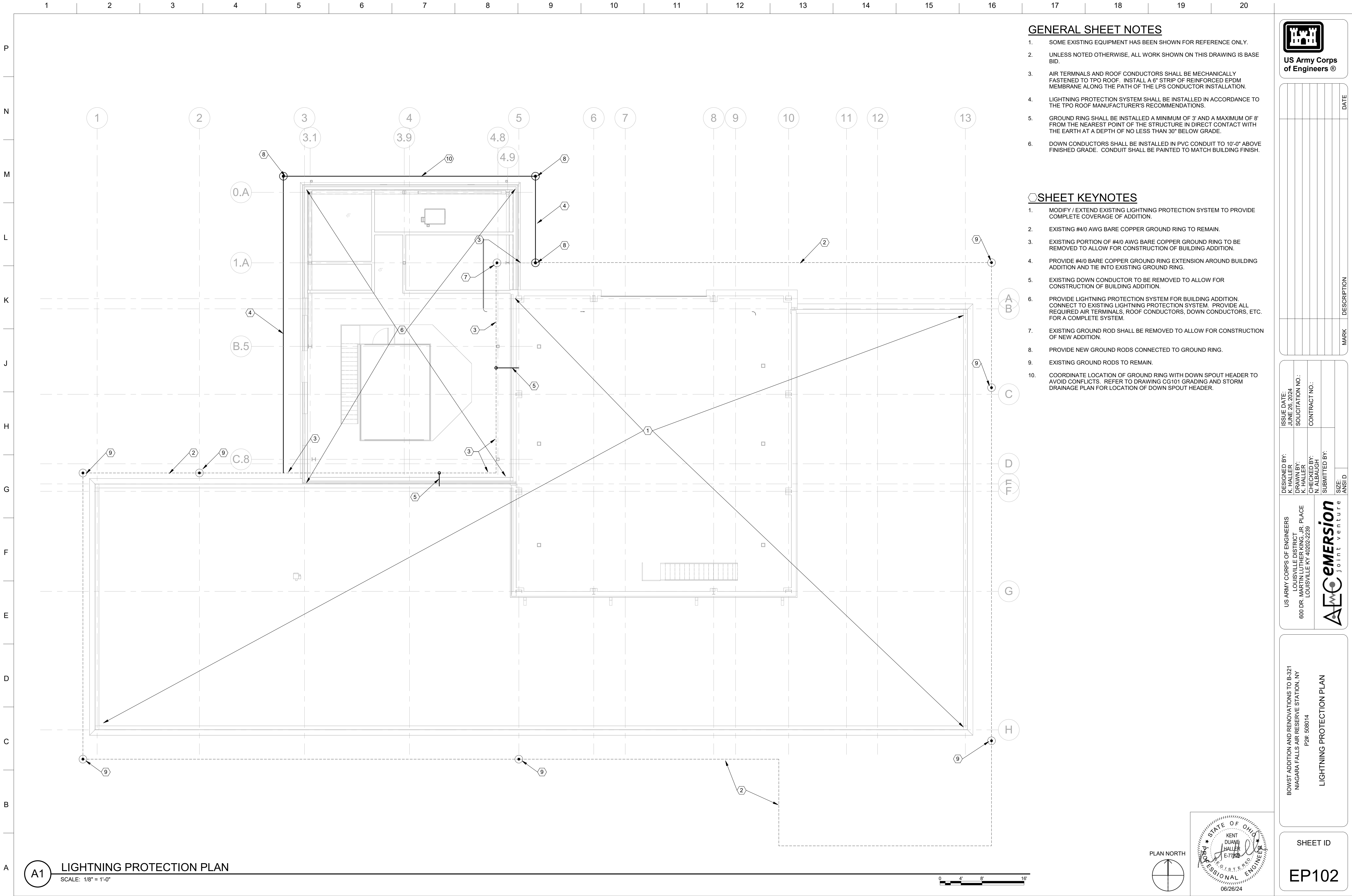
1. REFER TO ARCHITECTURAL REFLECTED CEILING PLANS FOR EXACT LOCATIONS OF LUMINAIRES. COORDINATE WITH OTHER TRADES CONTRACTORS IN ADVANCE OF INSTALLATION TO AVOID CONFLICTS OF SUFFICIENT SPACE ABOVE CEILINGS.
2. UNLESS NOTED OTHERWISE, ALL WORK SHOWN ON THIS DRAWING IS BASE BID.
3. WIRE EXIT SIGNS, NIGHT LIGHT AND EMERGENCY WALL PACKS AHEAD OF ALL SWITCHING.
4. OCCUPANCY SENSOR LOCATION - COORDINATE WITH MECHANICAL EQUIPMENT. COORDINATE LOCATION, TYPE AND QUANTITY PER MANUFACTURER'S RECOMMENDATIONS PRIOR TO INSTALLATION.
5. COORDINATE MOUNTING OF EXIT SIGNS WITH ARCHITECT PRIOR TO INSTALLATION OF BOXES AND CONDUIT. REFER TO REFLECTED CEILING PLANS, DOOR SCHEDULES, ELEVATIONS AND DETAILS.
6. MAINTAIN CONTINUITY OF BRANCH CIRCUITRY ASSOCIATED WITH ALL EXISTING TO REMAIN LIGHTING.
7. COORDINATE MOUNTING OF FIXTURE TYPE S2 AND S2A IN CORRIDOR, MECHANICAL AND COMPUTER ROOM WITH MECHANICAL DUCTWORK AND PIPING PRIOR TO INSTALLATION. FIXTURE SHALL BE CHAIN HUNG NO LOWER THAN 10'-0" AFF.
8. COORDINATE MOUNTING OF FIXTURE TYPE S3 WITH MECHANICAL DUCTWORK AND PIPING PRIOR TO INSTALLATION. FIXTURE SHALL BE CHAIN HUNG AT 24'-0" AFF.
9. COORDINATE MOUNTING OF TYPE S1, S1A AND S1B IN OFFICE AND BRIEF ROOM WITH ARCHITECT PRIOR TO INSTALLATION. FIXTURE SHALL BE PENDANT MOUNTED AT 7'-0" AFF.
10. OCCUPANCY SENSORS SHALL BE PROGRAMMED FOR MANUAL ON/ AUTO OFF AFTER 20 MIN. OF VACANCY UNO.
11. UNLESS NOTED OTHERWISE ALL W2A LUMINAIRES SHALL BE MOUNTED AT 9'-0" ABOVE FINISHED GRADE.

1. PROVIDE WALL MOUNTED OCCUPANCY SENSOR. MODIFY / EXTEND EXISTING WIRING AS REQUIRED FROM EXISTING LUMINAIRE TO OCCUPANCY SENSOR.
2. PROVIDE SINGLE POLE SWITCH AND INSTALL IN SAME LOCATION AS DEMOLISHED 3 WAY SWITCH. MODIFY WIRING AS REQUIRED FOR NEW SWITCHING SCHEME.
3. ALL WORK WITHIN THIS AREA IS PART OF OPTION 1.
4. CONNECT TO EXISTING LIGHTING CIRCUIT AND CONTROL WIRING IN CORRIDOR 105. MODIFY/ EXTEND EXISTING WIRING AS REQUIRED.
5. OCCUPANCY SENSORS IN BOWST BAY SHALL BE AUTO ON/ AUTO OFF AND SHALL BE CORNER MOUNTED AT 14'-0" AFF.
6. MOUNT THIS W2A LUMINAIRE AT 8'-0" ABOVE FINISHED GRADE TO COORDINATE WITH HEIGHT OF EXISTING CANOPY.
7. NEW LOCATION OF RELOCATED LUMINAIRE. MODIFY/ EXTEND CONDUIT AND WIRING AS REQUIRED. MAKE FINAL CONNECTIONS.
8. NEW LOCATION OF RELOCATED OCCUPANCY SENSOR. MODIFY/EXTEND CONDUIT AND WIRING AS REQUIRED. MAKE FINAL CONNECTIONS.
9. FIRE PUMP PEMB. ALL LUMINAIRES AND SWITCHES WITHIN THE FIRE PUMP PEMB SHALL BE PROVIDED AND WIRED BY THE FIRE PUMP PEMB MANUFACTURER.



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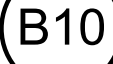




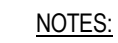
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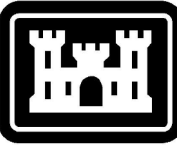


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US ARMY CORPS OF ENGINEERS
LOUISVILLE DISTRICT
600 DR. MARTIN LUTHER KING, JR. PLACE
LOUISVILLE KY 40202-2326

BOWST ADDITION AND RENOVATIONS TO B-321
 NIAGARA FALLS AIR RESERVE STATION, NY

P2#: 508014

DETAILS - ELECTRICAL

SHEET ID

E-501

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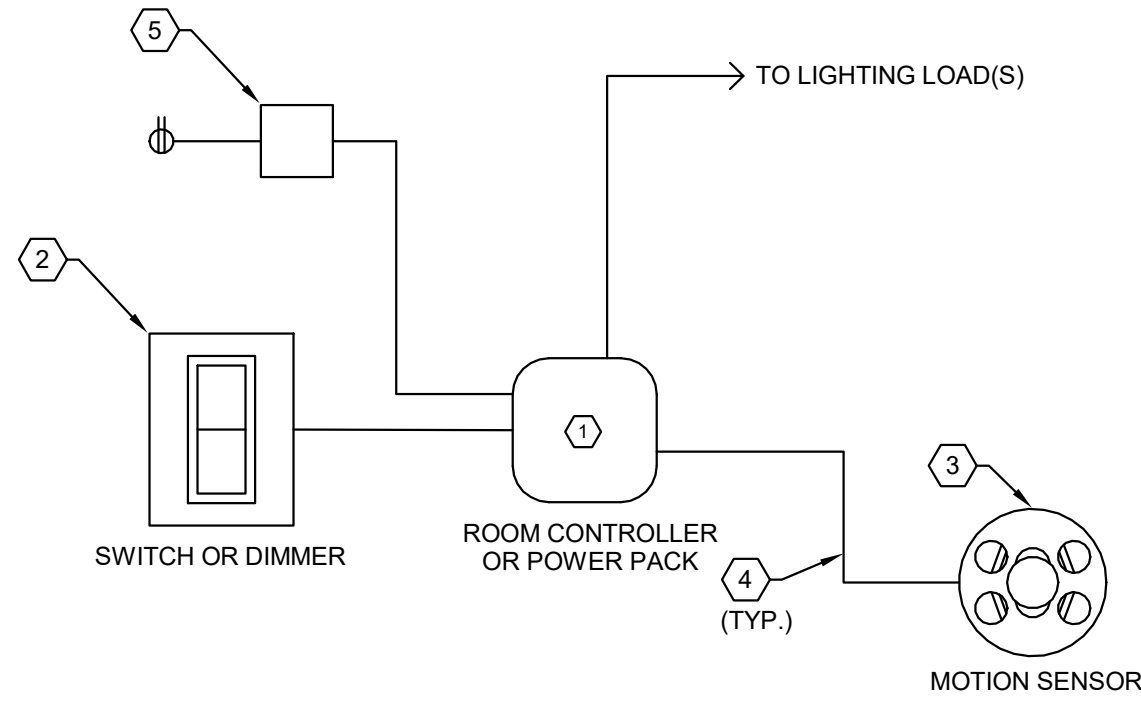
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DETAIL - DISTRIBUTED LIGHTING CONTROL

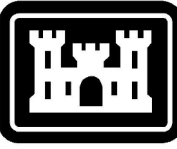
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GENERAL DETAIL NOTES

1. REFER TO FLOOR PLANS AND SPECIFICATION SECTION 26 51 00 INTERIOR LIGHTING. FOR ADDITIONAL LIGHT CONTROL SYSTEM REQUIREMENTS.

DETAIL KEYNOTES

1. ROOM CONTROLLER OR POWER PACK, PROVIDE PER SPECIFICATION 26 51 00 INTERIOR LIGHTING.
2. ROOMS SWITCH OR DIMMER, PROVIDE LOW VOLTAGE ROOM SWITCH OR DIMMER AS INDICATED ON FLOOR PLAN DRAWING. DIMMER SHALL BE 4 BUTTON ON, OFF, RAISE, AND LOWER. PROVIDE PER SPECIFICATION 26 51 00 INTERIOR LIGHTING.
3. OCCUPANCY SENSOR, PROVIDE OCCUPANCY SENSOR AS INDICATED ON FLOOR PLANS. ALL OCCUPANCY SENSORS ASSOCIATED WITH A ROOM CONTROLLER SHALL BE UTILIZED AS A VACANCY SENSOR. PROVIDE PER SPECIFICATION 26 51 00 INTERIOR LIGHTING.
4. PROVIDE CONTROL WIRING/CABLING FOR DISTRIBUTED CONTROL SYSTEM PER SPECIFICATION 26 51 00 INTERIOR LIGHTING.
5. POWER PACK FOR RECEPTACLE CONTROL, PROVIDE WHERE RECEPTACLES ARE INDICATED TO BE SWITCHED.

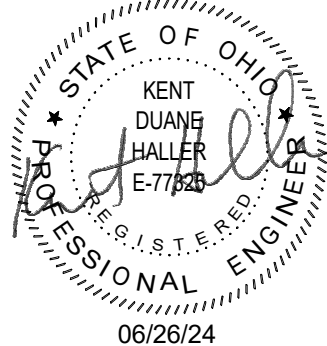
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 AECOM joint venture		DESIGNED BY: K. HALLER	ISSUE DATE: JUNE 26, 2024
		CHECKED BY: K. HALLER	SOLICITATION NO.:
		SUBMITTED BY: N. ALBAUGH	CONTRACT NO.:
US ARMY CORPS OF ENGINEERS LOUISVILLE DISTRICT 600 DR. MARTIN LUTHER KING, JR. PLACE LOUISVILLE KY 40202-2239			

**BOWST ADDITION AND RENOVATIONS TO B-321
NIAGARA FALLS AIR RESERVE STATION, NY**

P2#: 508014

DETAILS - ELECTRICAL





1. PROVIDE QUANTITY AND SIZE OF LUGS REQUIRED TO ACCOMMODATE CABLE SIZES SHOWN.
2. REFER TO SITE PLAN AND FLOOR PLAN DRAWINGS FOR ADDITIONAL INFORMATION.
3. UNLESS NOTED OTHERWISE, ALL WORK SHOWN ON THIS DRAWING IS BASE BID.

1. PROVIDE NEW CIRCUIT BREAKER IN LOCATION OF EXISTING "SPACE" FEEDER TO NEW PANELBOARD. NEW BREAKER SHALL MATCH MANUFACTURER TYPE AND AIC RATING FOR EXISTING PANEL. PANEL IS A GE SPECTRA SERIES WITH APBN HEAT RATED BOLT-ON STYLE BREAKERS, RATED AT 42 KAIC.
2. REMOVE AND REPLACE EXISTING 175 AMP FUSES AND PROVIDE 300 AMP FUSES.
3. EXISTING PANEL BOARD HAS BRANCH CIRCUIT MODIFICATIONS. REFER TO FLOOR PLANS AND PANEL SCHEDULE FOR ADDITIONAL INFORMATION.
4. COORDINATE EXACT FUSE SIZES WITH EQUIPMENT SUPPLIED.
5. PROVIDE CONDUIT AND WIRING FROM SECONDARY OF EXISTING PAD MOUNTED TRANSFORMER TO NEW FIRE PUMP CONTROLLER.
6. UTILIZE EXISTING SPARE CIRCUIT BREAKER FOR FEED TO NEW MINI-POWER ZONE IN FIRE PUMP PEMB.

NOTES:

- 1) IF CONDUIT OR CONDUCTOR SIZE ARE NOT INDICATED, SELECT FROM THE FEEDER SCHEDULE TABLE ABOVE.
- 2) NOT ALL SIZES SHOWN ARE USED.
- 3) "CONDUIT SIZES SHOWN ARE BASED ON EMT TYPE, AND CONDUCTORS WITH THHN, THWN, OR XHHW INSULATION. CONTRACT SHALL COMPLY WITH NEC FILL RATIOS IF ANY OTHER TYPE COMBINATIONS ARE USED."
- 4) AMPACITY SHOWN IS THE NOMINAL REQUIRED FOR CONDUCTORS INDICATED. CONDUCTORS WITH HIGHER CURRENT RATINGS MAY BE SHOWN ON THE DRAWINGS TO ACCOUNT FOR VOLTAGE DROP OR FOR FUTURE EXPANSION.



DESIGNED BY:	ISSUE DATE:
K. HALLER	JUNE 26, 2024
DRAWN BY:	SOLICITATION NO.:
K. HALLER	
CHECKED BY:	CONTRACT NO.:
N. ALBAUGH	
SUBMITTED BY:	
SIZE:	
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US ARMY CORPS OF ENGINEERS
LOUISVILLE DISTRICT
600 DR. MARTIN LUTHER KING, JR. PLACE
LOUISVILLE KY 40202-2239

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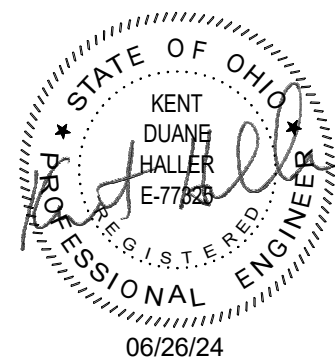
BOWST ADDITION AND RENOVATIONS TO B-321
 NIAGARA FALLS AIR RESERVE STATION, NY

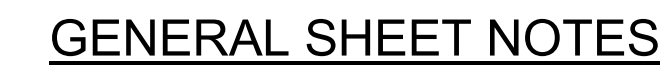
P2#: 508014

DIAGRAMS - ELECTRICAL

SHEET ID

E-701





- ## SHEET KEYNOTES

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BOWST ADDITION AND RENOVATIONS TO B-321
 NIAGARA FALLS AIR RESERVE STATION, NY
 PZH# 508014
 FIRST FLOOR PLAN - TELECOMMUNICATIONS -
 DEMOLITION

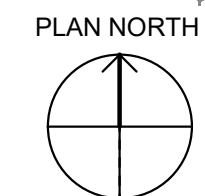
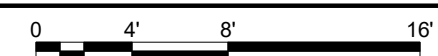
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FIRST FLOOR PLAN - TELECOMMUNICATIONS - DEMOLITION

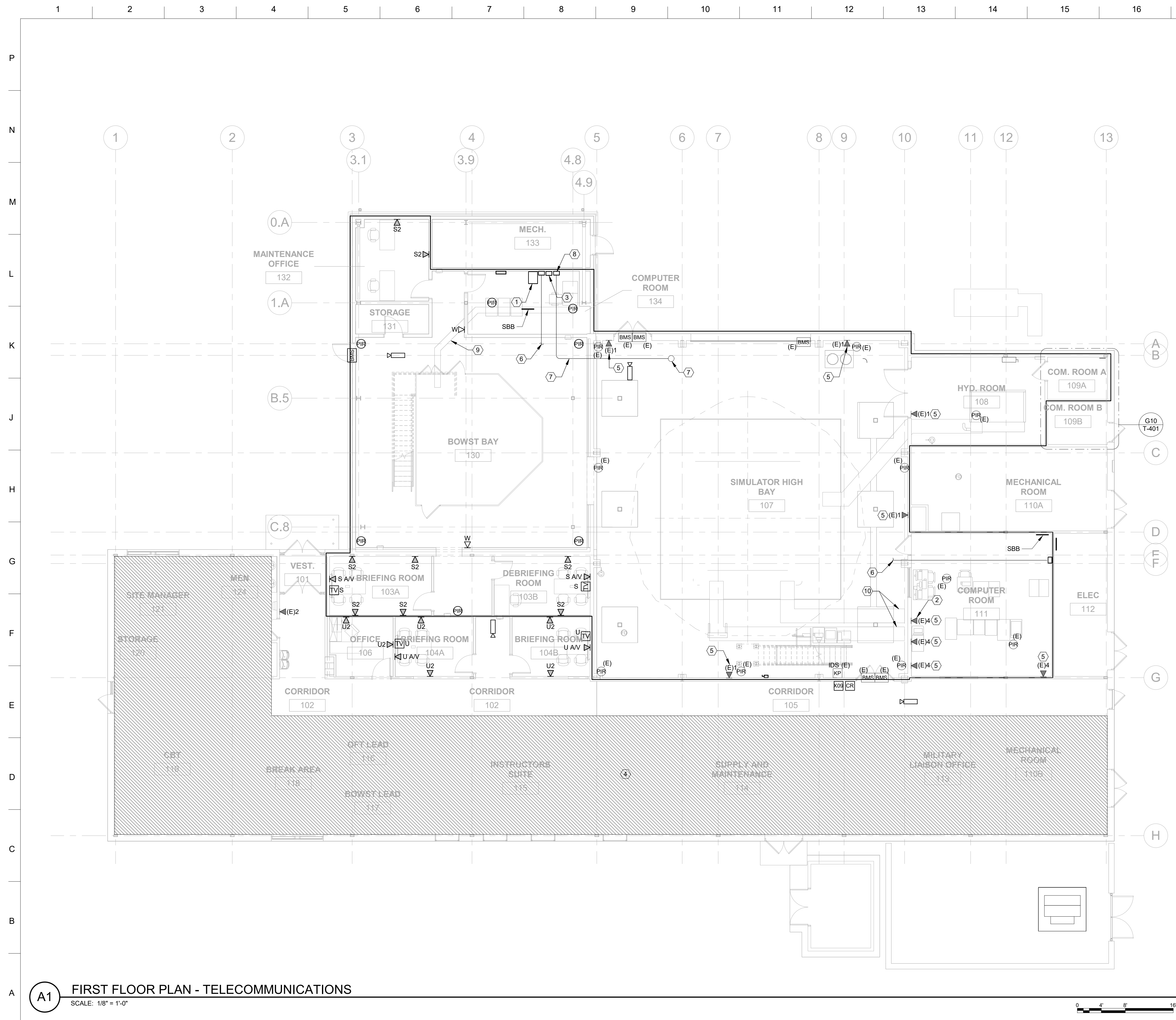
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PLAN NORTH



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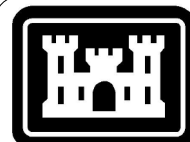


GENERAL SHEET NOTES

1. BOWST COMPUTER ROOM AND BOWST BAY EQUIPMENT IS PROVIDED BY BOWST EQUIPMENT VENDOR. CONTRACTOR SHALL PROVIDE DATA CONNECTION TO THE HOST CABINET IN THE COMPUTER ROOM. ANY CABLING AND TERMINATIONS AFTER THE HOST CABINET SHALL BE PROVIDED BY THE BOWST EQUIPMENT VENDOR.
2. COORDINATE THE EXACT LOCATION AND CABLING REQUIREMENTS FOR THE HOST CABINET WITH THE BOWST EQUIPMENT VENDOR.
3. ALL CABLING IN THE SECURED BOUNDARY, SEE SHEET A-101 SHALL BE SERVED FROM COM. ROOM A 109A. ALL CABLING OUTSIDE OF THE SECURED BOUNDARY SHALL BE SERVED FROM COM. ROOM 109B.
4. ALL IDS DEVICES SHALL BE ROUGH-IN WITH PULL STRINGS ONLY, WITH PATHWAYS ROUTED TO COM. ROOM A 109A.
5. ALL DATA JACKS AND CABLING WITHIN SECURED BOUNDARY SHALL BE GREEN.
6. ALL NEW DATA JACKS AND CABLING OUTSIDE THE SECURED BOUNDARY SHALL BE BLUE.

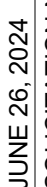
SHEET KEYNOTES

1. PROVIDE CAT6 CABLE FROM MEDIA CONVERTER IN COMPUTER ROOM 134 TO THE HOST CABINET IN BOWST COMPUTER ROOM. HOST CABINET PROVIDED BY BOWST EQUIPMENT VENDOR.
2. **OPTION 1:** UTILIZE EXISTING DATA OUTLET FOR CONNECTION TO OFT HOST CABINET. COORDINATE EXACT LOCATION WITH OFT EQUIPMENT VENDOR.
3. PROVIDE (1) SINGLE MODE FIBER OPTIC CABLE TO WALL MOUNTED FIBER TO CAT6 MEDIA CONVERTER FOR CONNECTION TO HOST CABINET IN BOWST COMPUTER ROOM 134.
4. NO SCOPE OF WORK IN THIS AREA.
5. PROVIDE NEW GREEN JACK AND GREEN CAT6 CABLING ROUTED FROM ROOM 109A AND TERMINATE AT MAINTAINED DEVICE.
6. PROVIDE 2" CONDUIT WITH (1) CAT6 CABLE AND 2-STRAND SINGLE MODE FIBER IN INNER DUCT, AND PULL STRING FROM COMPUTER ROOM 134 TO ROOM 111 FOR SIMULATOR CONNECTION. PROVIDE 6"x6" WALL MOUNTED BOX FOR TERMINATION OF CONDUIT AT BOTH ENDS. PROVIDE PULL BOXES AS NECESSARY PER SPECIFICATIONS AND FOR COMPLETE INSTALLATION. LEAVE CABLES COILED WITH 20' SERVICE LOOP AT EITHER END FOR TERMINATION BY SIMULATOR VENDORS.
7. PROVIDE 2" CONDUIT FOR GPS ANTENNA CABLING FROM COMPUTER ROOM, ROUTED TO SIMULATOR HIGH BAY AND THROUGH THE ROOF. STUB UP 30" ABOVE ROOF LINE AND PROVIDE WEATHERHEAD TO SEAL CONDUIT. SEE ARCHITECTURAL SHEET FOR ROOF FLASHING / SEALING REQUIREMENTS. COAXIAL CABLE AND ANTENNA SHALL BE PROVIDED BY SIMULATOR VENDOR.
8. PROVIDE 6" X 6" BOX ON WALL IN COMPUTER ROOM FOR TERMINATION OF 2" CONDUIT IN SHEET NOTE 7. PROVIDE #2 GROUND CONDUCTOR FROM BOX TO THE ROOM'S GROUND BAR AND TERMINATE PER SPECIFICATIONS.
9. FLOOR TRENCH FOR CABLING PATHWAY; SEE SHEET A-101, NOTE 03-040.
10. EXISTING FLOOR TRENCH FOR CABLING PATHWAY TO REMAIN; SEE SHEET A-101, NOTE 02-025.



**US Army Corps
of Engineers ®**

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	US ARMY CORPS OF ENGINEERS LOUISVILLE DISTRICT 600 DR. MARTIN LUTHER KING JR. PLACE LOUISVILLE KY 40202-2239	DESIGNED BY: C. BRIGGS CHECKED BY: C. BRIGGS	ISSUE DATE: JUNE 26, 2024
		SOLICITATION NO.: CONTRACT NO.:	

BOWST ADDITION AND RENOVATIONS TO B-321
NIAGARA FALLS AIR RESERVE STATION, NY

FIRST FLOOR PLAN - TELECOMMUNICATIONS

SHEET ID

T-101

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ENLARGED COM. ROOMS - TELECOMMUNICATIONS

SCALE: 1/2" = 1'-0"

SHEET KEYNOTES

- EXISTING 19" RACK 2-POST RACK WITH 6" WIDE VERTICAL CABLE MANAGEMENT ON BOTH SIDES OF THE RACK.
- EXISTING (2) 6" CONDUIT SLEEVES BETWEEN WALL, MOUNTED ABOVE EXISTING RACKS.
- EXISTING UNDERGROUND 2" CONDUIT STUBBED AND CAPPED 6" AFF.
- EXISTING UNDERGROUND 1" CONDUIT STUBBED AND CAPPED 6" AFF.
- EXISTING WALL MOUNTED FIRE RATED TELECOMMUNICATION PLYWOOD BACKBOARD.
- EXISTING TELECOM DUCTBANK, (4) 4" CONDUIT FLOOR PENETRATIONS, (3) OF THE CONDUITS ARE CURRENTLY SPARE.
- EXISTING WALL MOUNTED PROTECTED ENTRANCE TERMINAL.
- EXISTING 110 TYPE TERMINAL BLOCKS.
- EXISTING (2) 4" CONDUIT SLEEVES THROUGH WALL, MOUNTED NEAR CEILING.
- EXISTING LADDER RACK CABLE TRAY.
- EXISTING IDS CONTROL PANELS.
- EXISTING ACS CONTROL PANELS.
- PROVIDE (2) 24 PORT NETWORK SWITCH (CISCO 9300-24P, OR APPROVED EQUAL) AND (2) 24 PORT PATCH PANELS IN EXISTING RACK FOR CAT6 CABLES FOR THE BOWST PROGRAM. CABLES SHALL BE ROUTED IN 1" CONDUIT.
- PROVIDE 1" CONDUIT FROM COM. ROOM B TO COM. ROOM A FOR FIBER OPTIC CABLING TO CONNECT TO THE EXISTING RACK MOUNTED FIBER PATCH PANEL.
- EXISTING TECHNOLOGY BONDING BUSBAR. PROVIDE BONDING TO EXISTING RACKS AND CONNECT TO BONDING BUSBAR.

US ARMY CORPS OF ENGINEERS®

ISSUE DATE:
JUNE 26 2024

DESIGNED BY:
C. BRIGGS

DRAWN BY:
C. BRIGGS

CHECKED BY:
N. REED

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LOUISVILLE DISTRICT
600 DR. MARTIN LUTHER KING, JR. PLACE
LOUISVILLE KY 40202-2239

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BOWST ADDITION AND RENOVATIONS TO B-321
NIAGARA FALLS AIR RESERVE STATION, NY
P2#- 508014

ENLARGED PLANS - TELECOMMUNICATIONS

COMMUNICATIONS DISTRIBUTION

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Scale PLAN NORTH

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